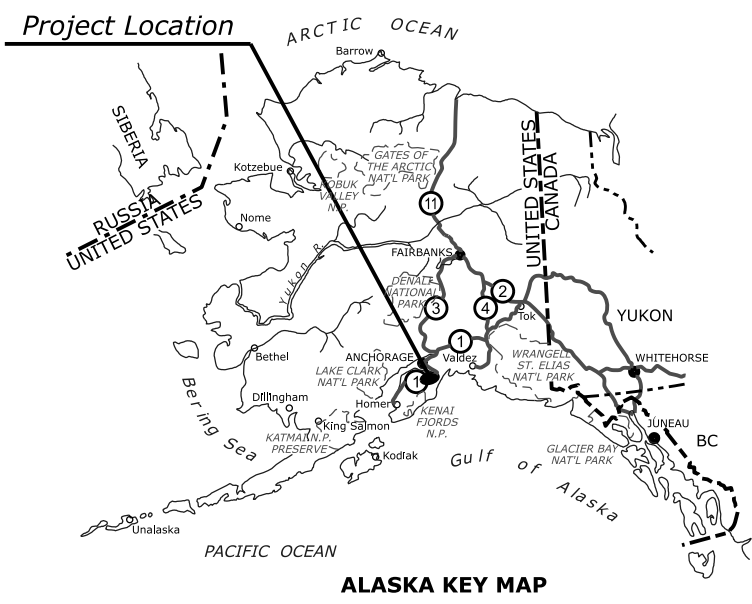


U. S. DEPARTMENT OF AGRICULTURE  
FOREST SERVICE  
CHUGACH NATIONAL FOREST



PLANS FOR PROPOSED PROJECT  
AK USFS CHUG1004 9(1)  
**VAGT LAKE TRAILHEAD**  
KENAI PENINSULA BOROUGH  
ALASKA

LENGTH 0.133 MILES



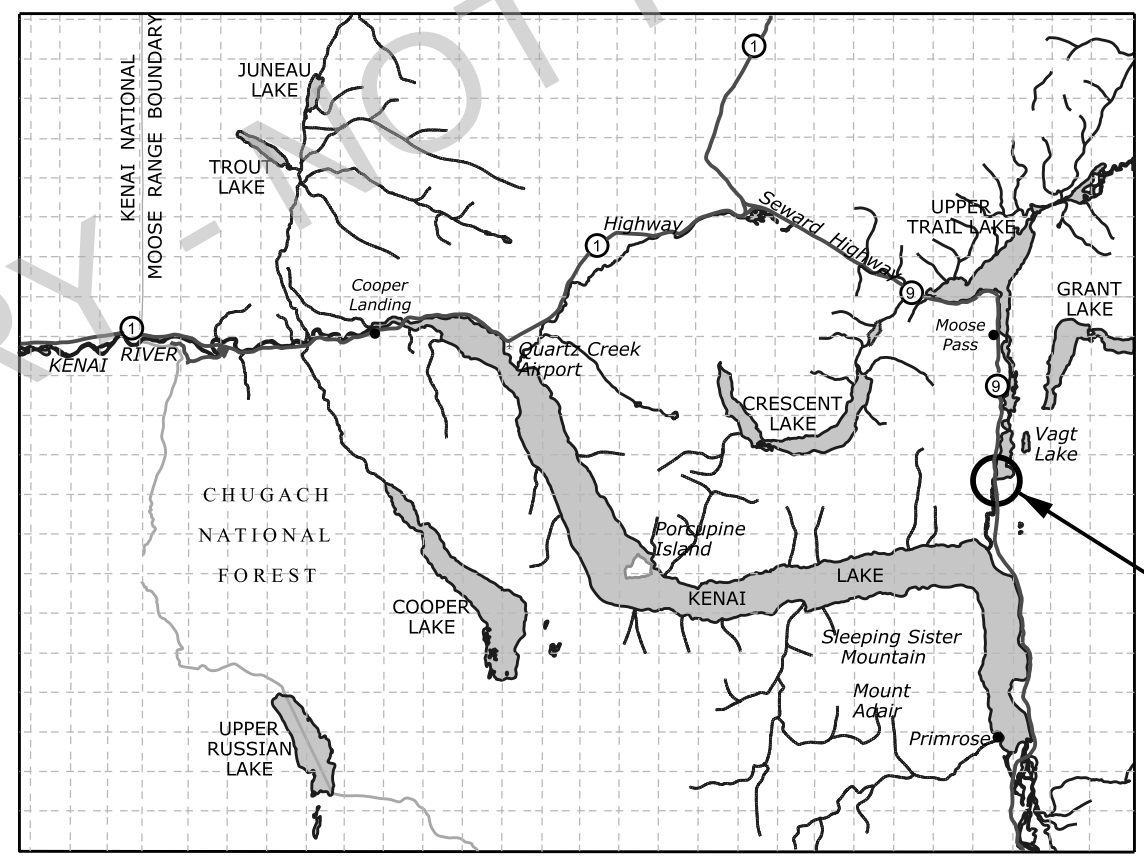
**TYPE OF CONSTRUCTION:**  
Roadway excavation, roadway aggregate,  
minor concrete pavement, drainage,  
signing, and striping

**DESIGN DESIGNATION:**  
ADT (2022) N/A  
ADT (2042) <400  
V 10 MPH

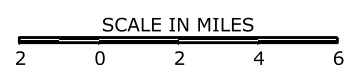
**SPECIFICATION:**  
Standard Specifications for  
Construction of Roads and Bridges  
on Federal Highway Projects, FP-14



PLANS PREPARED BY  
**U.S. DEPARTMENT OF TRANSPORTATION  
FEDERAL HIGHWAY ADMINISTRATION**  
WESTERN FEDERAL LANDS HIGHWAY DIVISION  
VANCOUVER, WASHINGTON



**PROJECT  
LOCATION**



|                       |              |
|-----------------------|--------------|
| PROJECT               | SHEET NUMBER |
| AK USFS CHUG1004 9(1) | A.1          |

| SECTION INDEX                        |
|--------------------------------------|
| A. GENERAL INFORMATION               |
| B. SUMMARY OF QUANTITIES             |
| C. SITE PLAN AND TYPICAL SECTIONS    |
| D. ROADWAY PLANS AND PROFILES        |
| E. TYPICAL DETAILS                   |
| G. SOIL EROSION AND SEDIMENT CONTROL |
| L. TEMPORARY TRAFFIC CONTROL         |
| M. PERMANENT TRAFFIC CONTROL         |

**RECOMMENDED BY:**

|                                                                        |      |
|------------------------------------------------------------------------|------|
| Seward District Ranger<br>U.S. Forest Service, Chugach National Forest | DATE |
| RLMH Staff Officer<br>U.S. Forest Service, Chugach National Forest     | DATE |
| Forest Engineer,<br>U.S. Forest Service, Chugach National Forest       | DATE |
| Forest Supervisor,<br>U.S. Forest Service, Chugach National Forest     | DATE |

**APPROVED:**

|                                                                    |      |
|--------------------------------------------------------------------|------|
| Regional Director of Recreation<br>U.S. Forest Service, Region 10  | DATE |
| Regional Director of Engineering<br>U.S. Forest Service, Region 10 | DATE |
| Chief of Engineering,<br>Western Federal Lands Highway Division    | DATE |

c:\pwwork\0511640\cnd00901.ea.dgn [Default] 23 July 2024 11:12 AM

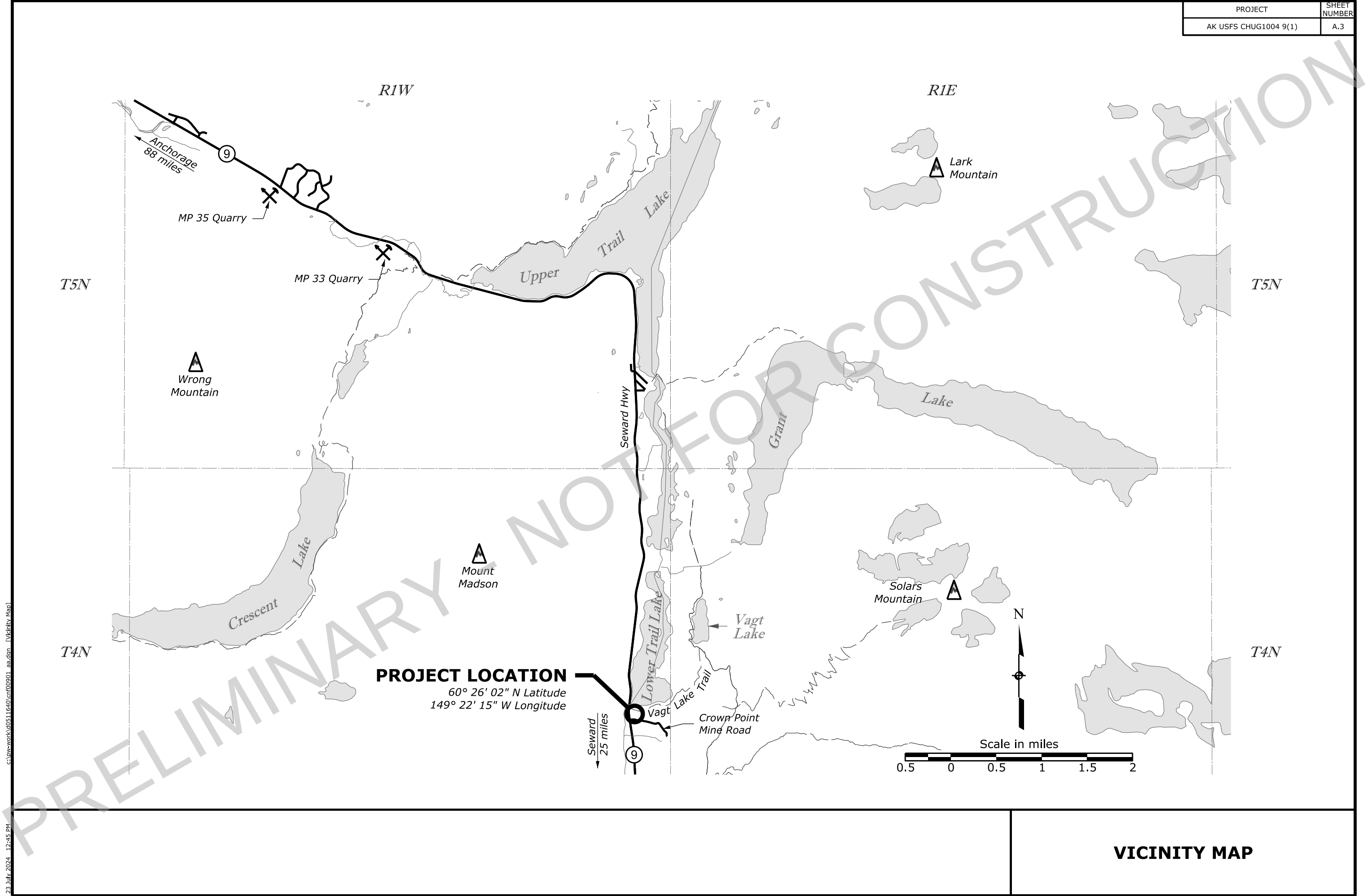
c:\pw-work\0511640\cnf00901\_aa.dgn [Sheet Index] 24 July 2024 10:00 AM

|                       |                 |
|-----------------------|-----------------|
| PROJECT               | SHEET<br>NUMBER |
| AK USFS CHUG1004 9(1) | A.2             |

| SHEET INDEX                                 |                                                  |
|---------------------------------------------|--------------------------------------------------|
| <b>A. GENERAL INFORMATION</b>               |                                                  |
| A.1                                         | TITLE SHEET                                      |
| A.2                                         | SHEET INDEX                                      |
| A.3                                         | VICINITY MAP                                     |
| A.4                                         | PLAN SYMBOLS AND ABBREVIATIONS                   |
| <b>B. SUMMARY OF QUANTITIES</b>             |                                                  |
| B.1-2                                       | SUMMARY OF QUANTITIES - SCHEDULE A               |
| <b>C. SITE PLAN AND TYPICAL SECTIONS</b>    |                                                  |
| C.1                                         | OVERALL SITE PLAN                                |
| C.2                                         | SURVEY CONTROL                                   |
| C.3                                         | TYPICAL ROADWAY SECTIONS                         |
| <b>D. ROADWAY PLANS AND PROFILES</b>        |                                                  |
| D.1                                         | TABULATION OF ROADWAY QUANTITIES                 |
| D.2                                         | VAGT LAKE ACCESS ROAD PLAN AND PROFILE           |
| D.3                                         | PARKING AREA PLAN                                |
| D.4-7                                       | VAGT LAKE TRAILHEAD LAYOUT AND GRADING PLANS 1-4 |
| D.8-10                                      | LAYOUT AND GRADING POINT DATA TABLES 1-3         |
| <b>E. TYPICAL DETAILS</b>                   |                                                  |
| E.1-2                                       | VAULT TOILET DETAILS                             |
| E.3                                         | PARKING AREA DETAILS                             |
| E.4-5                                       | MINOR CONCRETE PAVEMENT STD DETAILS              |
| G0-A8.4                                     | THREE POST DOUBLE KIOSK ARCHITECTURAL DRAWINGS   |
| G0                                          | COVER SHEET                                      |
| G1                                          | GENERAL NOTES, SHEET SCHEDULE                    |
| A2.1                                        | FLOOR PLAN                                       |
| A3.1                                        | EXTERIOR ELEVATIONS                              |
| A5.1                                        | FOUNDATION PLAN AND DETAILS                      |
| A7.1                                        | TRUSS DETAILS                                    |
| A7.2                                        | TRUSS DETAILS                                    |
| A7.3                                        | TRUSS PLATE DETAILS                              |
| A7.4                                        | COLUMN ANCHOR DETAILS                            |
| A8.3                                        | ROOF FRAMING DETAILS                             |
| A8.4                                        | ROOFING DETAILS                                  |
| <b>G. SOIL EROSION AND SEDIMENT CONTROL</b> |                                                  |
| G.1                                         | EROSION CONTROL PLAN                             |
| G.2-4                                       | EROSION CONTROL STD DETAILS                      |
| <b>L. TEMPORARY TRAFFIC CONTROL</b>         |                                                  |
| L.1                                         | TEMPORARY TRAFFIC CONTROL PLAN                   |
| L.2-3                                       | TEMPORARY TRAFFIC CONTROL STD DETAILS            |
| L.4                                         | SWPPP SIGN                                       |
| L.5                                         | BIPARTISAN INFRASTRUCTURE LAW SIGN               |

| SHEET INDEX                         |                                            |
|-------------------------------------|--------------------------------------------|
| <b>M. PERMANENT TRAFFIC CONTROL</b> |                                            |
| M.1                                 | PERMANENT TRAFFIC CONTROL PLAN             |
| M.2                                 | DETECTABLE WARNING PANEL DETAILS           |
| M.3                                 | VAGT LAKE TRAILHEAD SITE SIGN              |
| M.4                                 | SITE SIGN BASE DETAIL                      |
| M.5                                 | DIRECTIONAL SIGN DETAILS                   |
| M.6-7                               | PERMANENT TRAFFIC CONTROL STANDARD DETAILS |

SHEET INDEX



|                |                          |                                                                                                                   |                                      |                        |  |
|----------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------|------------------------|--|
|                |                          | PROJECT<br>AK USFS CHUG1004 9(1)                                                                                  |                                      | SHEET<br>NUMBER<br>A.4 |  |
| Δ              | total central angle      | M.L.                                                                                                              | main line                            | North Arrow            |  |
| Δc             | curve central angle      | M.P.                                                                                                              | mile post                            |                        |  |
| Ø              | diameter                 | matl.                                                                                                             | material                             |                        |  |
| θs             | spiral central angle     | max.                                                                                                              | maximum                              |                        |  |
| abut.          | abutment                 | MGAL                                                                                                              | thousand gallon                      |                        |  |
| ADT            | average daily traffic    | min.                                                                                                              | minimum                              |                        |  |
| AH             | ahead                    | mon.                                                                                                              | monument                             |                        |  |
| appr.          | approach                 | N                                                                                                                 | north                                |                        |  |
| BK             | back                     | NC                                                                                                                | normal crown                         |                        |  |
| BM             | bench mark               | o. c.                                                                                                             | on center                            |                        |  |
| BP             | balance point            | o. to o.                                                                                                          | out to out                           |                        |  |
| br.            | bridge                   | OD                                                                                                                | outside diameter                     |                        |  |
| brg.           | bearing                  | OG                                                                                                                | original ground                      |                        |  |
| cc or c. to c. | center to center         | PC                                                                                                                | point of curve                       |                        |  |
| £              | centerline               | PCC                                                                                                               | point of compound curve              |                        |  |
| clr.           | clear                    | PCS                                                                                                               | point of curve to spiral             |                        |  |
| CMP            | corrugated metal pipe    | PI                                                                                                                | point of intersection                |                        |  |
| col.           | column                   | pl.                                                                                                               | plate                                |                        |  |
| conc.          | concrete                 | POC                                                                                                               | point on curve                       |                        |  |
| conn.          | connection               | POS                                                                                                               | point on spiral                      |                        |  |
| constr. jt.    | construction joint       | POT                                                                                                               | point on tangent                     |                        |  |
| cont.          | continuous               | PS                                                                                                                | point of tangent to spiral           |                        |  |
| CS             | point of curve to spiral | PSC                                                                                                               | point of spiral to curve             |                        |  |
| ctrs.          | centers                  | PST                                                                                                               | point of spiral to tangent           |                        |  |
| CUFT           | cubic foot (feet)        | PT                                                                                                                | point of tangent                     |                        |  |
| culv.          | culvert                  | pvm.                                                                                                              | pavement                             |                        |  |
| CUYD           | cubic yard(s)            | R                                                                                                                 | radius                               |                        |  |
| D              | diameter                 | R.                                                                                                                | range                                |                        |  |
| DHV            | design hourly volume     | R/W                                                                                                               | right-of-way                         |                        |  |
| dia.           | diameter                 | rdwy.                                                                                                             | roadway                              |                        |  |
| diag.          | diagonal                 | reinf.                                                                                                            | reinforcement                        |                        |  |
| diaph.         | diaphragm                | reqd.                                                                                                             | required                             |                        |  |
| dist.          | distance                 | rt. or RT                                                                                                         | right                                |                        |  |
| drwg(s).       | drawing(s)               | rte.                                                                                                              | route                                |                        |  |
| E              | east                     | S                                                                                                                 | south                                |                        |  |
| e              | superelevation rate      | SADT                                                                                                              | seasonal average daily traffic       |                        |  |
| El. 94.16 ft   | elevation with number    | SC                                                                                                                | point of spiral to curve             |                        |  |
| elev.          | elevation                | sec.                                                                                                              | section                              |                        |  |
| emb.           | embankment               | shldr.                                                                                                            | shoulder                             |                        |  |
| EP             | edge of pavement         | SLRY                                                                                                              | slurry unit                          |                        |  |
| EQ or eq.      | equation                 | spa.                                                                                                              | spacing, spaces or spaced            |                        |  |
| ER             | edge of road             | SQFT                                                                                                              | square foot                          |                        |  |
| EW             | edge of water            | SQYD                                                                                                              | square yard                          |                        |  |
| exc.           | excavation               | SRS                                                                                                               | point of spiral to reverse spiral    |                        |  |
| exp. jt.       | expansion joint          | SS                                                                                                                | point of spiral to spiral (no curve) |                        |  |
| fin.           | finish                   | ST                                                                                                                | point of spiral to tangent           |                        |  |
| flg.           | flange                   | STA, Sta.                                                                                                         | station                              |                        |  |
| ft2            | square foot              | std.                                                                                                              | standard                             |                        |  |
| ft3            | cubic foot (feet)        | stgr.                                                                                                             | stringer                             |                        |  |
| ftg.           | footing                  | stiff.                                                                                                            | stiffener                            |                        |  |
| ga.            | gage (gauge)             | struc.                                                                                                            | structural                           |                        |  |
| galv.          | galvanized               | STS                                                                                                               | point of spiral to tangent spiral    |                        |  |
| hdwl.          | headwall                 | sym.                                                                                                              | symmetrical                          |                        |  |
| hex.           | hexagon                  | T                                                                                                                 | tangent distance                     |                        |  |
| HW             | high water               | T.                                                                                                                | township                             |                        |  |
| ID             | inside diameter          | TBM                                                                                                               | temporary bench mark                 |                        |  |
| jt.            | joint                    | thd.                                                                                                              | thread                               |                        |  |
| L              | length of curve          | TS                                                                                                                | point of tangent to spiral           |                        |  |
| lam.           | lamination               | Ts                                                                                                                | tangent distance (spiraled curve)    |                        |  |
| lat.           | latitude                 | typ.                                                                                                              | typical                              |                        |  |
| LNFT           | linear foot (feet)       | V                                                                                                                 | design speed                         |                        |  |
| long.          | longitudinal             | vph                                                                                                               | vehicles per hour                    |                        |  |
| LPSM           | lump sum                 | VPI                                                                                                               | vertical point of intersection       |                        |  |
| Ls             | length of spiral         | W                                                                                                                 | west                                 |                        |  |
| lt. or LT      | left                     | yd2                                                                                                               | square yard                          |                        |  |
| LW             | low water                | yd3                                                                                                               | cubic yard(s)                        |                        |  |
|                |                          | National Boundary                                                                                                 |                                      |                        |  |
|                |                          | State Boundary                                                                                                    |                                      |                        |  |
|                |                          | County Boundary                                                                                                   |                                      |                        |  |
|                |                          | City Boundary                                                                                                     |                                      |                        |  |
|                |                          | Township or Range Line                                                                                            |                                      |                        |  |
|                |                          | Section Line                                                                                                      |                                      |                        |  |
|                |                          | Section Corner (Found, Projected)                                                                                 |                                      |                        |  |
|                |                          | ¼ Section Line                                                                                                    |                                      |                        |  |
|                |                          | ¼ Section Corner (Found, Projected)                                                                               |                                      |                        |  |
|                |                          | ⅓ Section Line                                                                                                    |                                      |                        |  |
|                |                          | ⅓ Section Corner (Found, Projected)                                                                               |                                      |                        |  |
|                |                          | Property Line w/Found Property Corner                                                                             |                                      |                        |  |
|                |                          | Parcel Number                                                                                                     |                                      |                        |  |
|                |                          | National Park Boundary                                                                                            |                                      |                        |  |
|                |                          | National Forest Boundary                                                                                          |                                      |                        |  |
|                |                          | National Wildlife Refuge Boundary                                                                                 |                                      |                        |  |
|                |                          | BLM Lands Boundary                                                                                                |                                      |                        |  |
|                |                          | Indian Reservation Boundary                                                                                       |                                      |                        |  |
|                |                          | Existing Roadway (Road, Paved, Gravel)                                                                            |                                      |                        |  |
|                |                          | Railroad                                                                                                          |                                      |                        |  |
|                |                          | Trail                                                                                                             |                                      |                        |  |
|                |                          | Fiber Roll or Wattle                                                                                              |                                      |                        |  |
|                |                          | Silt Fence                                                                                                        |                                      |                        |  |
|                |                          | Intermittent Drainage or Small Creek                                                                              |                                      |                        |  |
|                |                          | Large Creek or River                                                                                              |                                      |                        |  |
|                |                          | Lake, Pond or Reservoir; Marshland                                                                                |                                      |                        |  |
|                |                          | Spring or Seep                                                                                                    |                                      |                        |  |
|                |                          | Treeline; Individual Trees                                                                                        |                                      |                        |  |
|                |                          | Material Source; Bore Hole; Test Pit                                                                              |                                      |                        |  |
|                |                          | Spot Elevation; Coordinate Grid Tick                                                                              |                                      |                        |  |
|                |                          | Above Ground Tank; Underground Tank                                                                               |                                      |                        |  |
|                |                          | Boulder; Well; Satellite Dish; Grave                                                                              |                                      |                        |  |
|                |                          | Cooking Grate; Garbage Can; Picnic Table                                                                          |                                      |                        |  |
|                |                          | Flagpole; Fire Hydrant                                                                                            |                                      |                        |  |
|                |                          | Gas & Water Meter; Gas & Water Valve                                                                              |                                      |                        |  |
|                |                          | Control Point (Terrestrial and GPS); Jump Hub                                                                     |                                      |                        |  |
|                |                          | EXISTING                                                                                                          |                                      |                        |  |
|                |                          | PROPOSED                                                                                                          |                                      |                        |  |
|                |                          | Slope Stake Limits                                                                                                |                                      |                        |  |
|                |                          | Fence                                                                                                             |                                      |                        |  |
|                |                          | Gate with Fence                                                                                                   |                                      |                        |  |
|                |                          | Cattleguard                                                                                                       |                                      |                        |  |
|                |                          | Guardrail                                                                                                         |                                      |                        |  |
|                |                          | Concrete Barrier                                                                                                  |                                      |                        |  |
|                |                          | Retaining Wall                                                                                                    |                                      |                        |  |
|                |                          | Signs (single, double post; portable)                                                                             |                                      |                        |  |
|                |                          | Delineators                                                                                                       |                                      |                        |  |
|                |                          | Pipe Culvert (arrow shows flow)                                                                                   |                                      |                        |  |
|                |                          | Pipe Culvert with End Section                                                                                     |                                      |                        |  |
|                |                          | Pipe Culvert with Headwall                                                                                        |                                      |                        |  |
|                |                          | Pipe Culvert with Drop Inlet                                                                                      |                                      |                        |  |
|                |                          | Box Culvert                                                                                                       |                                      |                        |  |
|                |                          | Underdrain                                                                                                        |                                      |                        |  |
|                |                          | Overhead/Above Ground Utilities                                                                                   |                                      |                        |  |
|                |                          | Underground Utilities                                                                                             |                                      |                        |  |
|                |                          | Poles (Power, Telephone, Joint Use, Light, Support w/Anchor)                                                      |                                      |                        |  |
|                |                          | Miscellaneous Utility Features                                                                                    |                                      |                        |  |
|                |                          | Building                                                                                                          |                                      |                        |  |
|                |                          | Right-of-Way Line with Monument                                                                                   |                                      |                        |  |
|                |                          | Permanent Easement                                                                                                |                                      |                        |  |
|                |                          | Construction Easement                                                                                             |                                      |                        |  |
|                |                          | Riprap                                                                                                            |                                      |                        |  |
|                |                          | NO SCALE                                                                                                          |                                      |                        |  |
|                |                          | U.S. DEPARTMENT OF TRANSPORTATION<br>FEDERAL HIGHWAY ADMINISTRATION<br>WESTERN FEDERAL LANDS AND HIGHWAY DIVISION |                                      |                        |  |
|                |                          | FLH STANDARD                                                                                                      |                                      |                        |  |
|                |                          | PLAN SYMBOLS<br>AND ABBREVIATIONS                                                                                 |                                      |                        |  |
|                |                          | DETAIL APPROVED FOR USE 11/2001                                                                                   |                                      |                        |  |
|                |                          | STANDARD                                                                                                          |                                      |                        |  |
|                |                          | REVISED: 9/2005 1/2007 10/2009 10/2014                                                                            |                                      |                        |  |
|                |                          | W101-1                                                                                                            |                                      |                        |  |

Δ

Δc

Ø

θs

abut.

ADT

AH

appr.

BK

BM

BP

br.

brg.

cc or c. to c.

£

clr.

CMP

col.

conc.

conn.

constr. jt.

cont.

CS

ctrs.

CUFT

culv.

CUYD

D

DHV

dia.

diag.

diaph.

dist.

drwg(s).

E

e

El. 94.16 ft

elev.

emb.

EP

EQ or eq.

ER

EW

exc.

exp. jt.

fin.

flg.

ft2

ft3

ftg.

ga.

galv.

hdwl.

hex.

HW

ID

jt.

L

lam.

lat.

LNFT

long.

LPSM

Ls

lt. or LT

LW

total central angle

curve central angle

diameter

spiral central angle

abutment

average daily traffic

ahead

approach

back

bench mark

balance point

bridge

bearing

center to center

centerline

clear

corrugated metal pipe

column

concrete

connection

construction joint

continuous

point of curve to spiral

centers

cubic foot (feet)

culvert

cubic yard(s)

diameter

design hourly volume

diameter

diagonal

diaphragm

distance

drawing(s)

east

superelevation rate

elevation with number

elevation

embankment

edge of pavement

equation

edge of road

edge of water

excavation

expansion joint

finish

flange

square foot

cubic foot (feet)

footing

gage (gauge)

galvanized

headwall

hexagon

high water

inside diameter

joint

length of curve

lamination

latitude

linear foot (feet)

longitudinal

lump sum

length of spiral

left

low water

M.L.

M.P.

matl.

max.

MGAL

min.

mon.

N

NC

o. c.

o. to o.

OD

OG

PC

PCC

PCS

PI

pl.

POC

POS

POT

PS

PSC

PST

PT

pvm.

R

R.

R/W

rdwy.

reinf.

reqd.

rt. or RT

rte.

S

SADT

SC

sec.

shldr.

SLRY

spa.

SQFT

SQYD

SRS

SS

ST

STA, Sta.

std.

stgr.

stiff.

struc.

STS

sym.

T

T.

TBM

thd.

TS

Ts

typ.

V

vph

VPI

W

yd2

yd3

main line

mile post

material

maximum

thousand gallon

minimum

monument

north

normal crown

on center

out to out

outside diameter

original ground

point of curve

point of compound curve

point of curve to spiral

point of intersection

plate

point on curve

point on spiral

point on tangent

point of tangent to spiral

point of spiral to curve

point of spiral to tangent

point of tangent

pavement

radius

range

right-of-way

roadway

reinforcement

required

right

route

south

seasonal average daily traffic

point of spiral to curve

section

shoulder

slurry unit

spacing, spaces or spaced

square foot

square yard

point of spiral to reverse spiral

point of spiral to spiral (no curve)

point of spiral to tangent

station

standard

stringer

stiffener

structural

point of spiral to tangent spiral

symmetrical

tangent distance

township

temporary bench mark

thread

point of tangent to spiral

tangent distance (spiraled curve)

typical

design speed

vehicles per hour

vertical point of intersection

west

square yard

cubic yard(s)

National Boundary

State Boundary

County Boundary

City Boundary

Township or Range Line

Section Line

Section Corner (Found, Projected)

¼ Section Line

¼ Section Corner (Found, Projected)

⅓ Section Line

⅓ Section Corner (Found, Projected)

Property Line w/Found Property Corner

Parcel Number

National Park Boundary

National Forest Boundary

National Wildlife Refuge Boundary

BLM Lands Boundary

Indian Reservation Boundary

Existing Roadway (Road, Paved, Gravel)

Railroad

Trail

Fiber Roll or Wattle

Silt Fence

Intermittent Drainage or Small Creek

Large Creek or River

Lake, Pond or Reservoir; Marshland

Spring or Seep

Treeline; Individual Trees

Material Source; Bore Hole; Test Pit

Spot Elevation; Coordinate Grid Tick

Above Ground Tank; Underground Tank

Boulder; Well; Satellite Dish; Grave

Cooking Grate; Garbage Can; Picnic Table

Flagpole; Fire Hydrant

Gas & Water Meter; Gas & Water Valve

Control Point (Terrestrial and GPS); Jump Hub

North Arrow

Slope Stake Limits

Fence

Gate with Fence

Cattleguard

Guardrail

Concrete Barrier

Retaining Wall

Signs (single, double post; portable)

Delineators

Pipe Culvert (arrow shows flow)

Pipe Culvert with End Section

Pipe Culvert with Headwall

Pipe Culvert with Drop Inlet

Box Culvert

Underdrain

Overhead/Above Ground Utilities

Underground Utilities

Poles (Power, Telephone, Joint Use, Light, Support w/Anchor)

Miscellaneous Utility Features

Building

Right-of-Way Line with Monument

Permanent Easement

Construction Easement

Riprap

Top of Cut

Toe of Fill

Transition

wall face

FM = force main, FO = fiber optic, G = gas, IRR = irrigation, O = oil, P = power, SA = sanitary sewer, SD = storm drain, SS = storm sewer, STEAM = steam, T = telephone, TV = CATV, W = water

EM = electric meter, T = telephone pedestal, TV = CATV pedestal, UP = transformer or junction box, WF = water fountain

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION  
FEDERAL HIGHWAY ADMINISTRATION  
WESTERN FEDERAL LANDS AND HIGHWAY DIVISION

FLH STANDARD

PLAN SYMBOLS  
AND ABBREVIATIONS

DETAIL APPROVED FOR USE 11/2001

STANDARD

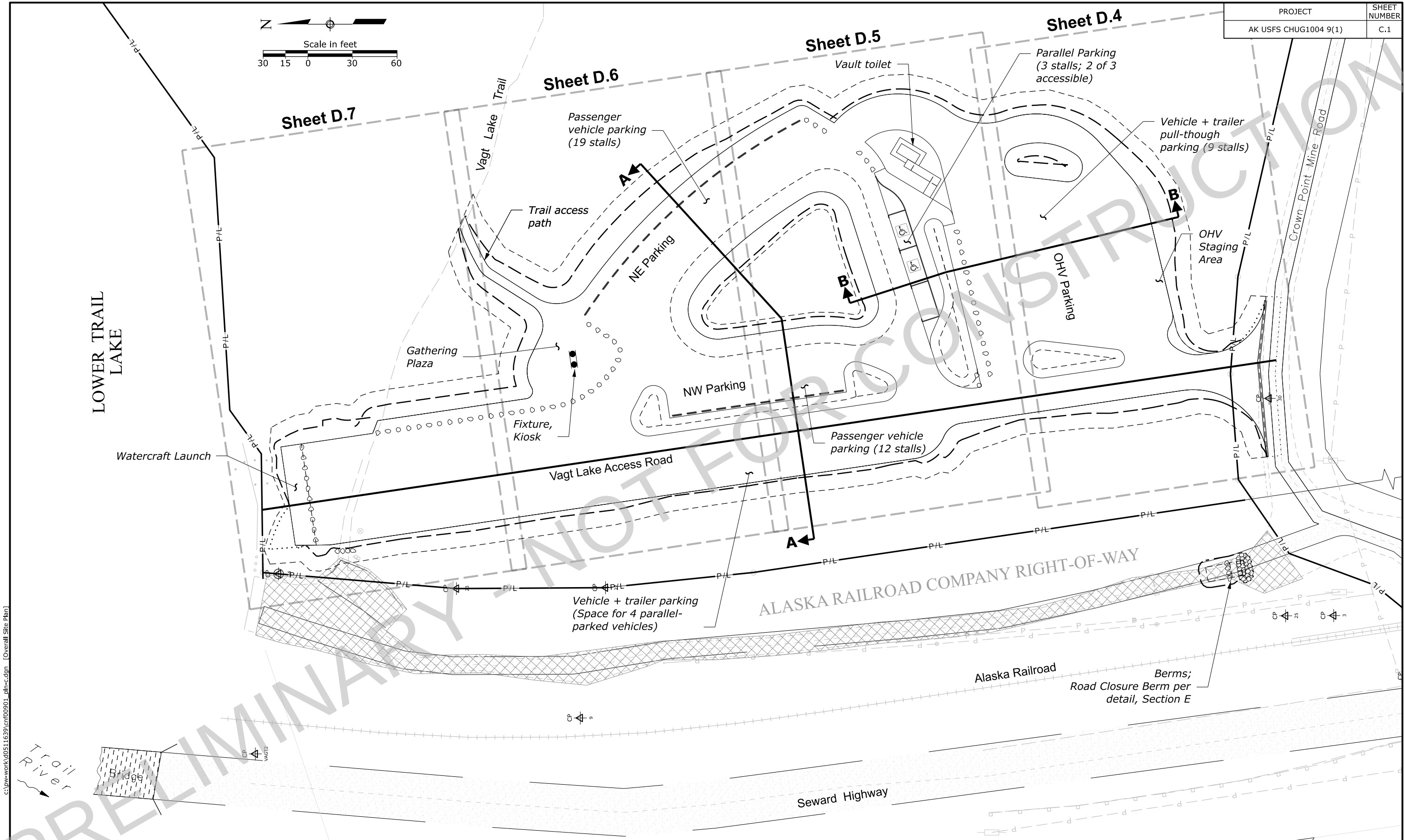
REVISED: 9/2005 1/2007 10/2009 10/2014

W101-1

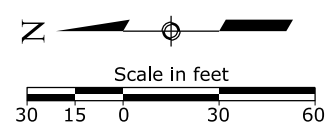
NOTE:

1. Other symbols used in the plans will be shown in a legend on the appropriate plan sheet.





| PROJECT               | SHEET NUMBER |
|-----------------------|--------------|
| AK USFS CHUG1004 9(1) | C.1          |



**Legend**

Roadway Obliteration, Method 2 <sup>[1]</sup>

**Footnote:**

<sup>[1]</sup> Coordinates for the limits of Roadway Obliteration, Method 2, will be provided as physical data for construction.

**OVERALL SITE PLAN**

c:\pw-work\0511639\cn00901\_plr-c.dgn [Overall Site Plan] 23 July 2024 1:17 PM

C:\MyFiles\Projects\_2024\Alaska\Vagt\_Lake\_Trailhead\pw\New folder\Vagt\_Lake\_Controlsheet.xlsx|Sheet

22-Jul-2024 2:27 PM

PROJECT : Vagt Lake Trailhead  
DATE OF FIELD WORK :Multiple entries  
DATE OF FINAL ADJUSTMENT : Control Provided by USFS April, 2015, wflhd pts updated 7/22/2024

COORDINATE SYSTEM : Alaska zone 4 SPCS NAD83 2011  
VERTICAL DATUM : Orthometric elevations based on the NAVD88 GEOID12b  
EPOCH DATE : 2010.000  
CADD SYSTEM COORDINATE SYSTEM : AK83/2011-4F NSRS11(NAD83/2011) datum, Alaska State Plane; Zone 4, US Foot

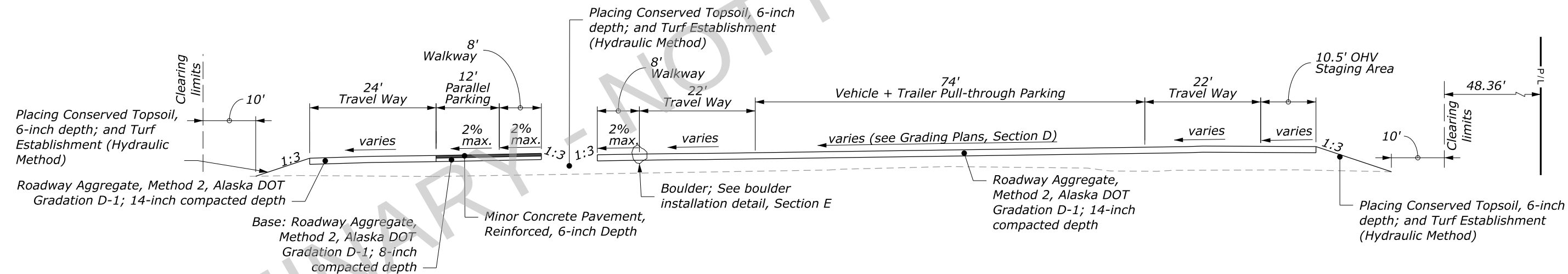
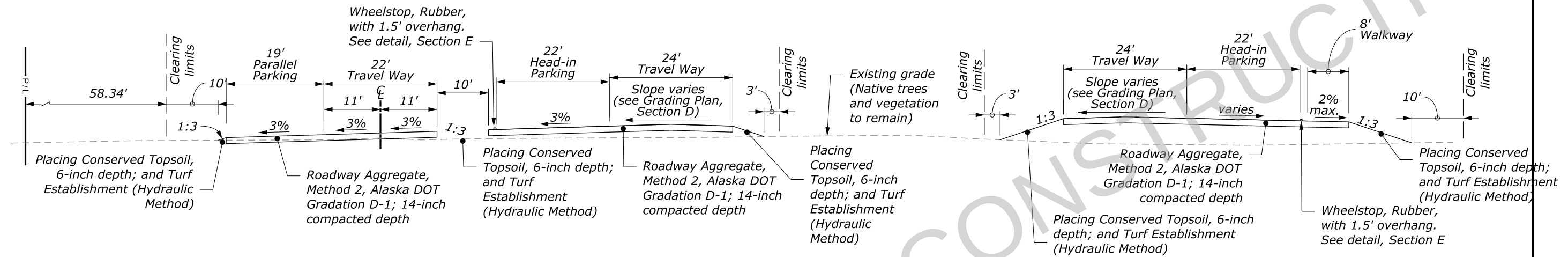
|                       |              |
|-----------------------|--------------|
| PROJECT               | SHEET NUMBER |
| AK USFS CHUG1004 9(1) | C.2          |

| POINT NUMBER | STATE PLANE COORDINATES |             |                       | GEO COORDINATES   |                    |                  |                 | DESCRIPTION               |
|--------------|-------------------------|-------------|-----------------------|-------------------|--------------------|------------------|-----------------|---------------------------|
|              | NORTH                   | EAST        | ORTHOMETRIC ELEVATION | LATITUDE          | LONGITUDE          | ELLIPSOID HEIGHT | COMBINED FACTOR |                           |
| 5            | 2350701.303             | 1754385.791 | 484.121               | 60°25'58.818920"N | 149°22'08.856412"W | 522.7728         | 0.999889845     | SPIKE Original FS control |
| 30           | 2350775.973             | 1754004.160 | 476.641               | 60°25'59.590250"N | 149°22'16.446704"W | 515.2868         | 0.999890104     | SPIKE Original FS control |
| 24101        | 2350731.024             | 1754185.257 | 479.897               | 60°25'59.130526"N | 149°22'12.846656"W | 518.5493         | 0.999889995     | 5/8" ir w/fhwa alum cap   |
| 24102        | 2350694.163             | 1754625.779 | 488.230               | 60°25'58.725945"N | 149°22'04.075702"W | 526.8894         | 0.999889711     | 5/8" ir w/fhwa alum cap   |
| 24103        | 2350610.324             | 1755003.735 | 497.049               | 60°25'57.864458"N | 149°21'56.560539"W | 535.722          | 0.999889388     | 5/8" ir w/fhwa alum cap   |

NOTE:

1. To precisely check distances between points as measured on the ground, inverse the state plane coordinates and divide the computed distance by a mean combined factor of the two points.

SURVEY CONTROL



## TYPICAL ROADWAY SECTIONS

NOT TO SCALE

24 July 2024 12:21 PM c:\pw-work\0511641\cnf00901\_pfr-pro\_d.dgn [Tabulation of Roadway Quantities]

|       |                    |              |
|-------|--------------------|--------------|
| STATE | PROJECT            | SHEET NUMBER |
| AK    | USFS CHUG1004 9(1) | D.1          |

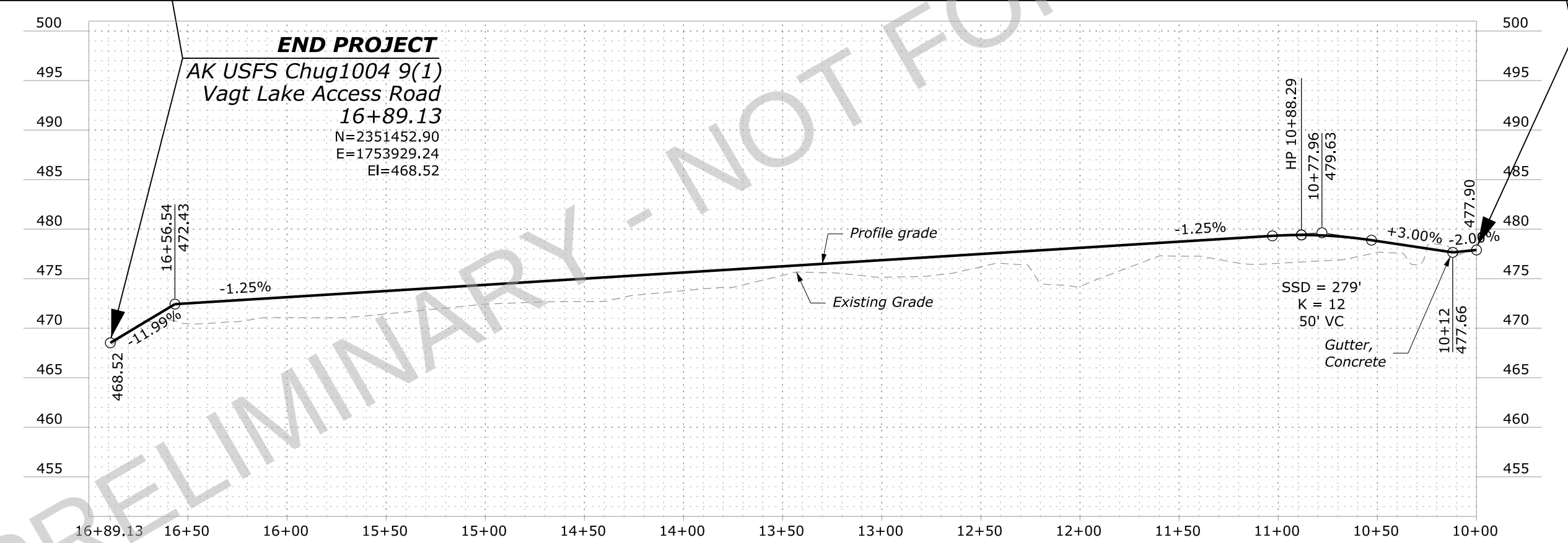
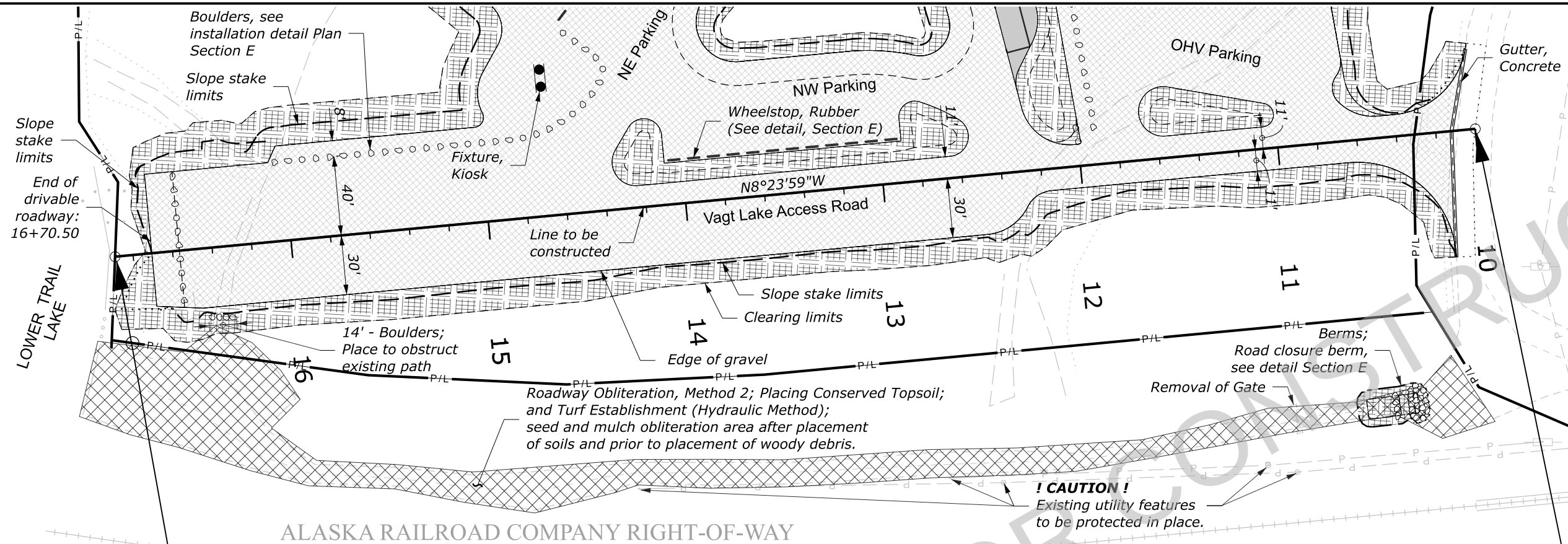
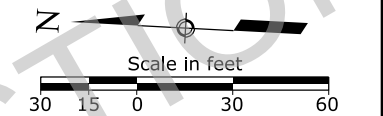
| ROADWAY QUANTITIES |                                                        |       |            |
|--------------------|--------------------------------------------------------|-------|------------|
| ITEM               | DESCRIPTION                                            | UNITS | QUANTITIES |
| 20101-0000         | CLEARING AND GRUBBING                                  | ACRE  | 2.9        |
| 20301-1100         | REMOVAL OF GATE                                        | EACH  | 1          |
| 20401-0000         | ROADWAY EXCAVATION                                     | CUYD  | 124        |
| 20403-0000         | UNCLASSIFIED BORROW                                    | CUYD  | 9,891      |
| 20443-0000         | BERMS                                                  | LNFT  | 20         |
| 21101-2000         | ROADWAY OBLITERATION, METHOD 2                         | SQYD  | 1,549      |
| 25125-0000         | BOULDER                                                | EACH  | 102        |
| 30201-2000         | ROADWAY AGGREGATE, METHOD 2 (ALASKA DOT GRADATION D-1) | CUYD  | 3,632      |
| 50101-0600         | MINOR CONCRETE PAVEMENT, REINFORCED, 6-INCH DEPTH      | SQYD  | 280        |
| 60905-1000         | GUTTER, CONCRETE                                       | LNFT  | 106        |
| 60915-4000         | WHEELSTOP, RUBBER                                      | EACH  | 31         |
| 61501-0100         | SIDEWALK, CONCRETE (4-INCH DEPTH)                      | SQYD  | 37         |
| 62405-0400         | PLACING CONSERVED TOPSOIL, 6-INCH DEPTH                | SQYD  | 2,667      |
| 62502-0000         | TURF ESTABLISHMENT (HYDRAULIC METHOD)                  | SQYD  | 2,667      |
| 64603-0800         | FIXTURE, KIOSK                                         | EACH  | 1          |
| 64603-1000         | FIXTURE, VAULT TOILET                                  | EACH  | 1          |

| EARTHWORK SUMMARY TABLE |                                       |                                                         |                                                 |                                                      |                                                                 |                                                                                            |                             |                                                              |                                                                        |                                    |                                                                           |                        |
|-------------------------|---------------------------------------|---------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------|------------------------|
| LOCATION                | CUT                                   |                                                         |                                                 |                                                      |                                                                 |                                                                                            | FILL                        |                                                              |                                                                        |                                    | TOTALS                                                                    |                        |
|                         | Item 20401-0000<br>ROADWAY EXCAVATION | (+) Additional Excavation (See Note 2)<br>for info only | (-) Topsoil Stripped from Cuts<br>for info only | (-) Estimate of Unsuitable Material<br>for info only | Total Unadjusted Excavation Available for Fill<br>for info only | Total Adjusted Available for Fill (adjusted based on shrink/swell factor)<br>for info only | Embankment<br>for info only | (+) Additional Backfill Needed (See Note 3)<br>for info only | (+) Topsoil stripped from embankment foundation areas<br>for info only | Total Fill Needed<br>for info only | Total Adjusted Available For Fill less Total Fill Needed<br>for info only | Waste<br>for info only |
| UNITS                   | CUYD                                  | CUYD                                                    | CUYD                                            | CUYD                                                 | CUYD                                                            | CUYD                                                                                       | CUYD                        | CUYD                                                         | CUYD                                                                   | CUYD                               | CUYD                                                                      | CUYD                   |
| VAGT LAKE TRAILHEAD     | 124                                   | -                                                       | 49                                              | -                                                    | 75                                                              | 68                                                                                         | 5,846                       |                                                              | 4,112                                                                  | 9,958                              | -9,891                                                                    |                        |
| TOTALS                  | 124                                   | 0                                                       | 49                                              | 0                                                    | 75                                                              | 68                                                                                         | 5,846                       | 0                                                            | 4,112                                                                  | 9,958                              | -9,891                                                                    | 0                      |

- NOTE:
- All volumes are in-place cubic yards (i.e. in situ or compacted in place).
  - Additional excavation that is suitable for use in embankment construction (Section 209 Structural excavation, Culverts).
  - Additional material needed for embankment construction (Structural backfill).
  - Volumes of existing topsoil not calculated for road closure berm.

TABULATION OF ROADWAY QUANTITIES

| PROJECT               | SHEET NUMBER |
|-----------------------|--------------|
| AK USFS CHUG1004 9(1) | D.2          |



**BEGIN PROJECT**  
AK USFS Chug1004 9(1)  
Vagt Lake Access Road  
10+00.00  
N=2350771.18  
E=1754029.90  
El=477.90

**Legend**

- |                                                                                       |                                                           |              |
|---------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------|
| Roadway Obliteration, Method 2<br>(see staking notes for obliteration limits)         | Roadway Aggregate, Method 2<br>(Alaska DOT Gradation D-1) | Grade breaks |
| Placing Conserved Topsoil, 6-inch Depth;<br>and Turf Establishment (Hydraulic Method) | Minor Concrete Pavement,<br>Reinforced, 6-inch Depth      |              |

**Notes:**

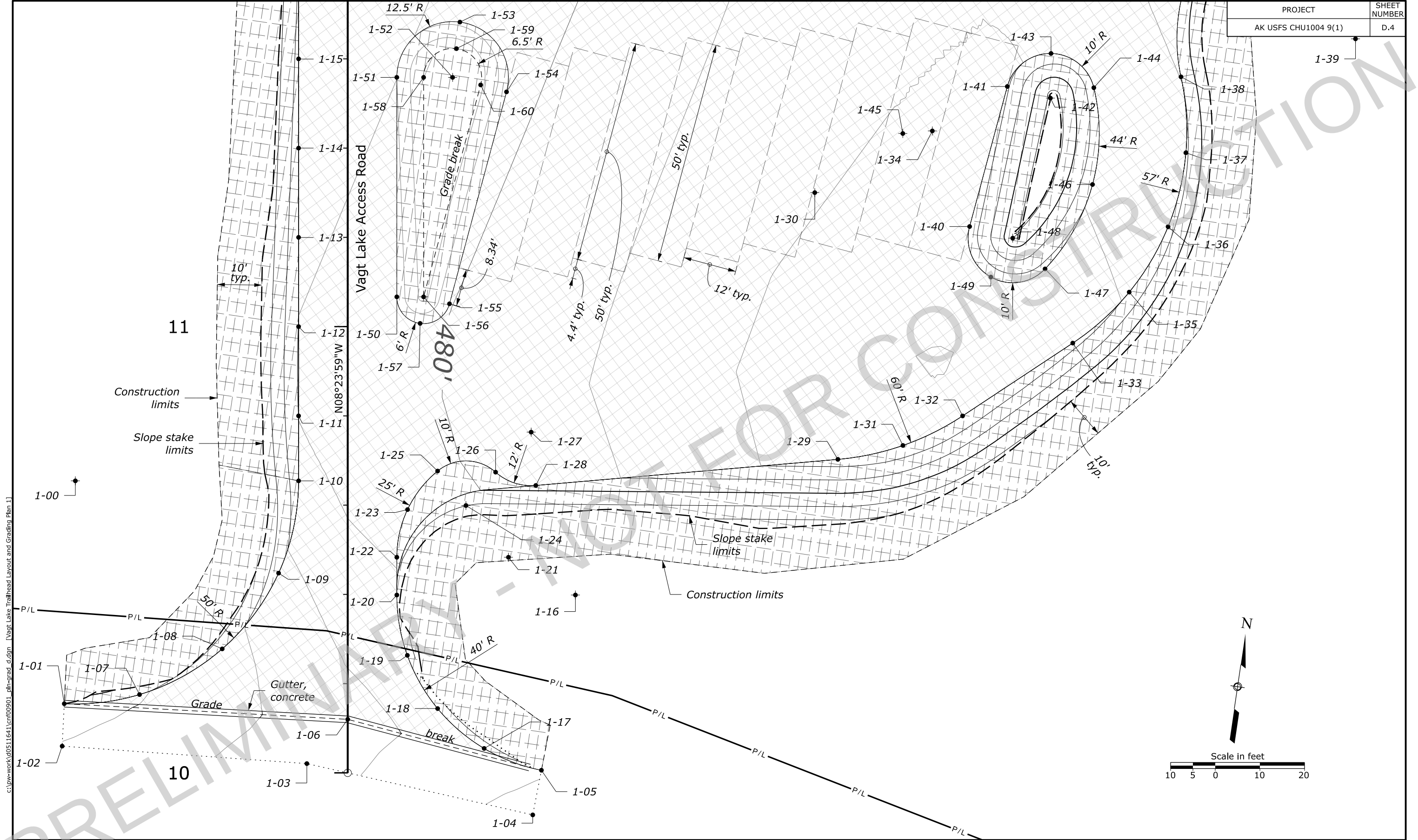
See Layout and Grading Plans and Data Tables,  
this Section, for detailed layout information with  
dimensions, point coordinates, and elevations.

**VAGT LAKE ACCESS ROAD  
PLAN AND PROFILE**

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**Legend**

Roadway Aggregate, Method 2 (Alaska DOT Gradation D-1)

Placing Conserved Topsoil, 6-inch Depth; and Turf Establishment (Hydraulic Method)

Grade breaks

Layout Lines (for reference only)

**Notes:**

See sheet D.8 for data point information tables.

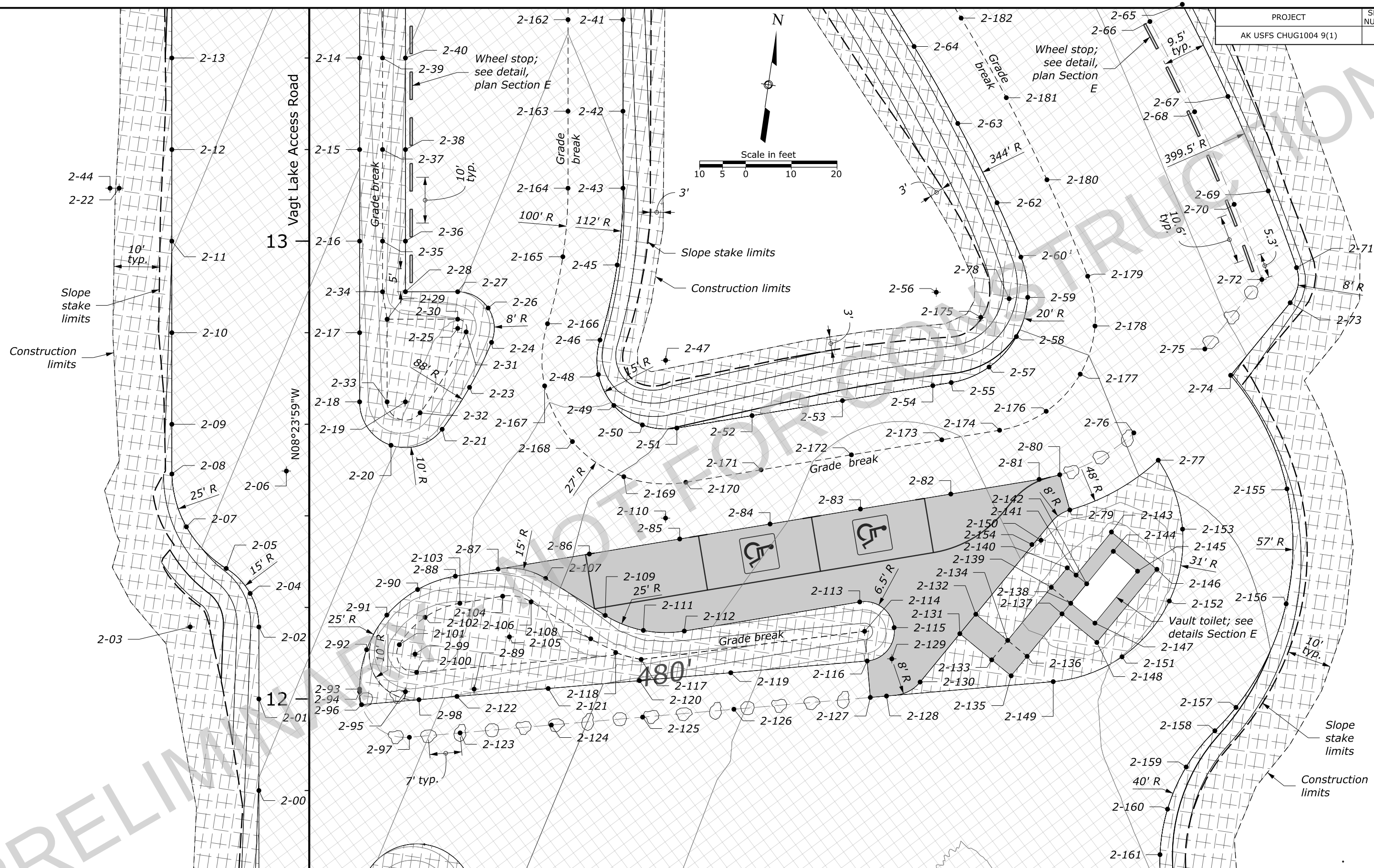
**VAGT LAKE TRAILHEAD LAYOUT AND GRADING PLAN 1**

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23 July 2024 6:34 PM



c:\pw-work\0511641\cn00901\_pln-grad.dgn [Vagt Lake Trailhead Layout and Grading Plan 2] 23 July 2024 6:34 PM

| PROJECT               | SHEET NUMBER |
|-----------------------|--------------|
| AK USFS CHUG1004 9(1) | D.5          |



#### Legend

|  |                                                        |  |                                                                                    |
|--|--------------------------------------------------------|--|------------------------------------------------------------------------------------|
|  | Roadway Aggregate, Method 2 (Alaska DOT Gradation D-1) |  | Placing Conserved Topsoil, 6-inch Depth; and Turf Establishment (Hydraulic Method) |
|  | Minor Concrete Pavement, Reinforced, 6-inch Depth      |  | Grade breaks                                                                       |
|  |                                                        |  | Layout Lines (for reference only)                                                  |

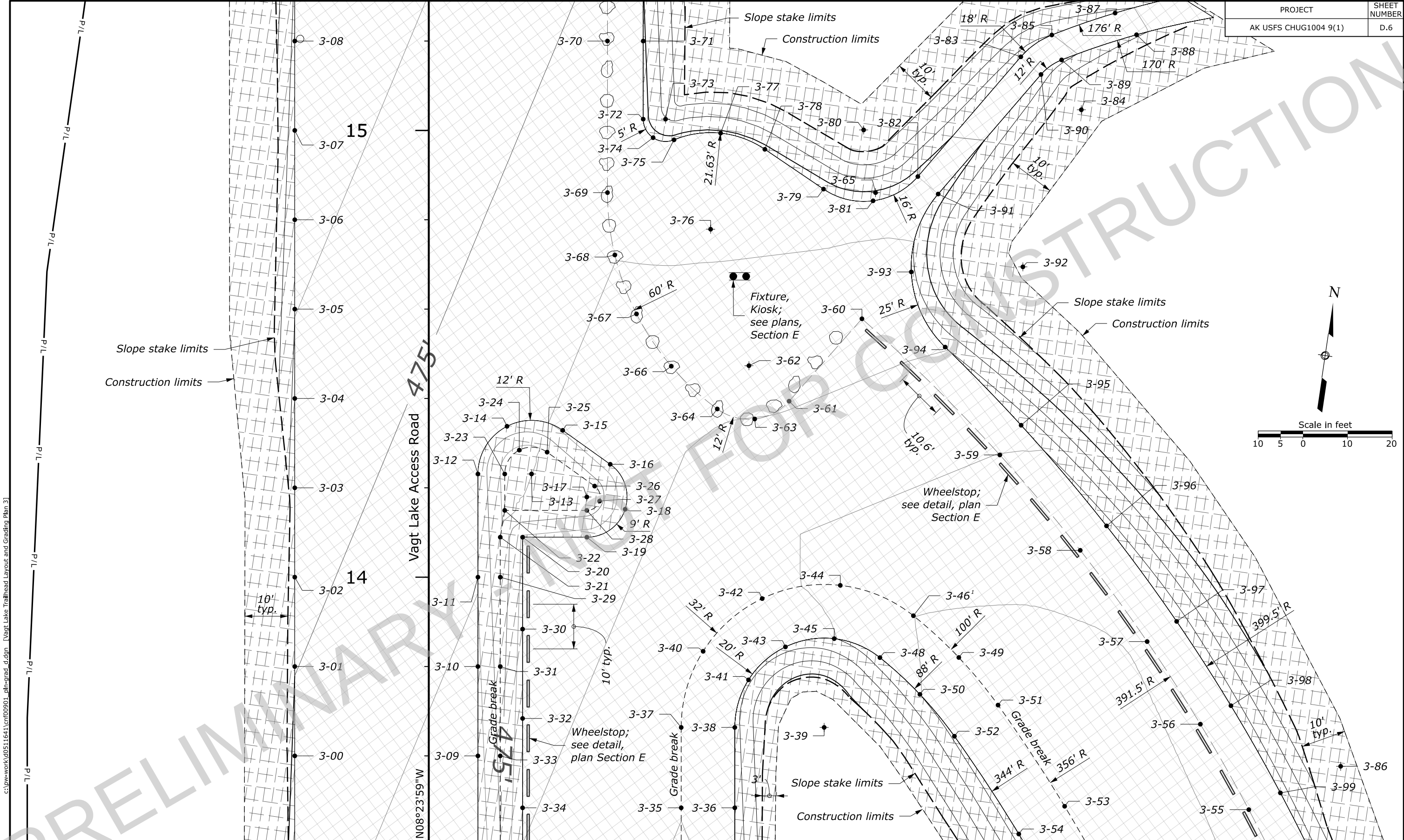
#### Notes:

See sheet D.8 and D.9 for data point information tables.

#### Footnotes:

<sup>1</sup> Point 2-61 not shown: it is an arc center point outside the display area. Coordinates are shown on the data point table, sheet D.8.

## VAGT LAKE TRAILHEAD LAYOUT AND GRADING PLAN 2



**Legend**

Roadway Aggregate, Method 2  
(Alaska DOT Gradation D-1)

Placing Conserved Topsoil; and  
Turf Establishment (Hydraulic Method)

Grade breaks

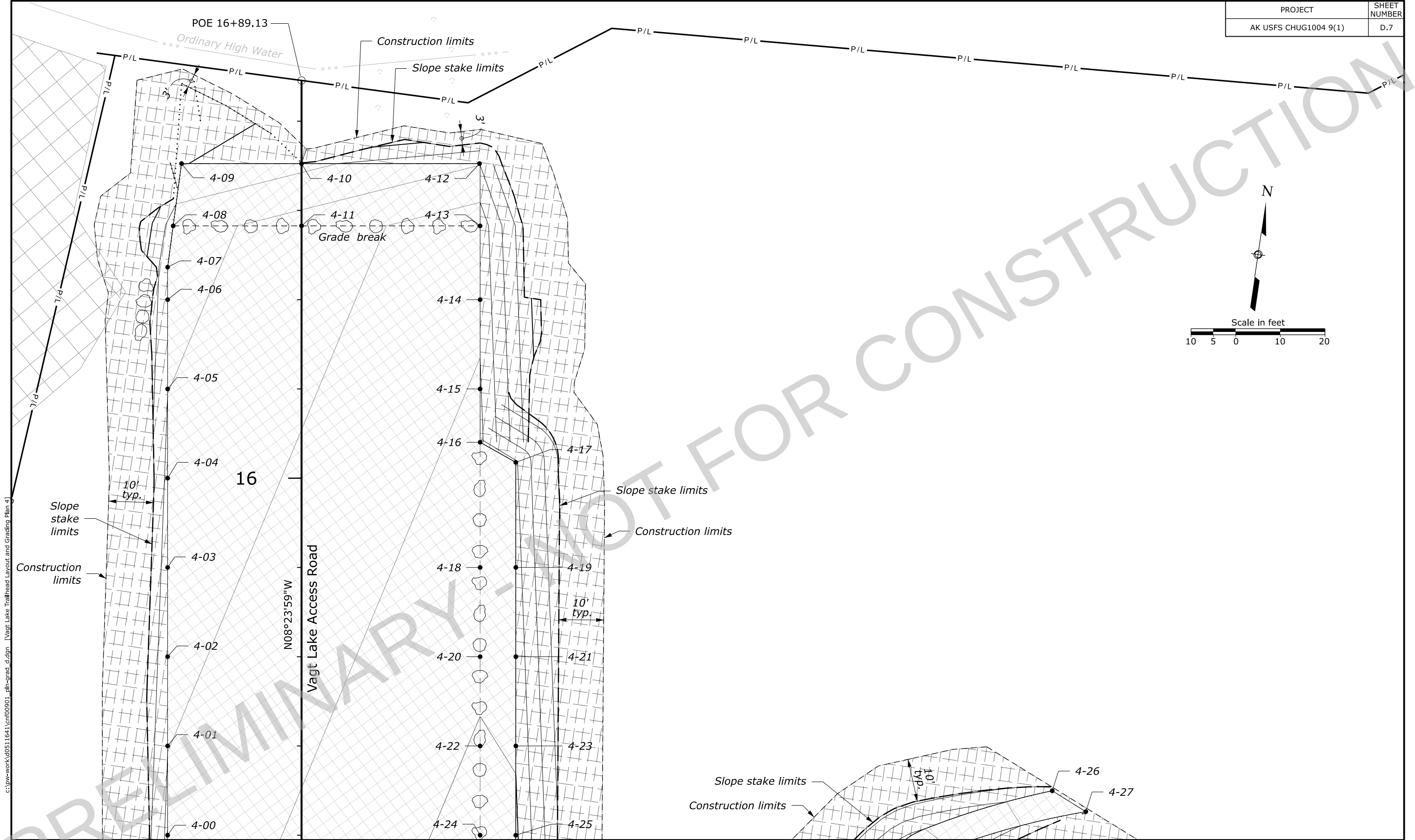
Layout Lines (for reference only)

**Notes:**  
See sheet D.10 for data point information tables.

**Footnotes:**  
<sup>1</sup> Point 3-47 not shown: it is an arc center point outside the display area.  
Coordinates are shown on the data point table, sheet D.10.

**VAGT LAKE TRAILHEAD LAYOUT  
AND GRADING PLAN 3**

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**Legend**

Roadway Aggregate, Method 2  
(Alaska DOT Gradation D-1)

Placement of Conserved Topsoil, 6-inch Depth;  
and Turf Establishment (Hydraulic Method)

**Notes:**  
See sheet D.10 for data point information tables.

**VAGT LAKE TRAILHEAD LAYOUT  
AND GRADING PLAN 4**

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23 July 2024 6:45 PM



c:\pwwork\0511641\m00901\_dhr-grad.dgn [Layout and Grading Point Data Tables - 1] 11 July 2023 12:31 PM

| GRADING PLAN 1 POINT DATA |            |            |        |                      |                        | GRADING PLAN 1 POINT DATA |            |            |        |                      |                       | GRADING PLAN 2 POINT DATA |            |            |        |                      |                         | PROJECT              | SHEET NUMBER |
|---------------------------|------------|------------|--------|----------------------|------------------------|---------------------------|------------|------------|--------|----------------------|-----------------------|---------------------------|------------|------------|--------|----------------------|-------------------------|----------------------|--------------|
|                           |            |            |        |                      |                        |                           |            |            |        |                      |                       |                           |            |            |        |                      |                         | AK USFS CHU1004 9(1) | D.8          |
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION            | PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION           | PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION             |                      |              |
| 1-00                      | 2350827.01 | 1753959.99 | -      | 10+65.44; 61.00' Lt. | Ctr                    | 1-45                      | 2350931.08 | 1754131.97 | -      | 11+43.28; 124.33' Rt | Ctr                   | 2-25                      | 2351053.84 | 1754020.93 | -      | 12+80.94; 32.41' Rt. | Ctr                     |                      |              |
| 1-01                      | 2350777.24 | 1753964.84 | 475.48 | 10+15.50; 63.48' Lt. | BEG Matchline / EOG-PC | 1-46                      | 2350925.98 | 1754175.67 | 481.88 | 11+31.85; 166.82' Rt | 1/2Δ                  | 2-26                      | 2351059.21 | 1754026.86 | 478.22 | 12+85.39; 39.06' Rt. | 1/2Δ                    |                      |              |
| 1-02                      | 2350767.76 | 1753965.76 | 476.06 | 10+05.99; 63.96' Lt. | HPI Matchline          | 1-47                      | 2350905.71 | 1754167.92 | 482.44 | 11+12.93; 156.19' Rt | PCC                   | 2-27                      | 2351061.76 | 1754019.76 | 477.98 | 12+88.94; 32.41' Rt. | PT                      |                      |              |
| 1-03                      | 2350771.90 | 1754020.55 | 477.57 | 10+02.08; 9.15' Lt.  | HPI Matchline          | 1-48                      | 2350911.48 | 1754159.75 | -      | 11+19.83; 148.95' Rt | Ctr                   | 2-28                      | 2351060.09 | 1754008.47 | 477.64 | 12+88.94; 21.00' Rt. | HPI                     |                      |              |
| 1-04                      | 2350767.92 | 1754072.28 | 479.37 | 9+90.58; 41.45' Rt.  | HPI Matchline          | 1-49                      | 2350902.14 | 1754156.18 | 482.73 | 11+11.11; 144.05' Rt | 1/2Δ                  | 2-29                      | 2351053.57 | 1754005.39 | 475.41 | 12+82.94; 17.00' Rt. | Grade break HPI         |                      |              |
| 1-05                      | 2350778.10 | 1754072.74 | 478.90 | 10+00.59; 43.39' Rt. | END Matchline / EOG-PC | 1-50                      | 2350878.31 | 1754025.20 | 479.61 | 11+06.67; 11.00' Rt. | BEG-END / PT          | 2-30                      | 2351055.82 | 1754020.63 | 475.98 | 12+82.94; 32.41' Rt. | Grade break PC          |                      |              |
| 1-06                      | 2350783.05 | 1754028.15 | 477.66 | 10+12; On Line       | Grade break @ CL       | 1-51                      | 2350926.96 | 1754018.02 | 478.99 | 11+55.85; 11.00' Rt. | PC                    | 2-31                      | 2351053.36 | 1754022.87 | 476.34 | 12+80.18; 34.26' Rt. | Grade break PCC         |                      |              |
| 1-07                      | 2350781.73 | 1753981.21 | 475.92 | 10+17.55; 46.62' Lt. | 1/4Δ                   | 1-52                      | 2350928.79 | 1754030.38 | -      | 11+55.85; 23.50' Rt. | Ctr                   | 2-32                      | 2351034.40 | 1754015.50 | 476.25 | 12+62.50; 24.21' Rt. | Grade break PCC         |                      |              |
| 1-08                      | 2350794.56 | 1753998.04 | 476.69 | 10+27.79; 28.11' Lt. | 1/2Δ                   | 1-53                      | 2350941.29 | 1754030.19 | 479.26 | 11+68.24; 25.14' Rt. | 1/2Δ                  | 2-33                      | 2351035.71 | 1754008.03 | 475.64 | 12+64.89; 17.00' Rt. | Grade break PT          |                      |              |
| 1-09                      | 2350813.20 | 1754008.05 | 477.70 | 10+44.76; 15.48' Lt. | 3/4Δ                   | 1-54                      | 2350927.34 | 1754042.80 | 479.77 | 11+52.60; 35.57' Rt. | PT                    | 2-34                      | 2351059.36 | 1754003.52 | 475.67 | 12+88.94; 16.00' Rt. | Grade break HPI         |                      |              |
| 1-10                      | 2350834.31 | 1754009.46 | 478.68 | 10+65.44; 11.00' Lt. | PT                     | 1-55                      | 2350878.49 | 1754037.10 | 479.98 | 11+05.11; 22.79' Rt. | PC                    | 2-35                      | 2351070.30 | 1754001.91 | 475.53 | 13+00; 16.00' Rt.    | Grade break @ 13+00     |                      |              |
| 1-11                      | 2350848.71 | 1754007.33 | 479.01 | 10+80; 11.00' Lt.    | EOG @ 10+80            | 1-56                      | 2350879.19 | 1754031.14 | 477.61 | 11+06.67; 17.00' Rt. | Ctr / Grade break HPI | 2-36                      | 2351071.03 | 1754006.85 | 477.50 | 13+00; 21.00' Rt.    | EOG @ 13+00 Rt.         |                      |              |
| 1-12                      | 2350868.50 | 1754004.41 | 479.03 | 11+00; 11.00' Lt.    | EOG @ 11+00            | 1-57                      | 2350873.19 | 1754031.23 | 479.83 | 11+00.72; 16.21' Rt. | 1/2Δ                  | 2-37                      | 2351090.08 | 1753998.99 | 475.28 | 13+20; 16.00' Rt.    | Grade break @ 13+20     |                      |              |
| 1-13                      | 2350888.29 | 1754001.49 | 478.78 | 11+20; 11.00' Lt.    | EOG @ 11+20            | 1-58                      | 2350927.84 | 1754023.95 | 476.99 | 11+55.85; 17.00' Rt. | Grade break PC        | 2-38                      | 2351090.81 | 1754003.93 | 477.25 | 13+20; 21.00' Rt.    | EOG @ 13+20 Rt.         |                      |              |
| 1-14                      | 2350908.07 | 1753998.57 | 478.53 | 11+40; 11.00' Lt.    | EOG @ 11+40            | 1-59                      | 2350935.29 | 1754030.29 | 477.26 | 11+62.29; 24.35' Rt. | Grade break 1/2Δ      | 2-39                      | 2351109.87 | 1753996.06 | 475.04 | 13+40; 16.00' Rt.    | Grade break @ 13+40     |                      |              |
| 1-15                      | 2350927.86 | 1753995.65 | 478.28 | 11+60; 11.00' Lt.    | EOG @ 11+60            | 1-60                      | 2350928.03 | 1754036.84 | 477.77 | 11+54.16; 29.78' Rt  | Grade break PT        | 2-40                      | 2351110.60 | 1754001.01 | 477.00 | 13+40; 21.00' Rt.    | EOG @ 13+40 Rt.         |                      |              |
| 1-16                      | 2350818.06 | 1754074.53 | -      | 10+39.86; 51.00' Rt. | Ctr                    | GRADING PLAN 2 POINT DATA |            |            |        |                      |                       | 2-41                      | 2351125.82 | 1754046.78 | 477.60 | 13+48.37; 68.50' Rt. | EOG                     |                      |              |
| 1-17                      | 2350781.06 | 1754059.34 | 478.54 | 10+05.48; 30.56' Rt. | 1/4Δ                   | PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION           | 2-42                      | 2351106.04 | 1754049.70 | 477.85 | 13+28.37; 68.50' Rt. | EOG                     |                      |              |
| 1-18                      | 2350788.38 | 1754047.72 | 478.40 | 10+14.41; 20.14' Rt. | 1/2Δ                   | 2-00                      | 2350947.64 | 1753992.73 | 478.03 | 11+80; 11.00' Lt.    | EOG @ 11+80           | 2-43                      | 2351089.40 | 1754052.16 | 478.06 | 13+11.56; 68.50' Rt. | PC                      |                      |              |
| 1-19                      | 2350799.19 | 1754039.27 | 478.50 | 10+26.34; 13.36' Rt. | 3/4Δ                   | 2-01                      | 2350967.43 | 1753989.81 | 477.78 | 12+00; 11.00' Lt.    | EOG @ 12+00           | 2-44                      | 2351073.04 | 1753941.36 | -      | 13+11.56; 43.5' Lt.  | Ctr                     |                      |              |
| 1-20                      | 2350812.22 | 1754034.96 | 478.82 | 10+39.86; 11.00' Rt. | PT                     | 2-02                      | 2350982.99 | 1753987.51 | 477.59 | 12+15.74; 11.00' Lt. | PC                    | 2-45                      | 2351072.63 | 1754053.36 | 478.22 | 12+94.79; 67.24' Rt. | 1/2Δ                    |                      |              |
| 1-21                      | 2350824.23 | 1754058.46 | -      | 10+48.32; 36.00' Rt. | Ctr                    | 2-03                      | 2350980.80 | 1753972.67 | -      | 12+15.74; 26.00' Lt. | Ctr                   | 2-46                      | 2351055.87 | 1754052.03 | 478.31 | 12+78.40; 63.48' Rt. | PRC                     |                      |              |
| 1-22                      | 2350820.58 | 1754033.73 | 479.07 | 10+48.32; 11.00' Rt. | PC                     | 2-04                      | 2350989.95 | 1753984.56 | 477.44 | 12.90' Rt.           | 1/2Δ                  | 2-47                      | 2351053.57 | 1754066.86 | -      | 12+73.96; 77.81' Rt. | Ctr                     |                      |              |
| 1-23                      | 2350831.51 | 1754034.54 | 479.42 | 10+59.00; 13.40' Rt. | 1/2Δ                   | 2-05                      | 2350994.58 | 1753978.59 | 477.22 | 12+28.50; 18.13' Lt. | PRC                   | 2-48                      | 2351048.36 | 1754052.79 | 478.52 | 12+70.86; 63.13' Rt. | 1/4Δ                    |                      |              |
| 1-24                      | 2350834.31 | 1754047.35 | -      | 10+59.91; 26.48' Rt. | Ctr                    | 2-06                      | 2351017.55 | 1753988.47 | -      | 12+49.78; 5.00' Lt.  | Ctr                   | 2-49                      | 2351042.16 | 1754057.11 | 479.08 | 12+64.10; 66.50' Rt. | 1/2Δ                    |                      |              |
| 1-25                      | 2350841.03 | 1754039.94 | 479.80 | 10+67.64; 20.14' Rt. | PCC                    | 2-07                      | 2351002.31 | 1753968.65 | 476.84 | 26.83' Rt.           | 1/2Δ                  | 2-50                      | 2351038.86 | 1754063.91 | 479.69 | 12+59.85; 72.74' Rt. | 3/4Δ                    |                      |              |
| 1-26                      | 2350842.68 | 1754052.83 | 480.19 | 10+67.39; 33.12' Rt. | PRC                    | 2-08                      | 2351013.90 | 1753963.74 | 476.59 | 12+49.78; 30.00' Lt. | PT                    | 2-51                      | 2351039.29 | 1754071.45 | 480.08 | 12+59.17; 80.26' Rt. | PT                      |                      |              |
| 1-27                      | 2350852.72 | 1754059.40 | -      | 10+76.36; 41.09' Rt. | Ctr                    | 2-09                      | 2351024.01 | 1753962.24 | 476.47 | 12+60; 30.00' Lt.    | EOG @ 12+60 Lt.       | 2-52                      | 2351044.38 | 1754087.30 | 480.41 | 12+61.89; 96.69' Rt. | EOG                     |                      |              |
| 1-28                      | 2350841.04 | 1754062.17 | 480.39 | 10+64.40; 42.13' Rt  | PT                     | 2-10                      | 2351043.79 | 1753959.32 | 476.22 | 12+80; 30.00' Lt.    | EOG @ 12+80 Lt.       | 2-53                      | 2351050.50 | 1754106.34 | 480.77 | 12+65.17; 116.42' Rt | EOG                     |                      |              |
| 1-29                      | 2350856.71 | 1754128.25 | 482.55 | 10+70.25; 109.78' Rt | PC                     | 2-11                      | 2351063.58 | 1753956.40 | 475.97 | 13+00; 30.00' Lt.    | EOG @ 13+00 Lt.       | 2-54                      | 2351056.63 | 1754125.38 | 480.72 | 12+68.44; 136.15' Rt | EOG                     |                      |              |
| 1-30                      | 2350915.09 | 1754114.40 | -      | 11+30.03; 104.62' Rt | Ctr                    | 2-12                      | 2351083.36 | 1753953.48 | 475.72 | 13+20; 30.00' Lt.    | EOG @ 13+20 Lt.       | 2-55                      | 2351057.87 | 1754129.26 | 480.64 | 12+69.11; 140.17' Rt | PC                      |                      |              |
| 1-31                      | 2350861.93 | 1754142.21 | 482.89 | 10+73.37; 124.36' Rt | 1/2Δ                   | 2-13                      | 2351103.15 | 1753950.56 | 475.47 | 13+40; 30.00' Lt.    | EOG @ 13+40 Lt.       | 2-56                      | 2351076.92 | 1754123.14 | -      | 12+88.84; 136.90' Rt | Ctr                     |                      |              |
| 1-32                      | 2350870.42 | 1754154.46 | 482.94 | 10+79.99; 137.72' Rt | PT                     | 2-14                      | 2351109.14 | 1753991.12 | 476.70 | 13+40; 11.00' Rt.    | EOG @ 13+40 Rt.       | 2-57                      | 2351062.43 | 1754136.94 | 480.36 | 12+72.50; 148.43' Rt | 1/4Δ                    |                      |              |
| 1-33                      | 2350890.17 | 1754176.48 | 482.42 | 10+96.31; 162.39' Rt | PC                     | 2-15                      | 2351089.35 | 1753994.04 | 476.95 | 13+20; 11.00' Rt.    | EOG @ 13+20 Rt.       | 2-58                      | 2351069.88 | 1754141.86 | 480.05 | 12+79.14; 154.39' Rt | 1/2Δ                    |                      |              |
| 1-34                      | 2350932.61 | 1754138.43 | -      | 11+43.85; 130.94' Rt | Ctr                    | 2-16                      | 2351069.57 | 1753996.96 | 477.20 | 13+00; 11.00' Rt.    | EOG @ 13+00 Rt.       | 2-59                      | 2351078.72 | 1754143.06 | 479.81 | 12+87.72; 156.87' Rt | 3/4Δ                    |                      |              |
| 1-35                      | 2350903.33 | 1754187.33 | 481.93 | 11+07.73; 175.04' Rt | 1/4Δ                   | 2-17                      | 2351049.78 | 1753999.88 | 477.45 | 12+80; 11.00' Rt.    | EOG @ 12+80 Rt.       | 2-60                      | 2351087.21 | 1754140.29 | 479.63 | 12+96.52; 155.37' Rt | PCC                     |                      |              |
| 1-36                      | 2350919.09 | 1754193.80 | 481.49 | 11+22.39; 183.75' Rt | 1/2Δ                   | 2-18                      | 2351034.84 | 1754002.09 | 477.64 | 12+64.89; 11.00' Rt. | PC                    | 2-61                      | 2350910.17 | 1753845.34 | -      | 11+64.46; 162.28' Lt | Ctr - Not shown on plan |                      |              |
| 1-37                      | 2350936.07 | 1754195.32 | 481.17 | 11+38.96; 187.73' Rt | 3/4Δ                   | 2-19                      | 2351036.30 | 1754011.98 | -      | 12+64.89; 21.00' Rt. | Ctr                   | 2-62                      | 2351098.19 | 1754133.42 | 479.48 | 13+08.38; 150.17' Rt | EOG                     |                      |              |
| 1-38                      | 2350952.74 | 1754191.76 | 480.99 | 11+55.97; 186.64' Rt | PRC                    | 2-20                      | 2351026.45 | 1754010.25 | 477.96 | 12+55.40; 17.85' Rt. | 1/2Δ                  | 2-63                      | 2351114.07 | 1754122.40 | 479.47 | 13+25.70; 141.60' Rt | EOG                     |                      |              |
| 1-39                      | 2350966.87 | 1754229.18 | -      | 11+64.48; 225.72' Rt | Ctr                    | 2-21                      | 2351031.55 | 1754020.78 | 478.25 | 12+58.91; 29.01' Rt. | PCC                   | 2-64                      | 2351129.30 | 1754110.52 | 479.45 | 13+42.51; 478.58' Rt | EOG                     |                      |              |
| 1-40                      | 2350912.64 | 1754149.82 | 482.67 | 11+22.43; 139.29' Rt | BEG-END / PT           | 2-22                      | 2351073.33 | 1753943.34 | -      | 13+11.56; 41.50' Lt. | Ctr                   | 2-65                      | 2351146.96 | 1754167.10 | 478.53 | 13+51.71; 190.62' Rt | EOG                     |                      |              |
| 1-41                      | 2350944.91 | 1754153.59 | 482.14 | 11+53.80; 147.74' Rt | PC                     | 2-23                      | 2351041.48 | 1754025.37 | 478.32 | 12+68.06; 35.00' Rt. | 1/2Δ                  | 2-66                      | 2351142.22 | 1754160.66 | 478.69 | 13+47.96; 183.56' Rt | LL                      |                      |              |
| 1-42                      | 2350943.75 | 1754163.52 | -      | 11+51.21; 157.39' Rt | Ctr                    | 2-24                      | 2351051.89 | 1754028.68 | 478.34 | 12+77.88; 39.80' Rt. | PCC                   | 2-67                      | 2351128.52 | 1754179.89 | 478.48 | 13+31.60; 200.58' Rt | EOG                     |                      |              |
| 1-43                      | 2350953.67 | 1754162.19 | 481.80 | 11+61.21; 157.53' Rt | 1/2Δ                   |                           |            |            |        |                      |                       |                           |            |            |        |                      |                         |                      |              |
| 1-44                      | 2350947.48 | 1754172.80 | 481.60 | 11+53.54; 167.11' Rt | PCC                    |                           |            |            |        |                      |                       |                           |            |            |        |                      |                         |                      |              |

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| GRADING PLAN 2 POINT DATA |            |            |        |                      |                  |
|---------------------------|------------|------------|--------|----------------------|------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION      |
| 2-68                      | 2351124.15 | 1754173.19 | 478.64 | 13+28.25; 193.31' Rt | LL               |
| 2-69                      | 2351109.39 | 1754191.62 | 478.48 | 13+10.96; 209.39' Rt | EOG              |
| 2-70                      | 2351105.40 | 1754184.69 | 478.64 | 13+08.03; 201.95' Rt | LL               |
| 2-71                      | 2351093.72 | 1754200.18 | 478.50 | 12+94.21; 215.57' Rt | PCC              |
| 2-72                      | 2351090.05 | 1754193.07 | 478.66 | 12+91.61; 208.00' Rt | Ctr / LL HPI     |
| 2-73                      | 2351085.93 | 1754199.93 | 478.61 | 12+86.54; 214.18' Rt | PT               |
| 2-74                      | 2351068.36 | 1754189.38 | 479.14 | 12+70.69; 201.18' Rt | EOG HPI          |
| 2-75                      | 2351073.21 | 1754182.97 | 479.22 | 12+76.43; 195.54' Rt | LL               |
| 2-76                      | 2351052.83 | 1754170.31 | 479.91 | 12+58.12; 180.05' Rt | LL               |
| 2-77                      | 2351047.70 | 1754176.45 | 479.87 | 12+52.15; 185.37' Rt | EOG HPI          |
| 2-78                      | 2351077.84 | 1754139.09 | -      | 12+87.42; 152.82' Rt | Ctr              |
| 2-79                      | 2351034.14 | 1754158.94 | 480.54 | 12+41.29; 166.07' Rt | EOP HPI          |
| 2-80                      | 2351041.42 | 1754155.63 | 480.51 | 12+48.98; 163.86' Rt | EOP HPI          |
| 2-81                      | 2351039.76 | 1754151.33 | 480.66 | 12+47.96; 159.36' Rt | PT               |
| 2-82                      | 2351033.78 | 1754132.72 | 481.15 | 12+44.76; 140.08' Rt | EOP              |
| 2-83                      | 2351027.66 | 1754113.68 | 481.24 | 12+41.49; 120.35' Rt | EOP              |
| 2-84                      | 2351021.54 | 1754094.64 | 480.92 | 12+38.22; 100.62' Rt | EOP              |
| 2-85                      | 2351015.41 | 1754075.60 | 480.52 | 12+34.94; 80.89' Rt. | EOP              |
| 2-86                      | 2351009.29 | 1754056.56 | 480.10 | 12+31.67; 61.16' Rt. | EOP              |
| 2-87                      | 2351003.12 | 1754037.35 | 479.32 | 12+28.36; 41.25' Rt. | EOP END / PC     |
| 2-88                      | 2351000.22 | 1754028.34 | 478.86 | 12+26.82; 31.91' Rt. | PC               |
| 2-89                      | 2350976.42 | 1754035.99 | -      | 12+02.15; 36.00' Rt. | Ctr              |
| 2-90                      | 2350996.13 | 1754020.61 | 478.52 | 12+23.90; 23.66' Rt. | 1/4Δ             |
| 2-91                      | 2350989.63 | 1754014.76 | 478.34 | 12+18.32; 16.93' Rt. | 1/2Δ             |
| 2-92                      | 2350981.51 | 1754011.51 | 478.32 | 12+10.76; 12.53' Rt. | 3/4Δ             |
| 2-93                      | 2350972.77 | 1754011.25 | 478.42 | 12+02.15; 11.00' Rt. | PT               |
| 2-94                      | 2350972.22 | 1754011.34 | 478.43 | 12+01.59; 11.00' Rt. | PC               |
| 2-95                      | 2350973.68 | 1754021.23 | -      | 12+01.59; 21.00' Rt. | Ctr              |
| 2-96                      | 2350969.47 | 1754012.16 | 478.47 | 11+98.76; 11.41' Rt. | HPI              |
| 2-97                      | 2350963.95 | 1754023.56 | 478.87 | 11+91.63; 21.86' Rt. | LL PT            |
| 2-98                      | 2350972.38 | 1754024.41 | 478.84 | 11+99.84; 23.96' Rt. | Grade break PC   |
| 2-99                      | 2350982.11 | 1754022.11 | -      | 12+09.80; 23.10' Rt. | Ctr              |
| 2-100                     | 2350978.21 | 1754023.03 | 476.84 | 12+05.82; 23.44' Rt. | Grade break PC   |
| 2-101                     | 2350983.62 | 1754018.40 | 476.31 | 12+11.85; 19.66' Rt. | Grade break PCC  |
| 2-102                     | 2350990.37 | 1754023.08 | 476.46 | 12+17.83; 25.27' Rt. | Grade break 1/2Δ |
| 2-103                     | 2350994.51 | 1754030.17 | 476.86 | 12+20.90; 32.89' Rt. | Grade break PT   |
| 2-104                     | 2350997.41 | 1754039.18 | 477.32 | 12+22.45; 42.23' Rt. | Grade break PC   |
| 2-105                     | 2350988.84 | 1754041.94 | -      | 12+13.57; 43.70' Rt. | Ctr              |
| 2-106                     | 2350997.10 | 1754045.51 | 477.84 | 12+21.22; 48.44' Rt. | Grade break PT   |
| 2-107                     | 2351002.61 | 1754047.89 | 479.84 | 12+26.32; 51.60' Rt. | EOP PT           |
| 2-108                     | 2350991.02 | 1754059.58 | 478.41 | 12+13.15; 61.47' Rt. | Grade break PC   |
| 2-109                     | 2350996.52 | 1754061.96 | 480.41 | 12+18.25; 64.63' Rt. | EOP PC           |
| 2-110                     | 2351019.47 | 1754071.88 | -      | 12+39.50; 77.80' Rt. | Ctr              |
| 2-111                     | 2350994.50 | 1754070.67 | 480.66 | 12+14.97; 72.95' Rt. | EOP 1/2Δ         |
| 2-112                     | 2350995.67 | 1754079.53 | 480.87 | 12+14.83; 81.89' Rt. | EOP PT           |

| GRADING PLAN 2 POINT DATA |            |            |        |                      |                           |
|---------------------------|------------|------------|--------|----------------------|---------------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION               |
| 2-113                     | 2351007.55 | 1754116.49 | 481.61 | 12+21.19; 120.19' Rt | EOP PC                    |
| 2-114                     | 2351001.36 | 1754118.48 | 479.39 | 12+14.78; 121.26' Rt | Ctr / BEG Grade break     |
| 2-115                     | 2351003.11 | 1754124.74 | 481.64 | 12+15.59; 127.70' Rt | EOP 1/2Δ                  |
| 2-116                     | 2350995.04 | 1754119.98 | 481.56 | 12+08.30; 121.82' Rt | EOP HPI / PT              |
| 2-117                     | 2350988.17 | 1754071.12 | 478.58 | 12+08.64; 72.47' Rt. | Gradebreak PC             |
| 2-118                     | 2350988.84 | 1754065.39 | 478.29 | 12+10.14; 66.90' Rt. | Grade break HPI           |
| 2-119                     | 2350988.14 | 1754090.89 | 480.80 | 12+05.73; 92.03' Rt. | EOG                       |
| 2-120                     | 2350983.53 | 1754071.43 | 480.23 | 12+04; 72.10' Rt.    | EOG                       |
| 2-121                     | 2350978.91 | 1754051.97 | 479.65 | 12+02.28; 52.16' Rt. | EOG                       |
| 2-122                     | 2350974.30 | 1754032.51 | 479.08 | 12+00.56; 32.25' Rt. | EOG                       |
| 2-123                     | 2350966.51 | 1754034.36 | 479.20 | 11+92.59; 32.94' Rt. | LL                        |
| 2-124                     | 2350971.13 | 1754053.82 | 479.77 | 11+94.31; 52.86' Rt. | LL                        |
| 2-125                     | 2350975.74 | 1754073.28 | 480.35 | 11+96.03; 72.79' Rt. | LL                        |
| 2-126                     | 2350980.36 | 1754092.74 | 480.92 | 11+97.76; 92.72' Rt. | LL                        |
| 2-127                     | 2350987.26 | 1754121.83 | 481.66 | 12+00.33; 122.50' Rt | EOP HPI                   |
| 2-128                     | 2350988.50 | 1754127.09 | 481.69 | 12+00.64; 126.03' Rt | EOP PC                    |
| 2-129                     | 2350996.29 | 1754125.24 | -      | 12+08.77; 127.20' Rt | Ctr                       |
| 2-130                     | 2350992.17 | 1754132.10 | 481.69 | 12+03.69; 133.38' Rt | EOP PT                    |
| 2-131                     | 2351003.88 | 1754139.13 | 481.48 | 12+14.25; 142.05' Rt | EOP HPI                   |
| 2-132                     | 2351008.60 | 1754141.96 | 481.37 | 12+18.51; 145.54' Rt | EOP HPI                   |
| 2-133                     | 2350999.25 | 1754146.85 | 481.71 | 12+08.55; 149.01' Rt | EOP @ Grade Break         |
| 2-134                     | 2351003.97 | 1754149.68 | 481.82 | 12+12.80; 152.50' Rt | EOP @ Grade Break         |
| 2-135                     | 2350996.42 | 1754151.56 | 481.60 | 12+05.06; 153.26' Rt | EOP                       |
| 2-136                     | 2351001.14 | 1754154.39 | 481.71 | 12+09.31; 156.75' Rt | EOP @ Grade Break         |
| 2-137                     | 2351011.43 | 1754160.57 | 482.31 | 12+18.58; 164.36' Rt | EOP HPI                   |
| 2-138                     | 2351014.00 | 1754162.11 | 482.46 | 12+20.90; 166.26' Rt | Vault Toilet Finish Floor |
| 2-139                     | 2351016.83 | 1754157.40 | 482.35 | 12+24.39; 162.01' Rt | EOP @ Grade Break         |
| 2-140                     | 2351021.54 | 1754160.23 | 482.41 | 12+28.64; 165.50' Rt | EOP                       |
| 2-141                     | 2351020.26 | 1754162.37 | 482.46 | 12+27.06; 167.43' Rt | EOP HPI                   |
| 2-142                     | 2351018.71 | 1754164.94 | 482.52 | 12+25.16; 169.75' Rt | Vault Toilet Finish Floor |
| 2-143                     | 2351030.69 | 1754168.63 | 482.58 | 12+36.46; 175.15' Rt | EOP                       |
| 2-144                     | 2351026.57 | 1754169.66 | 482.61 | 12+32.24; 175.57' Rt | Vault Toilet Finish Floor |
| 2-145                     | 2351023.06 | 1754175.52 | 482.68 | 12+27.91; 180.85' Rt | Vault Toilet Finish Floor |
| 2-146                     | 2351024.09 | 1754179.64 | 482.58 | 12+28.32; 185.07' Rt | EOP                       |
| 2-147                     | 2351010.48 | 1754167.97 | 482.53 | 12+16.57; 171.55' Rt | Vault Toilet Finish Floor |
| 2-148                     | 2351006.37 | 1754169.00 | 482.47 | 12+12.35; 171.96' Rt | EOP                       |
| 2-149                     | 2350996.51 | 1754160.83 | 481.19 | 12+03.78; 162.44' Rt | PC                        |
| 2-150                     | 2351026.67 | 1754153.68 | -      | 12+34.67; 159.77' Rt | Ctr                       |
| 2-151                     | 2351004.07 | 1754174.90 | 480.96 | 12+09.21; 177.46' Rt | 1/4Δ                      |
| 2-152                     | 2351017.62 | 1754183.33 | 480.69 | 12+21.39; 187.78' Rt | 1/2Δ                      |
| 2-153                     | 2351033.58 | 1754183.90 | 480.27 | 12+37.09; 190.68' Rt | 3/4Δ                      |
| 2-154                     | 2351025.45 | 1754151.86 | -      | 12+33.72; 157.79' Rt | Ctr                       |
| 2-155                     | 2351045.62 | 1754205.17 | 479.35 | 12+45.89; 213.48' Rt | EOG                       |
| 2-156                     | 2351020.74 | 1754208.67 | 479.80 | 12+20.77; 213.30' Rt | EOG                       |
| 2-157                     | 2350996.77 | 1754201.12 | 480.28 | 11+98.16; 202.34' Rt | EOG                       |

| GRADING PLAN 2 POINT DATA |            |            |        |                      |                  |
|---------------------------|------------|------------|--------|----------------------|------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION      |
| 2-158                     | 2350991.05 | 1754197.31 | 480.42 | 11+93.06; 197.73' Rt | PRC              |
| 2-159                     | 2350982.30 | 1754192.27 | 480.61 | 11+85.14; 191.47' Rt | 1/4Δ             |
| 2-160                     | 2350972.60 | 1754189.59 | 480.78 | 11+75.93; 187.40' Rt | 1/2Δ             |
| 2-161                     | 2350962.53 | 1754189.41 | 480.90 | 11+66; 477.04' Rt.   | 3/4Δ             |
| 2-162                     | 2351124.07 | 1754034.91 | 477.96 | 13+48.37; 56.50' Rt. | Grade Break      |
| 2-163                     | 2351104.28 | 1754037.83 | 478.21 | 13+28.37; 56.50' Rt. | Grade Break      |
| 2-164                     | 2351087.65 | 1754040.28 | 478.42 | 13+11.56; 56.50' Rt. | Grade Break PC   |
| 2-165                     | 2351072.67 | 1754041.36 | 478.58 | 12+96.59; 55.37' Rt  | Grade Break 1/2Δ |
| 2-166                     | 2351057.71 | 1754040.17 | 478.67 | 12+81.96; 52.02' Rt. | Grade Break PRC  |
| 2-167                     | 2351044.18 | 1754041.54 | 478.88 | 12+68.43; 51.38' Rt. | Grade Break 1/4Δ |
| 2-168                     | 2351033.04 | 1754049.32 | 479.44 | 12+56.22; 57.46' Rt. | Grade Break 1/2Δ |
| 2-169                     | 2351027.10 | 1754061.55 | 480.05 | 12+48.56; 68.68' Rt. | Grade Break 3/4Δ |
| 2-170                     | 2351027.86 | 1754075.12 | 480.44 | 12+47.33; 82.21' Rt. | Grade Break PT   |
| 2-171                     | 2351032.96 | 1754090.97 | 480.77 | 12+50.05; 98.64' Rt. | Grade Break      |
| 2-172                     | 2351039.08 | 1754110.01 | 481.13 | 12+53.33; 118.37' Rt | Grade Break      |
| 2-173                     | 2351045.20 | 1754129.05 | 481.08 | 12+56.60; 138.10' Rt | Grade Break      |
| 2-174                     | 2351049.11 | 1754141.22 | 480.77 | 12+58.69; 150.73' Rt | Grade Break PC   |
| 2-175                     | 2351072.91 | 1754133.57 | -      | 12+83.35; 146.62' Rt | Grade Break Ctr  |
| 2-176                     | 2351054.70 | 1754150.69 | 480.43 | 12+62.83; 160.90' Rt | Grade Break 1/4Δ |
| 2-177                     | 2351063.81 | 1754156.85 | 480.11 | 12+70.94; 168.33' Rt | Grade Break 1/2Δ |
| 2-178                     | 2351074.68 | 1754158.51 | 479.88 | 12+81.45; 171.56' Rt | Grade Break 3/4Δ |
| 2-179                     | 2351085.21 | 1754155.34 | 479.79 | 12+92.34; 169.96' Rt | Grade Break PCC  |
| 2-180                     | 2351104.75 | 1754143.46 | 479.80 | 13+13.40; 161.07' Rt | Grade Break      |
| 2-181                     | 2351121.18 | 1754132.07 | 479.83 | 13+31.32; 152.20' Rt | Grade Break      |
| 2-182                     | 2351136.95 | 1754119.77 | 479.81 | 13+48.72; 142.33' Rt | Grade Break      |

Abbreviations

BEG = Begin  
CL = Centerline  
Ctr = Center Point of Arc  
END = End

EOG = Edge of Gravel  
EOP = Edge of Pavement  
HPI = Horizontal Point of Intersection  
LL = Layout Line

Lt. = Left  
PC = Point of Curvature  
PCC = Point of Compound Curvature  
POC = Point on Curve

PRC = Point of Reverse Curvature  
PT = Point of Tangency  
Rt. = Right  
X/XΔ = Fraction of Included Angle

LAYOUT AND GRADING POINT  
DATA TABLES - 2

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| GRADING PLAN 3 POINT DATA |            |            |        |                      |                     |
|---------------------------|------------|------------|--------|----------------------|---------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION         |
| 3-00                      | 2351122.93 | 1753947.64 | 475.22 | 13+60; 30.00' Lt.    | EOG @ 13+60 Lt.     |
| 3-01                      | 2351142.72 | 1753944.72 | 474.97 | 13+80; 30.00' Lt.    | EOG @ 13+80 Lt.     |
| 3-02                      | 2351162.50 | 1753941.79 | 474.72 | 14+00; 30.00' Lt.    | EOG @ 14+00 Lt.     |
| 3-03                      | 2351182.29 | 1753938.87 | 474.48 | 14+20; 30.00' Lt.    | EOG @ 14+20 Lt.     |
| 3-04                      | 2351202.08 | 1753935.95 | 474.23 | 14+40; 30.00' Lt.    | EOG @ 14+40 Lt.     |
| 3-05                      | 2351221.86 | 1753933.03 | 473.98 | 14+60; 30.00' Lt.    | EOG @ 14+60 Lt.     |
| 3-06                      | 2351241.65 | 1753930.11 | 473.73 | 14+80; 30.00' Lt.    | EOG @ 14+80 Lt.     |
| 3-07                      | 2351261.43 | 1753927.19 | 473.48 | 15+00; 30.00' Lt.    | EOG @ 15+00 Lt.     |
| 3-08                      | 2351281.22 | 1753924.26 | 473.23 | 15+20; 30.00' Lt.    | EOG @ 15+20 Lt.     |
| 3-09                      | 2351128.92 | 1753988.20 | 476.45 | 13+60; 11.00' Rt.    | EOG @ 13+60 Rt.     |
| 3-10                      | 2351148.71 | 1753985.28 | 476.20 | 13+80; 11.00' Rt.    | EOG @ 13+80 Rt.     |
| 3-11                      | 2351168.49 | 1753982.35 | 475.95 | 14+00; 11.00' Rt.    | EOG @ 14+00 Rt.     |
| 3-12                      | 2351191.35 | 1753978.98 | 475.67 | 14+23.10; 11.00' Rt. | PC                  |
| 3-13                      | 2351193.10 | 1753990.85 | -      | 14+23.10; 23.00' Rt. | Ctr                 |
| 3-14                      | 2351202.85 | 1753983.86 | 475.73 | 14+33.77; 17.51' Rt. | 1/2Δ                |
| 3-15                      | 2351203.78 | 1753996.32 | 476.11 | 14+32.86; 29.97' Rt. | PT                  |
| 3-16                      | 2351197.82 | 1754007.95 | 476.53 | 14+25.27; 40.61' Rt. | PC                  |
| 3-17                      | 2351189.81 | 1754003.85 | -      | 14+17.94; 35.38' Rt. | Ctr                 |
| 3-18                      | 2351188.34 | 1754012.73 | 476.75 | 14+15.20; 43.95' Rt. | 1/2Δ                |
| 3-19                      | 2351180.90 | 1754005.16 | 476.57 | 14+08.94; 35.38' Rt. | PT                  |
| 3-20                      | 2351178.80 | 1753990.94 | 476.14 | 14+08.94; 21.00' Rt. | HPI                 |
| 3-21                      | 2351178.07 | 1753985.99 | 474.18 | 14+08.94; 16.00' Rt. | Grade break HPI     |
| 3-22                      | 2351184.16 | 1753986.11 | 473.77 | 14+14.94; 17.00' Rt. | Grade break HPI     |
| 3-23                      | 2351192.22 | 1753984.92 | 473.67 | 14+23.10; 17.00' Rt. | Grade break PC      |
| 3-24                      | 2351197.98 | 1753987.36 | 473.73 | 14+23.10; 23.00' Rt. | Grade break 1/2Δ    |
| 3-25                      | 2351198.44 | 1753993.59 | 474.11 | 14+27.98; 26.49' Rt. | Grade break PT      |
| 3-26                      | 2351192.48 | 1754005.22 | 474.53 | 14+20.39; 37.12' Rt. | Grade break PC      |
| 3-27                      | 2351189.32 | 1754006.81 | 474.75 | 14+17.03; 38.24' Rt. | Grade break 1/2Δ    |
| 3-28                      | 2351186.84 | 1754004.29 | 474.57 | 14+14.94; 35.38' Rt. | Grade break PT      |
| 3-29                      | 2351169.22 | 1753987.30 | 474.29 | 14+00; 16.00' Rt.    | Grade break @ 14+00 |
| 3-30                      | 2351158.45 | 1753993.94 | 476.40 | 13+88.37; 21.00' Rt. | EOG                 |
| 3-31                      | 2351149.44 | 1753990.22 | 474.54 | 13+80; 16.00' Rt.    | Grade break @ 13+80 |
| 3-32                      | 2351138.67 | 1753996.87 | 476.65 | 13+68.37; 21.00' Rt. | EOG                 |
| 3-33                      | 2351129.65 | 1753993.14 | 474.79 | 13+60; 16.00' Rt.    | Grade break @ 13+60 |
| 3-34                      | 2351118.88 | 1753999.79 | 476.90 | 13+48.37; 21.00' Rt. | EOG                 |
| 3-35                      | 2351214.07 | 1754034.91 | 477.96 | 13+48.37; 56.50' Rt. | Grade break         |
| 3-36                      | 2351125.82 | 1754046.78 | 477.60 | 13+48.37; 68.50' Rt. | EOG                 |
| 3-37                      | 2351141.87 | 1754032.28 | 477.74 | 13+66.37; 56.50' Rt. | Grade break PC      |
| 3-38                      | 2351143.62 | 1754044.15 | 477.38 | 13+66.37; 68.50' Rt. | EOG PC              |
| 3-39                      | 2351146.54 | 1754063.93 | -      | 13+66.37; 88.50' Rt. | Ctr                 |
| 3-40                      | 2351159.44 | 1754034.65 | 477.57 | 13+83.40; 61.41' Rt. | Grade break 1/4Δ    |
| 3-41                      | 2351154.60 | 1754045.63 | 477.21 | 13+77.01; 71.57' Rt. | EOG 1/4Δ            |
| 3-42                      | 2351173.05 | 1754046.00 | 477.76 | 13+95.21; 74.63' Rt. | Grade break 1/2Δ    |
| 3-43                      | 2351163.11 | 1754052.73 | 477.40 | 13+84.39; 79.83' Rt. | EOG 1/2Δ            |
| 3-44                      | 2351178.52 | 1754062.86 | 478.32 | 13+98.16; 92.11' Rt. | Grade break 3/4Δ    |
| 3-45                      | 2351166.53 | 1754063.26 | 477.96 | 13+86.24; 90.75' Rt. | EOG 3/4Δ            |

| GRADING PLAN 3 POINT DATA |            |            |        |                      |                         |
|---------------------------|------------|------------|--------|----------------------|-------------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION             |
| 3-46                      | 2351174.19 | 1754080.05 | 478.99 | 13+91.37; 108.48' Rt | Grade break PCC         |
| 3-47                      | 2351087.79 | 1754029.70 | -      | 13+85.28; 115.17' Rt | Ctr - Not shown on plan |
| 3-48                      | 2351163.82 | 1754074.01 | 478.63 | 13+81.99; 100.99' Rt | EOG PCC                 |
| 3-49                      | 2351166.43 | 1754091.47 | 479.34 | 13+82.02; 118.64' Rt | Grade break 1/2Δ        |
| 3-50                      | 2351157.00 | 1754084.06 | 478.98 | 13+73.77; 109.93' Rt | EOG 1/2Δ                |
| 3-51                      | 2351157.17 | 1754101.71 | 479.58 | 13+71.37; 127.42' Rt | Grade break PCC         |
| 3-52                      | 2351148.85 | 1754093.07 | 479.22 | 13+64.39; 117.66' Rt | EOG PCC                 |
| 3-53                      | 2351136.94 | 1754119.77 | 479.81 | 13+48.72; 142.33' Rt | Grade break             |
| 3-54                      | 2351129.30 | 1754110.52 | 479.45 | 13+42.51; 132.06' Rt | EOG                     |
| 3-55                      | 2351142.22 | 1754160.66 | 478.69 | 13+47.96; 183.56' Rt | LL                      |
| 3-56                      | 2351159.56 | 1754147.13 | 478.78 | 13+67.09; 172.71' Rt | LL                      |
| 3-57                      | 2351176.11 | 1754132.65 | 478.89 | 13+85.58; 160.80' Rt | LL                      |
| 3-58                      | 2351194.13 | 1754114.85 | 478.64 | 14+06.01; 145.83' Rt | LL                      |
| 3-59                      | 2351212.62 | 1754093.93 | 478.01 | 14+27.36; 127.82' Rt | LL                      |
| 3-60                      | 2351238.27 | 1754058.94 | 476.52 | 14+57.84; 96.96' Rt. | LL HPI                  |
| 3-61                      | 2351217.63 | 1754045.51 | 476.93 | 14+39.39; 80.65' Rt. | LL PC                   |
| 3-62                      | 2351224.18 | 1754035.45 | -      | 14+47.33; 71.66' Rt. | LL Ctr                  |
| 3-63                      | 2351212.57 | 1754038.47 | 475.18 | 14+35.38; 72.68' Rt. | LL 1/2Δ                 |
| 3-64                      | 2351213.56 | 1574029.86 | 476.94 | 14+37.35; 65.01' Rt. | LL PCC                  |
| 3-65                      | 2351266.65 | 1754057.82 | -      | 14+86.08; 100.00' Rt | LL Ctr                  |
| 3-66                      | 2351221.53 | 1754018.27 | 476.66 | 14+47.22; 54.28' Rt. | LL 1/4Δ                 |
| 3-67                      | 2351231.97 | 1754008.86 | 476.29 | 14+58.92; 46.50' Rt. | LL 1/2Δ                 |
| 3-68                      | 2351244.31 | 1754002.14 | 474.72 | 14+73.02; 41.44' Rt. | LL 3/4Δ                 |
| 3-69                      | 2351257.88 | 1753998.47 | 475.75 | 14+86.08; 40.00' Rt. | LL PT                   |
| 3-70                      | 2351291.44 | 1753993.51 | 475.33 | 15+20; 40.00' Rt.    | LL @ 15+20 Rt.          |
| 3-71                      | 2351292.61 | 1754001.43 | 475.17 | 15+20; 48.00' Rt.    | EOG @ 15+20 Rt.         |
| 3-72                      | 2351275.32 | 1754003.98 | 475.39 | 15+02.52; 48.00' Rt. | PC                      |
| 3-73                      | 2351276.05 | 1754008.93 | -      | 15+02.52; 53.00' Rt. | Ctr                     |
| 3-74                      | 2351271.53 | 1754006.78 | 475.41 | 14+98.10; 50.66' Rt. | 1/2Δ                    |
| 3-75                      | 2351271.75 | 1754011.48 | 475.34 | 14+97.89; 54.89' Rt. | PRC                     |
| 3-76                      | 2351253.14 | 1754022.50 | -      | 14+77.88; 63.09' Rt. | Ctr                     |
| 3-77                      | 2351274.75 | 1754021.58 | 475.20 | 14+99.39; 65.33' Rt. | 1/2Δ                    |
| 3-78                      | 2351272.62 | 1754031.90 | 475.30 | 14+95.77; 75.23' Rt. | PT                      |
| 3-79                      | 2351265.73 | 1754046.19 | 475.60 | 14+86.86; 88.36' Rt. | PC                      |
| 3-80                      | 2351280.14 | 1754053.15 | -      | 15+00.11; 97.34' Rt. | Ctr                     |
| 3-81                      | 2351264.76 | 1754057.55 | 475.71 | 14+84.25; 99.45' Rt. | 1/2Δ                    |
| 3-82                      | 2351271.59 | 1754066.67 | 475.42 | 14+89.67; 109.47' Rt | PT                      |
| 3-83                      | 2351301.47 | 1754085.56 | 474.42 | 15+16.47; 132.52' Rt | PC                      |
| 3-84                      | 2351291.72 | 1754100.69 | -      | 15+04.62; 146.06' Rt | Ctr                     |
| 3-85                      | 2351307.34 | 1754091.75 | 474.01 | 15+21.36; 139.45' Rt | PCC                     |
| 3-86                      | 2351154.82 | 1754179.58 | -      | 13+57.67; 204.11' Rt | Ctr                     |
| 3-87                      | 2351314.28 | 1754105.09 | 473.22 | 15+26.30; 153.72' Rt | 1/2Δ                    |
| 3-88                      | 2351310.14 | 1754110.46 | 473.16 | 15+21.41; 158.42' Rt | 1/2Δ                    |
| 3-89                      | 2351302.11 | 1754094.69 | 474.23 | 15+15.78; 141.66' Rt | PCC                     |
| 3-90                      | 2351298.22 | 1754090.60 | 474.53 | 15+12.52; 137.03' Rt | PT                      |
| 3-91                      | 2351268.38 | 1754071.74 | 475.54 | 14+85.76; 114.02' Rt | PC                      |

| GRADING PLAN 3 POINT DATA |            |            |        |                      |             |
|---------------------------|------------|------------|--------|----------------------|-------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION |
| 3-92                      | 2351255.03 | 1754092.88 | -      | 14+69.46; 132.98' Rt | Ctr         |
| 3-93                      | 2351250.25 | 1754068.34 | 476.26 | 14+68.32; 108.00' Rt | 1/2Δ        |
| 3-94                      | 2351234.72 | 1754078.30 | 476.97 | 14+51.50; 115.59' Rt | PRC         |
| 3-95                      | 2351219.90 | 1754097.67 | 477.80 | 14+34.01; 132.59' Rt | EOG         |
| 3-96                      | 2351200.40 | 1754119.87 | 478.47 | 14+11.48; 151.71' Rt | EOG         |
| 3-97                      | 2351181.55 | 1754138.52 | 478.73 | 13+90.10; 167.40' Rt | EOG         |
| 3-98                      | 2351164.66 | 1754153.30 | 478.62 | 13+71.23; 179.55' Rt | EOG         |
| 3-99                      | 2351146.96 | 1754167.10 | 478.53 | 13+51.71; 190.62' Rt | EOG         |

| GRADING PLAN 4 POINT DATA |            |            |        |                      |                       |
|---------------------------|------------|------------|--------|----------------------|-----------------------|
| PT.                       | NORTHING   | EASTING    | ELEV.  | STATION - OFFSET     | DESCRIPTION           |
| 4-00                      | 2351281.22 | 1753924.26 | 473.23 | 15+20; 30.00' Lt.    | EOG @ 15+20 Lt.       |
| 4-01                      | 2351301.00 | 1753921.34 | 472.98 | 15+40; 30.00' Lt.    | EOG @ 15+40 Lt.       |
| 4-02                      | 2351320.79 | 1753918.42 | 472.73 | 15+60; 30.00' Lt.    | EOG @ 15+60 Lt.       |
| 4-03                      | 2351340.57 | 1753915.50 | 472.48 | 15+80; 30.00' Lt.    | EOG @ 15+80 Lt.       |
| 4-04                      | 2351360.36 | 1753912.58 | 472.23 | 16+00; 30.00' Lt.    | EOG @ 16+00 Lt.       |
| 4-05                      | 2351380.15 | 1753909.66 | 471.99 | 16+20; 30.00' Lt.    | EOG @ 16+20 Lt.       |
| 4-06                      | 2351399.93 | 1753906.74 | 471.74 | 16+40; 30.00' Lt.    | EOG @ 16+40 Lt.       |
| 4-07                      | 2351407.13 | 1753905.67 | 471.65 | 16+47.28; 30.00' Lt. | HPI                   |
| 4-08                      | 2351416.47 | 1753905.55 | 471.57 | 16+56.54; 28.76' Lt. | EOG @ Grade break Lt. |
| 4-09                      | 2351430.56 | 1753905.36 | 469.95 | 16+70.50; 26.89' Lt. | END EOG               |
| 4-10                      | 2351434.48 | 1753931.96 | 470.76 | 16+70.50             | END EOG @ CL          |
| 4-11                      | 2351420.68 | 1753934.00 | 472.43 | 16+56.54             | Grade break @ CL      |
| 4-12                      | 2351440.33 | 1753971.53 | 471.96 | 16+70.50; 40.00' Rt. | END EOG               |
| 4-13                      | 2351426.52 | 1753973.57 | 473.63 | 16+56.54; 40.00' Rt. | EOG @ Grade break Rt. |
| 4-14                      | 2351410.16 | 1753975.98 | 473.84 | 16+40; 40.00' Rt.    | EOG @ 16+40 Rt.       |
| 4-15                      | 2351390.37 | 1753978.91 | 474.09 | 16+20; 40.00' Rt.    | EOG @ 16+20 Rt.       |
| 4-16                      | 2351378.55 | 1753980.65 | 474.23 | 16+08.05; 40.00' Rt. | EOG HPI               |
| 4-17                      | 2351375.26 | 1753989.22 | 474.13 | 16+03.54; 48.00' Rt. | EOG HPI               |
| 4-18                      | 2351350.80 | 1753984.75 | 474.58 | 15+80; 40.00' Rt.    | LL @ 15+80 Rt.        |
| 4-19                      | 2351351.97 | 1753992.66 | 474.42 | 15+80; 48.00' Rt.    | EOG @ 15+80 Rt.       |
| 4-20                      | 2351331.01 | 1753987.67 | 474.83 | 15+60; 40.00' Rt.    | LL @ 15+60 Rt.        |
| 4-21                      | 2351332.18 | 1753995.58 | 474.67 | 15+60; 48.00' Rt.    | EOG @ 15+60 Rt.       |
| 4-22                      | 2351311.23 | 1753990.59 | 475.08 | 15+40; 40.00' Rt.    | LL @ 15+40 Rt.        |
| 4-23                      | 2351312.40 | 1753998.51 | 474.92 | 15+40; 48.00' Rt.    | EOG @ 15+40 Rt.       |
| 4-24                      | 2351291.44 | 1753993.51 | 475.33 | 15+20; 40.00' Rt.    | LL @ 15+20 Rt.        |
| 4-25                      | 2351292.61 | 1754001.43 | 475.17 | 15+20; 48.00' Rt.    | EOG @ 15+20 Rt.       |
| 4-26                      | 2351320.01 | 1754118.85 | Match  | 15+29.96; 168.16' Rt | HPI / Match Existing  |
| 4-27                      | 2351316.48 | 1754126.97 | Match  | 15+25.27; 175.68' Rt | HPI / Match Existing  |

Abbreviations

BEG = Begin  
CL = Centerline  
Ctr = Center Point of Arc  
END = End

EOG = Edge of Gravel  
EOP = Edge of Pavement  
HPI = Horizontal Point of Intersection  
LL = Layout Line

Lt. = Left  
PC = Point of Curvature  
PCC = Point of Compound Curvature  
POC = Point on Curve

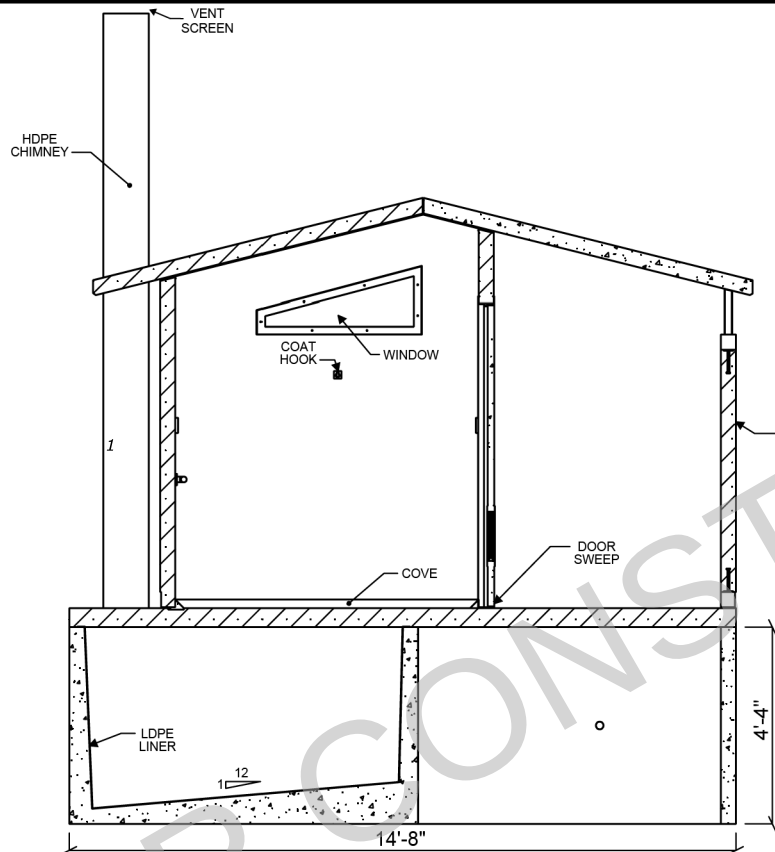
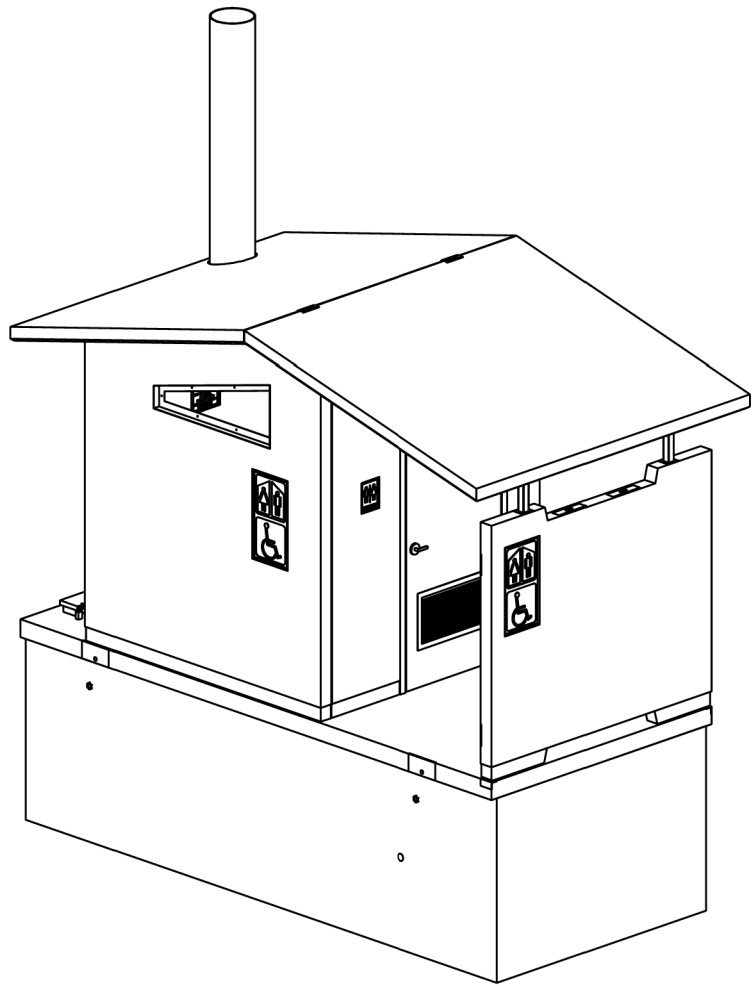
PRC = Point of Reverse Curvature  
PT = Point of Tangency  
Rt. = Right  
X/XΔ = Fraction of Included Angle

LAYOUT AND GRADING POINT  
DATA TABLES - 3

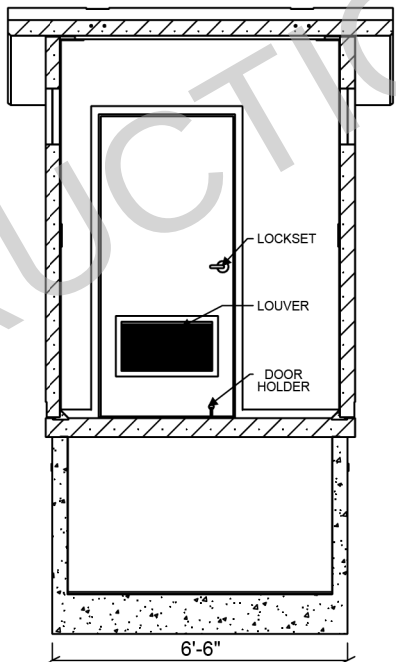


c:\pw-work\0548524\cnf00901\_det1.dgn [Vault Toilet Detail 1 of 2]  
16 May 2024 11:49 AM

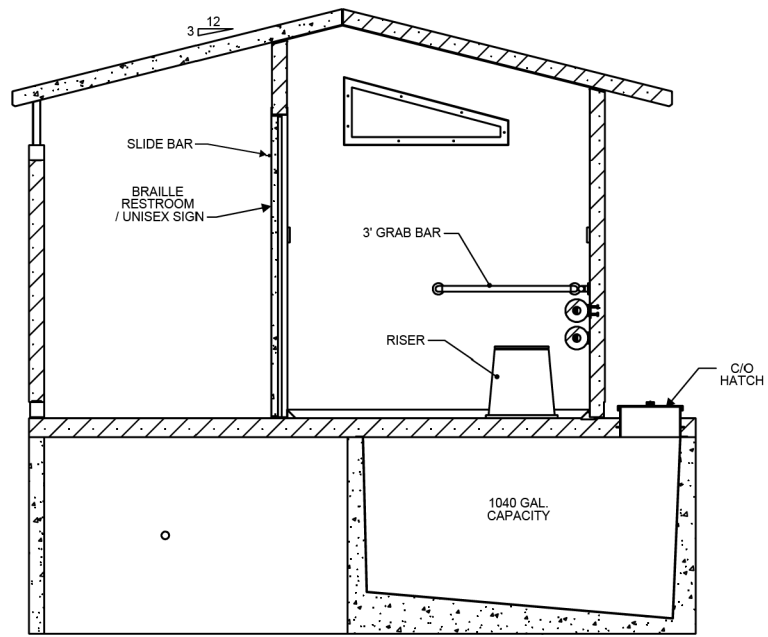
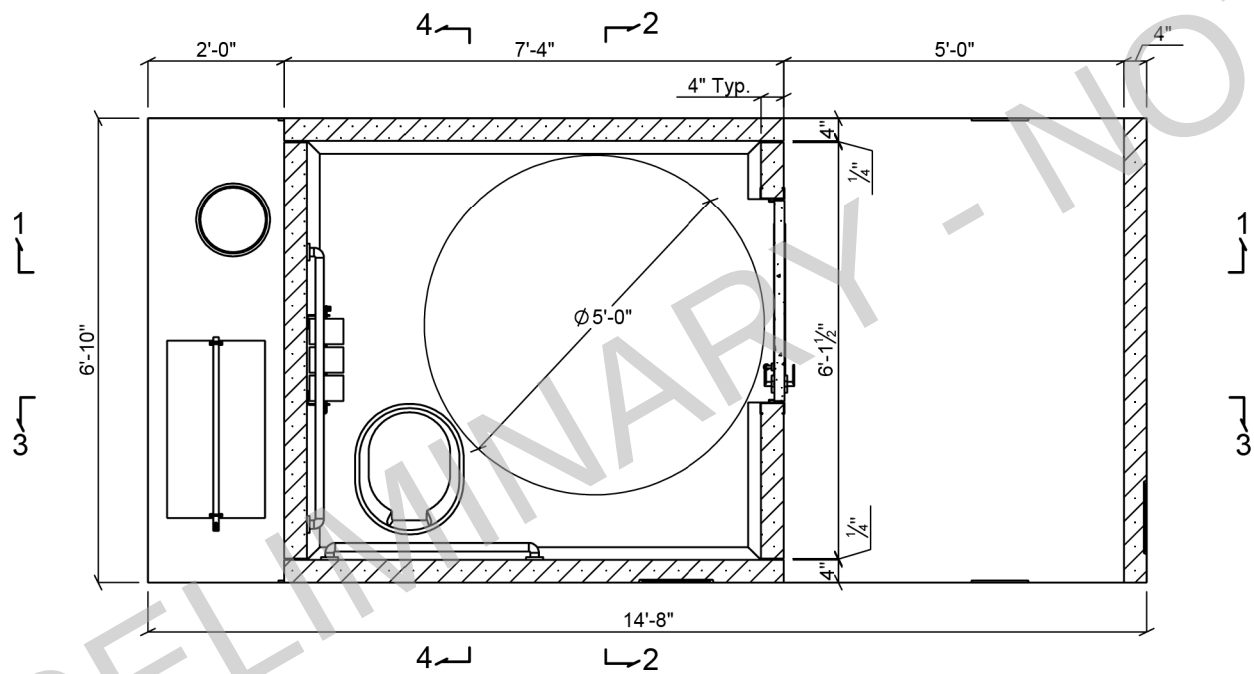
| PROJECT                | SHEET NUMBER |
|------------------------|--------------|
| AK USFS CHUG 1004 9(1) | E.1          |



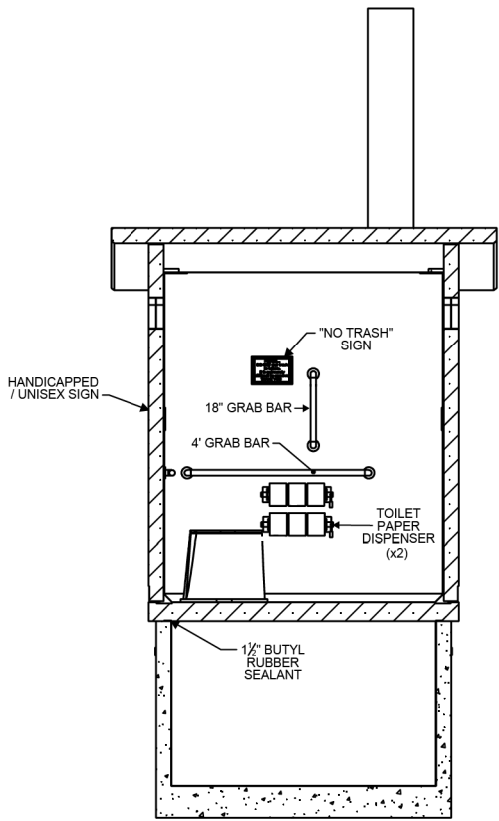
SECTION 1-1



SECTION 2-2



SECTION 3-3



SECTION 4-4

**Notes:**

Procure commercial product vault toilet with corrosion resistant package that includes corrosion resistant coating on all concrete surfaces and stainless steel doors, door & window frames, clean-out riser and hatch, and other hardware.

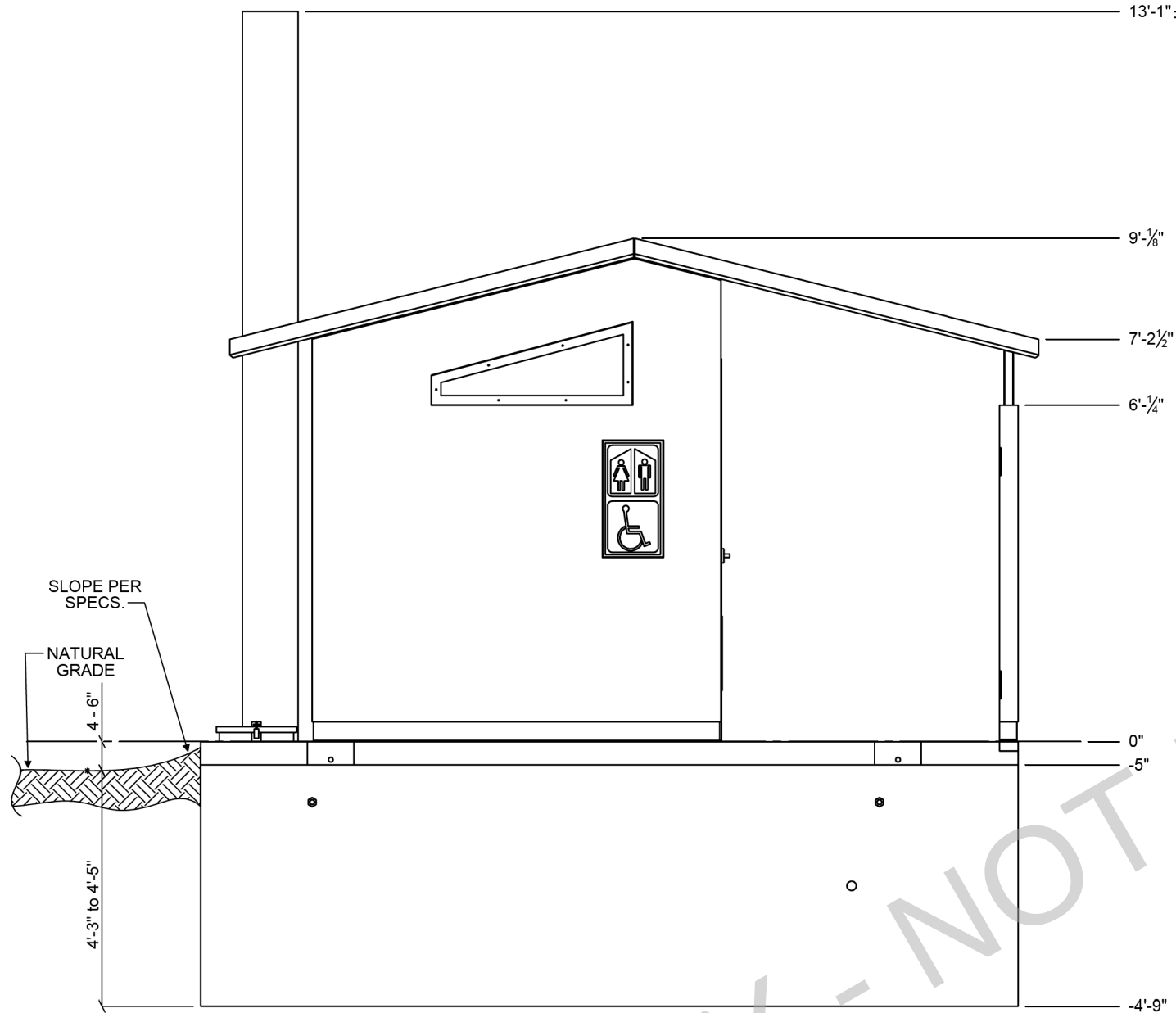
**Footnotes:**

<sup>1</sup> Clean-out hatch to be raised above finish floor grade and be secured with a freeze-resistant paddle lock.

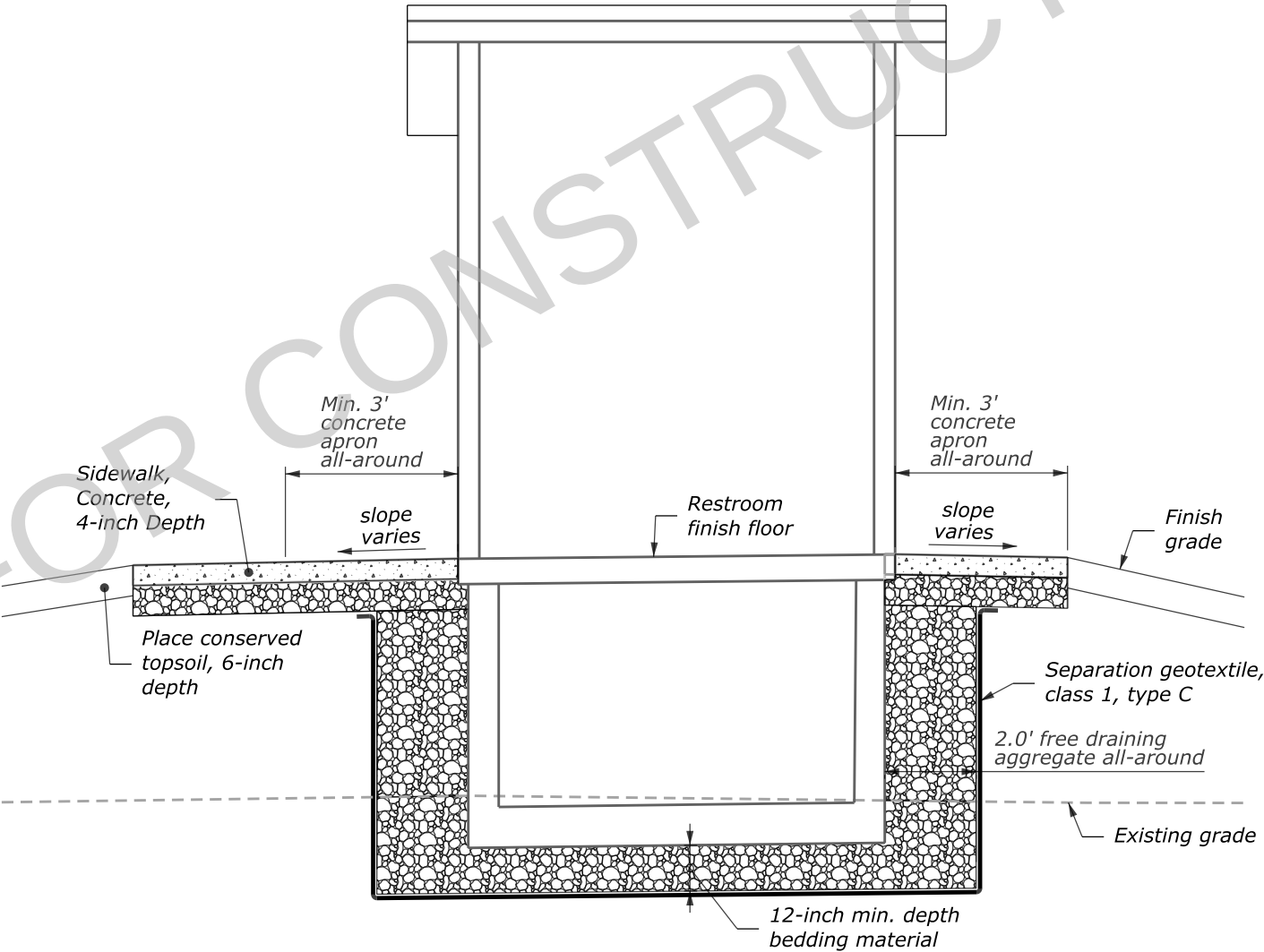
Not to Scale

**VAULT TOILET DETAIL 1 OF 2**

| STATE | PROJECT            | SHEET NUMBER |
|-------|--------------------|--------------|
| AK    | USFS CHUG1004 9(1) | E.2          |



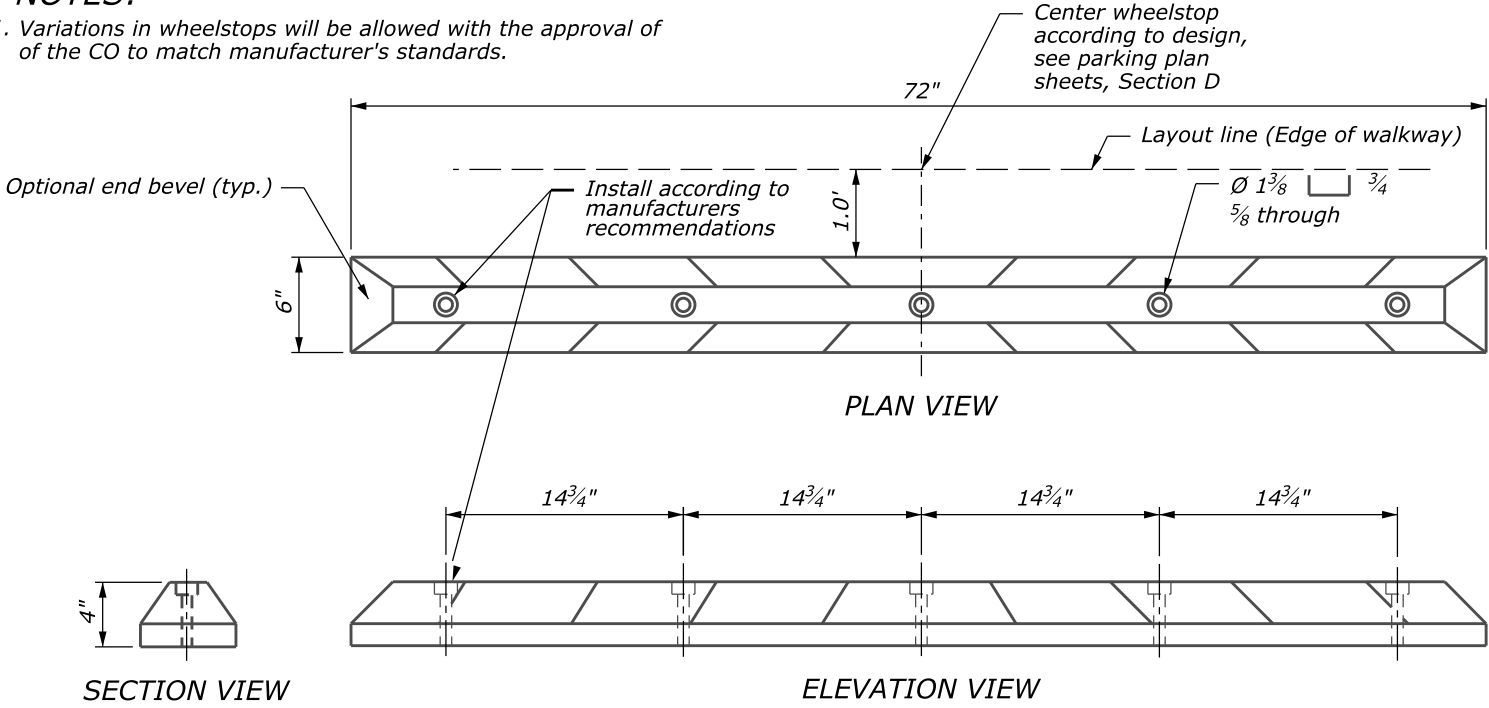
- WEIGHT:
- Building - 24,000 lbs.
  - Vault - 16,000 lbs.
- EXCAVATION:
- Vault Hole - 12' wide x 20' long x 5'-1" below desired finish floor elevation
  - See specifications for additional requirements.



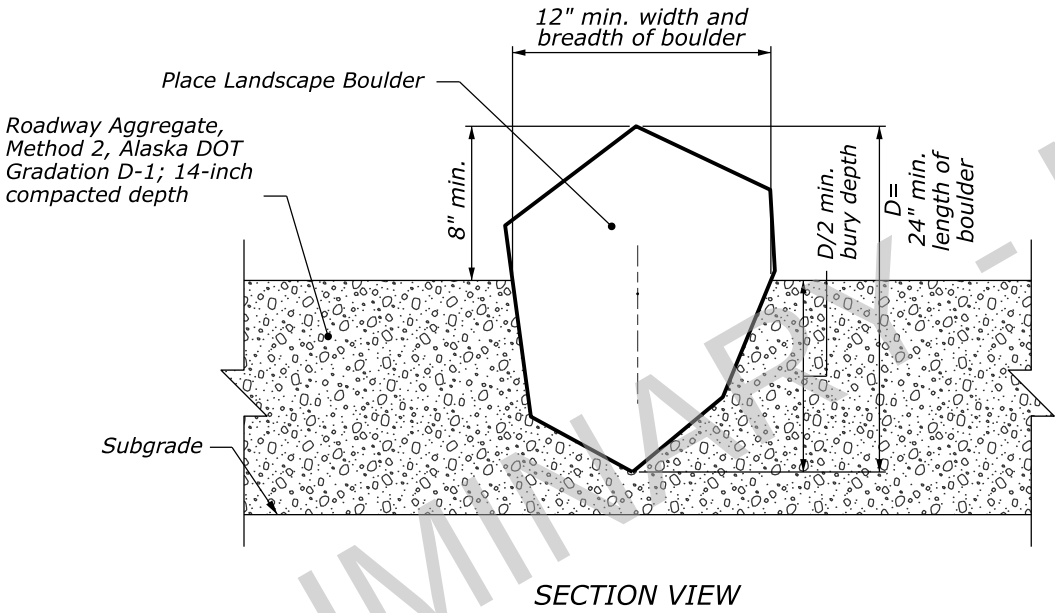
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NOTES:

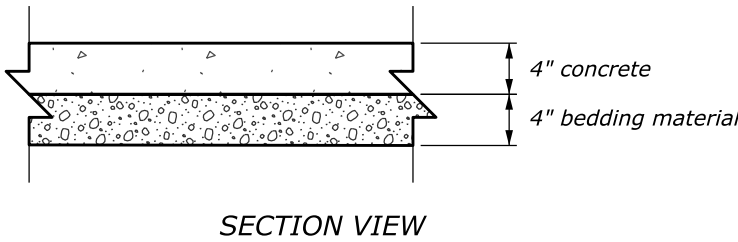
1. Variations in wheelstops will be allowed with the approval of of the CO to match manufacturer's standards.



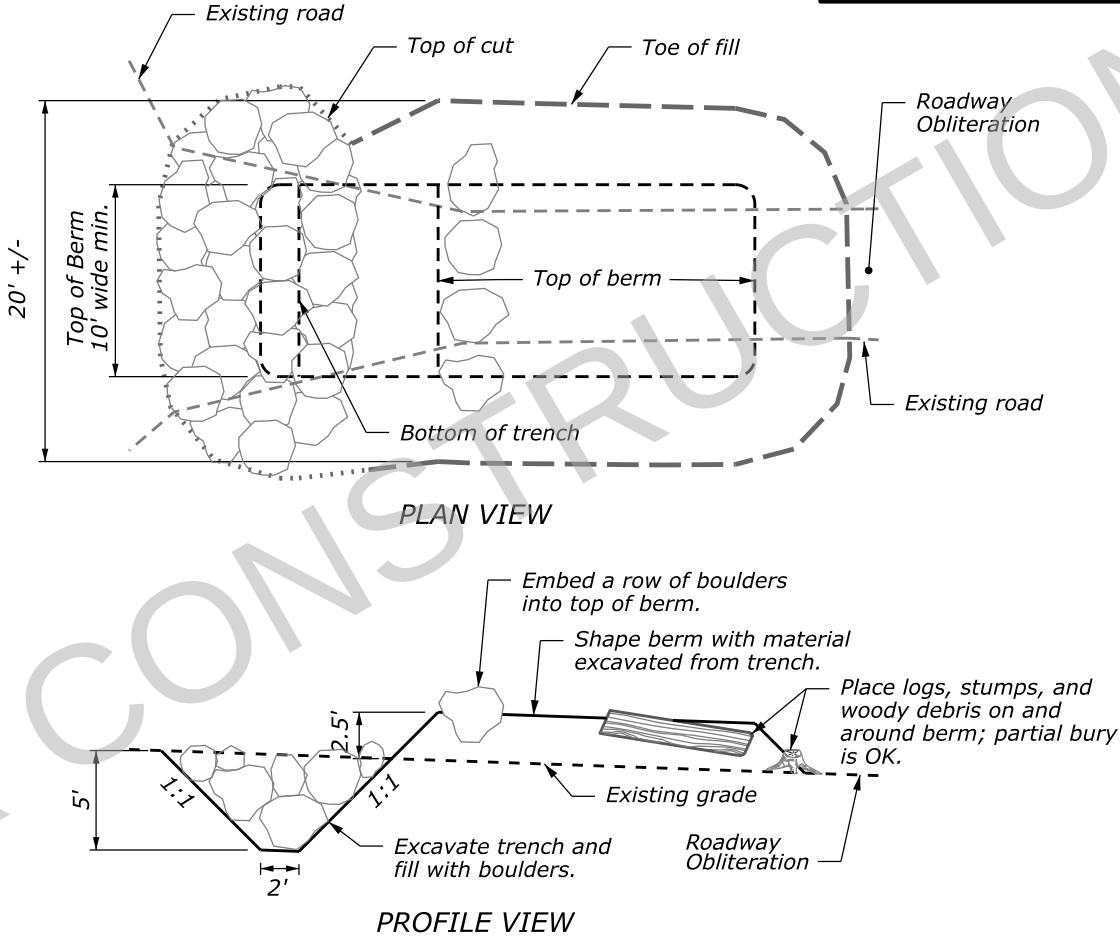
RUBBER WHEELSTOP DETAIL



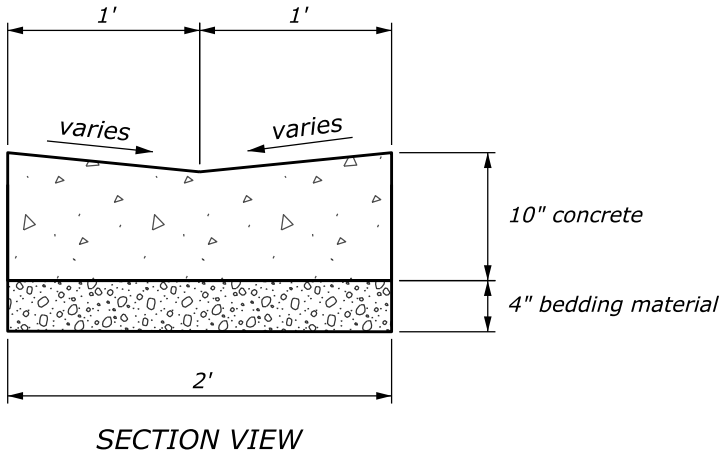
BOULDER INSTALLATION DETAIL



SIDEWALK, CONCRETE, 4-INCH DEPTH



ROAD CLOSURE BERM DETAIL

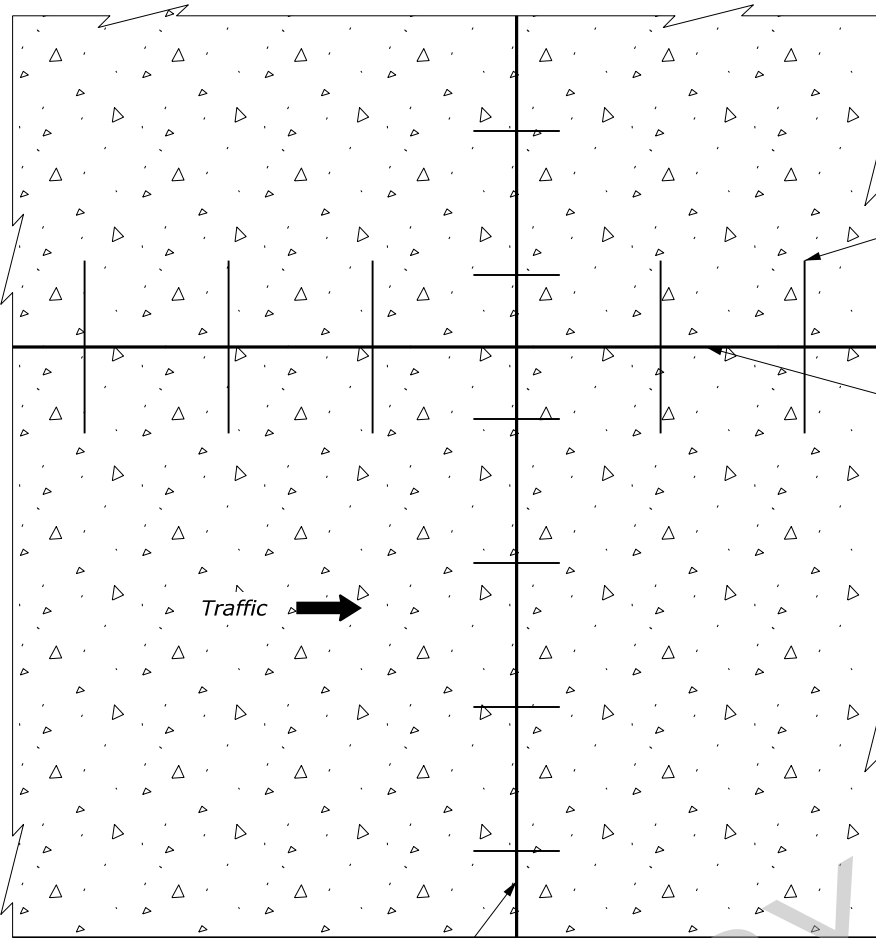


CONCRETE GUTTER DETAIL

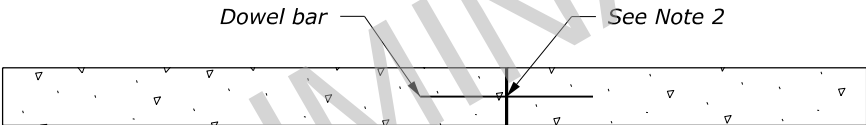
NOTE:

- 1. Provide the same type of dowel assemblies and tie bars for joints in plain portland cement concrete pavement as shown for joints in reinforced pavement.
- 2. See Standard 501-2 for joint and joint sealing details.
- 3. Lap longitudinal and transverse reinforcement not less than 15 inches.

| PAVEMENT THICKNESS (in) | TRANSVERSE JOINT SPACING (ft) |
|-------------------------|-------------------------------|
| $T < 6$                 | 10                            |
| $6 \leq T < 12$         | 15                            |

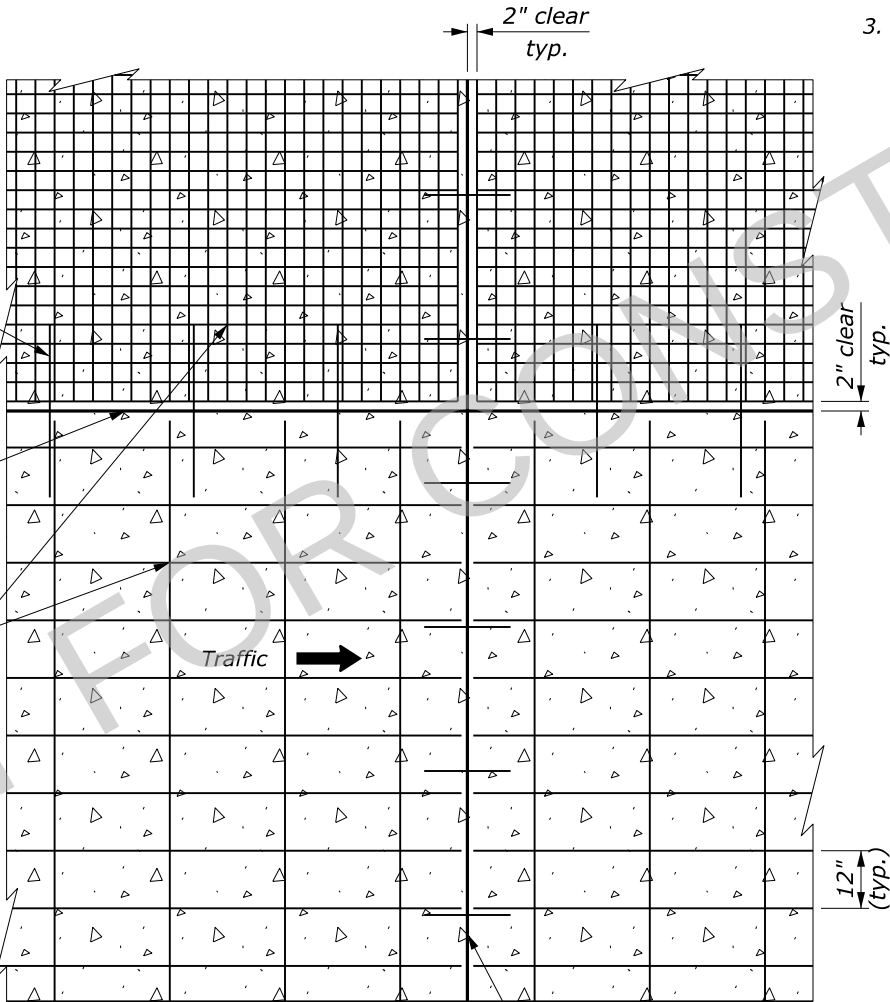


PLAN

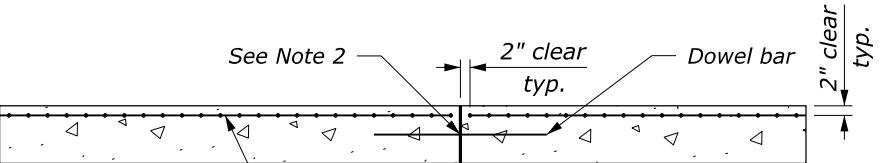


PROFILE

PLAIN MINOR CONCRETE PAVEMENT



PLAN



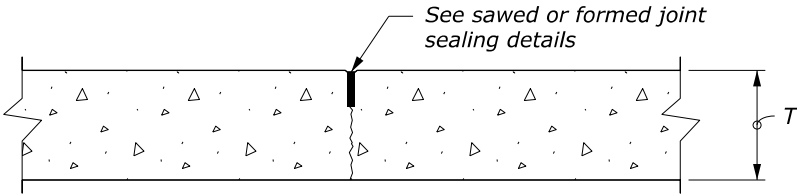
PROFILE

REINFORCED MINOR CONCRETE PAVEMENT

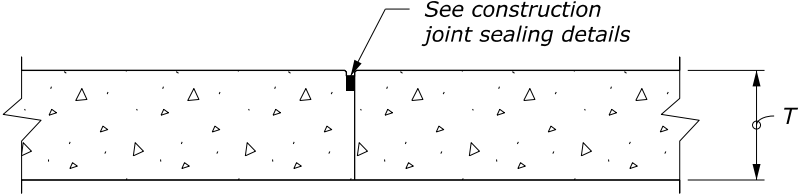
NO SCALE

NOTE:

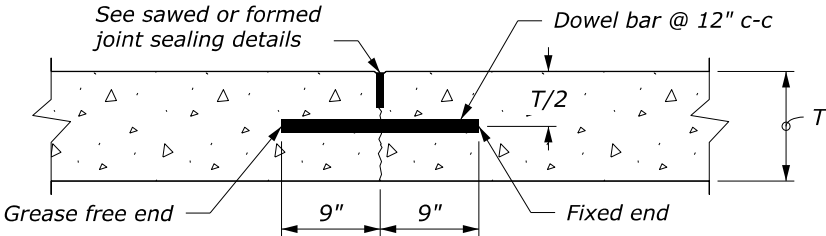
1. Use epoxy-coated material for all tie bars, dowels, and other steel used in the construction of concrete pavement.
2. Deformed reinforcing bars or hook bolts may be used for tie bars.
3. Do not place tie bars within 15 inches of transverse joints.
4. Install isolation joints when abutting a fixed structure. Use expansion joint material extending the full depth and length of the concrete surface.
5. Transverse and longitudinal construction joints are not included in the joint layout plan. Use transverse and longitudinal construction joints sparingly. Submit planned construction joint locations for approval.
6. For construction joints, if tie bars and dowels are not set into concrete during placement, drill and anchor the tie bars and dowels into the existing concrete construction with epoxy resin.
7. Maintain joint sealant shape factor of 1:1; except when silicone sealant is used maintain the width to depth shape factor of 2:1 or as recommended by sealant manufacturer.
8. See Section 712 for joint material requirements.
9. See Standards 501-1 or 502-1 for reinforcement details.



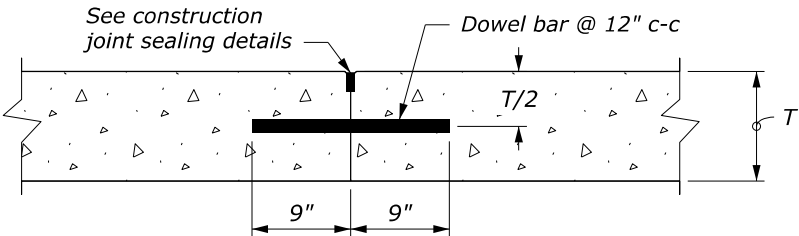
**CONTRACTION JOINT**  
UNDOWELED - TRANSVERSE and  
UNTIED - LONGITUDINAL



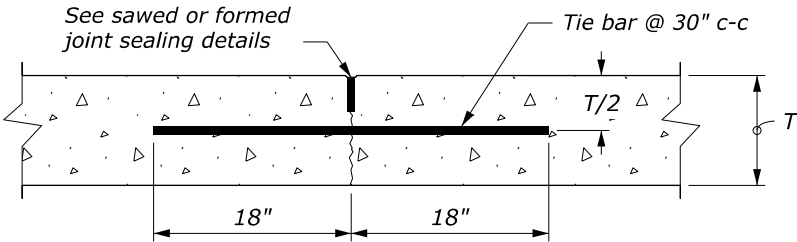
**CONSTRUCTION JOINT**  
PLAIN - TRANSVERSE or LONGITUDINAL



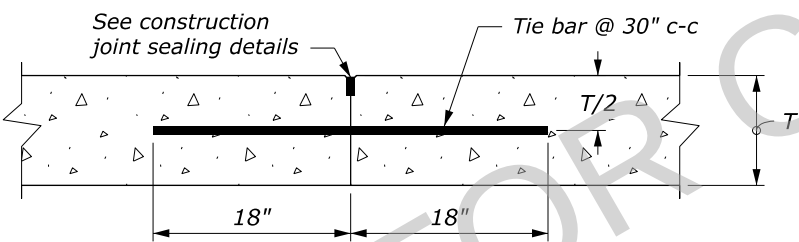
**CONTRACTION JOINT**  
DOWELED - TRANSVERSE



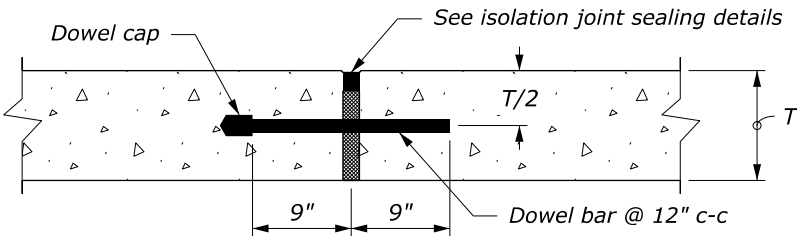
**CONSTRUCTION JOINT**  
DOWEL BUTT - TRANSVERSE



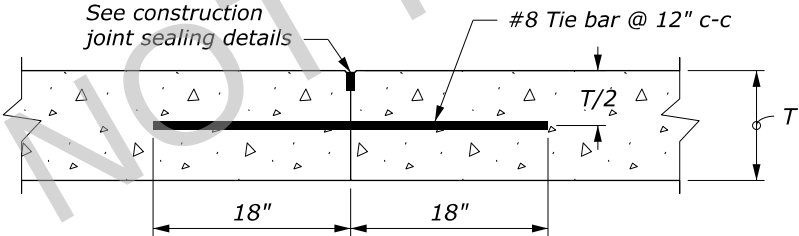
**CONTRACTION JOINT**  
TIED - LONGITUDINAL



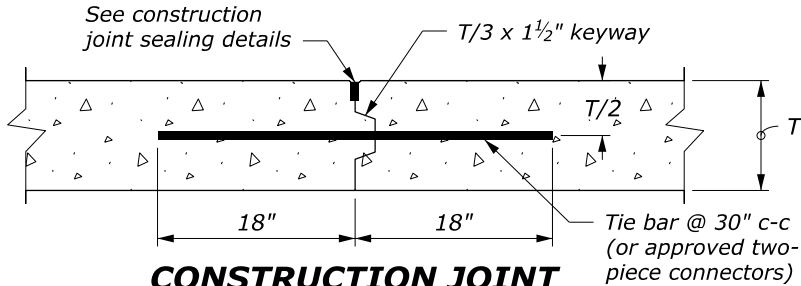
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TIED BUTT - LONGITUDINAL



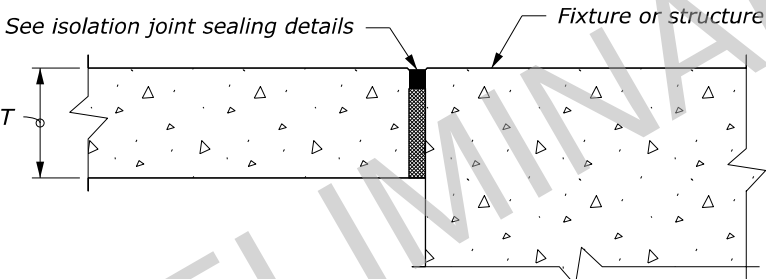
**ISOLATION/EXPANSION JOINT**  
DOWELED - TRANSVERSE



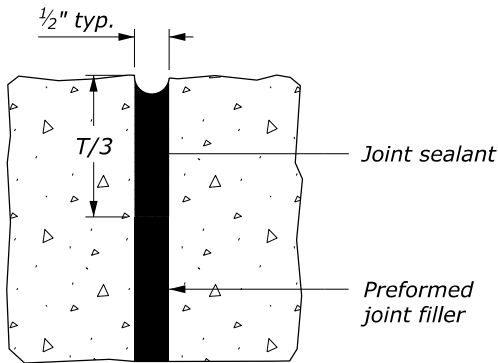
**CONSTRUCTION JOINT**  
TIED BUTT - TRANSVERSE



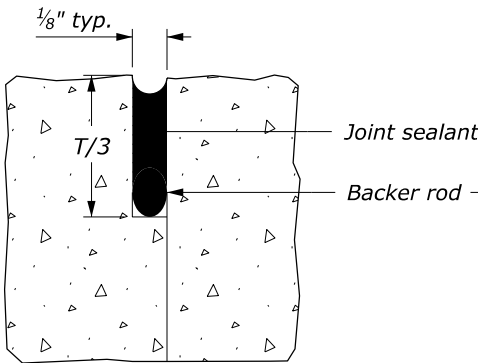
**CONSTRUCTION JOINT**  
KEYWAY - LONGITUDINAL



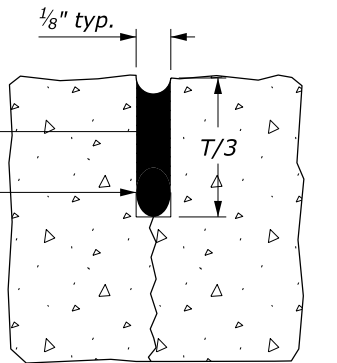
**ISOLATION JOINT**  
UNDOWELED - LONGITUDINAL



ISOLATION JOINT



CONSTRUCTION JOINT



SAWED OR FORMED JOINT

| BAR SIZES                   |         |                         |
|-----------------------------|---------|-------------------------|
| PAVEMENT THICKNESS (T) (in) | TIE BAR | DOWEL BAR DIAMETER (in) |
| $T \leq 8$                  | #5      | 1                       |
| $8 < T \leq 10$             | #5      | 1 1/4                   |
| $10 < T \leq 12$            | #6      | 1 1/2                   |

U.S. DEPARTMENT OF TRANSPORTATION  
FEDERAL HIGHWAY ADMINISTRATION  
OFFICE OF FEDERAL LANDS HIGHWAY

FLH STANDARD

**MINOR CONCRETE  
PAVEMENT JOINTS**

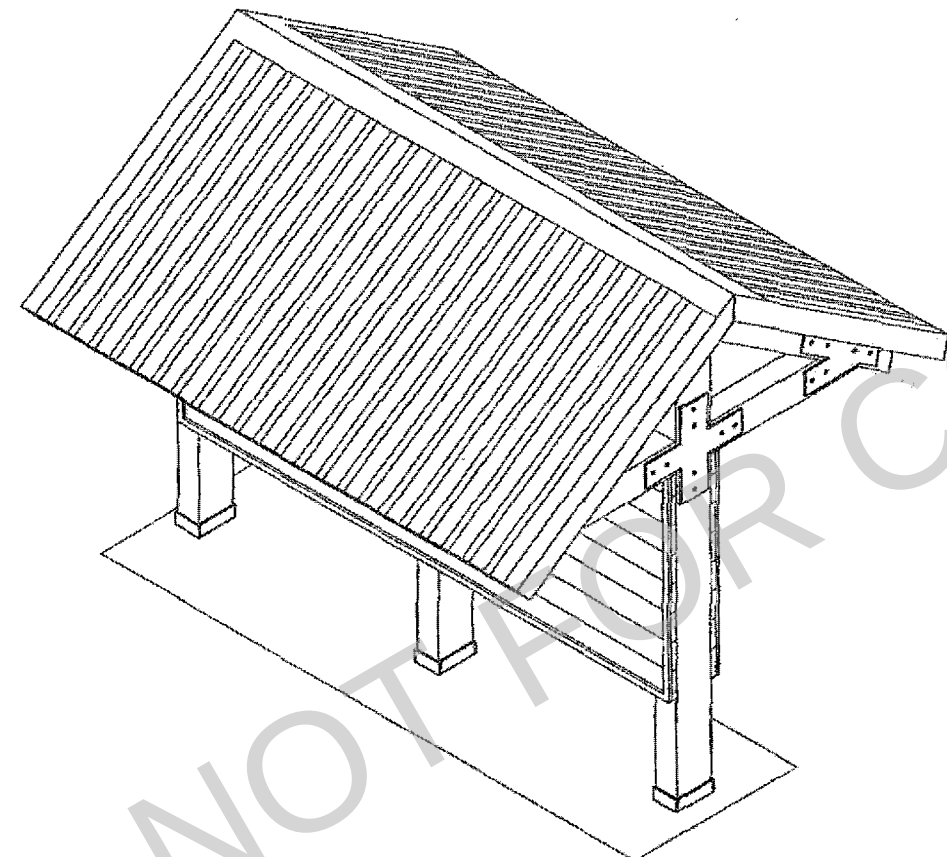
STANDARD APPROVED FOR USE --/----  
REVISED: 9/2016

STANDARD  
501-2

NO SCALE

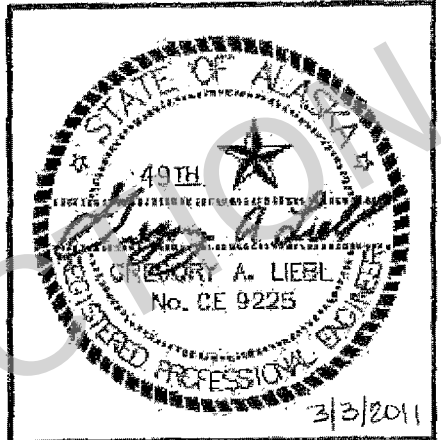
**MINOR CONCRETE PAVEMENT JOINT SEALING DETAILS**





THREE POST DOUBLE KIOSK

KIOSK #0003 & #0005  
ANCHORAGE, ALASKA



This drawing is  
intended to be  
plotted at full size  
11 x 17 paper size.

|                          |          |
|--------------------------|----------|
| Date Stamped: XXXXX/2010 |          |
| By                       | WKS      |
| Check                    | WKS      |
| Revision                 | WKS      |
| Date                     | WKS      |
| 1                        | 10/15/10 |
| 2                        | 10/19/10 |
| 3                        | 10/19/10 |
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| 99                       | 10/19/10 |
| 100                      | 10/19/10 |

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KIOSK #0003 & #0005  
THREE POST  
DOUBLE KIOSK  
Anchorage, AK

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 08/21/2009 |

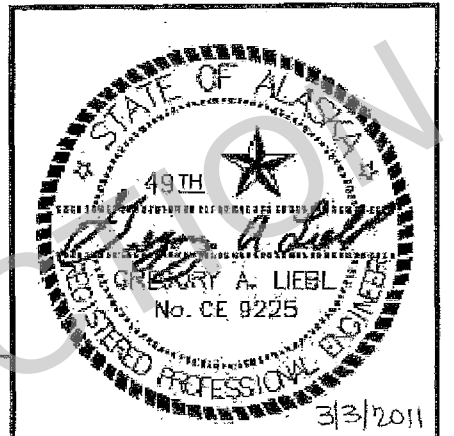
Sheet Content:  
COVER SHEET

Sheet No.:  
GO

FINAL

ROOF: SNOW LOAD 168 PSF  
 ROOF CLADDING DESIGN PRESSURE 30 PSF  
 DESIGN WIND SPEED 130 MPH (3 SEC. GUST),  
 EXPOSURE C  
 SEISMIC  
 SPS = 1.17  
 SD1 = 0.65

**NOTE:**  
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1. ALL WORK SHALL CONFORM TO STATE & LOCAL CODES AND ORDINANCES, AND SHALL COMPLY WITH THE AMERICANS WITH DISABILITIES ACT ACCESSIBILITY GUIDELINES (ADAAG), IBC GENERAL FASTENING SCHEDULE TABLE 2304.4.1 AND THE 2009 IBC STANDARDS. ALL WORK SHALL MEET OR EXCEED INDUSTRY STANDARDS FOR MATERIALS, WORKMANSHIP, ETC.
2. CONTRACTOR SHALL REVIEW THE DRAWINGS THOROUGHLY BEFORE PROCEEDING WITH ANY WORK. ANY DISCREPANCIES FOUND WITHIN THESE DOCUMENTS SHALL BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTING OFFICER. CONTRACTOR SHALL NOT PROCEED WITH ANY WORK HE KNOWS TO BE IN CONFLICT WITH OTHER WORK, OR IS NOT APPROVED BY CODE, UNTIL RESOLVED BY THE FOREST SERVICE.
3. CONSTRUCT FOOTINGS ONLY ON COMPACTED FILL OR FIRM NATIVE MINERAL SOIL. BACKFILL AROUND BUILDING FOUNDATION ACCORDING TO SPECIFICATIONS SECTION 203 - STRUCTURE EXCAVATION AND BACKFILL.
4. CAST-IN-PLACE CONCRETE: 4000 PSI MINIMUM STRENGTH AT 28 DAYS. BROOM FINISH INTERIOR SURFACES AND EXTERIOR SLABS.
5. ANCHOR AND MACHINE BOLTS SHALL BE ASTM A307. SCREWS AND MACHINE BOLT CALLOUTS ARE MINIMUM SIZE ALLOWED, ACTUAL SIZE MAY VARY. STEEL PLATES & SHAPES SHALL BE ASTM A36,  $F_y = 36 \text{ ksi}$ . CONCRETE REINFORCING STEEL (REBAR): 60 ksi. (GRADE 60). WOOD FRAMING SHALL BE ALASKA YELLOW CEDAR NO. 2 OR EQUAL, UNO.
6. THE SPECIFICATIONS ARE THE AUTHORITY FOR DEFINING WHO IS RESPONSIBLE FOR WORK PERFORMANCE AND SUPPLYING ALL ITEMS ASSOCIATED WITH THE INSTALLATION OF THIS BUILDING.
7. ASSUMED ALLOWABLE BEARING CAPACITY IS 1500 PSF.



2

REVISION NUMBER SEE  
TITLEBLOCK FOR DATE  
& REVISED AREA



1  
A7.3



1  
AB3.00

DETAIL REFERENCE



1  
A4.1

BUILDING SECTION



1  
AB3.00

1  
A7.3

ENLARGED DETAIL  
REFERENCE

|      |                             |      |                         |      |                             |
|------|-----------------------------|------|-------------------------|------|-----------------------------|
| AB   | ANCHOR BOLT                 | MIR  | MIRROR                  | TSCD | TOILET SEAT COVER DISPENSER |
| AFF  | ABOVE FINISHED FLOOR        | MO   | MASONRY OPENING         | TYP  | TYPICAL                     |
| ATS  | AUTOMATIC TRANSFER SWITCH   | MR   | METAL ROOFING           | UNO  | UNLESS NOTED OTHERWISE      |
| BOT  | BOTTOM                      | ND   | NAPKIN DISPOSAL         | VB   | VAPOR BARRIER               |
| BP   | BREAKER PANEL               | NTS  | NOT TO SCALE            | VTR  | VENT THROUGH ROOF           |
| CJ   | CONTROL JOINT               | OC   | ON CENTER               | WH   | WATER HEATER                |
| CO   | CLEAN OUT                   | OCEW | ON CENTER EACH WAY      | WWM  | WOVEN WIRE MESH             |
| CMU  | CONCRETE MASONRY UNIT       | OSB  | ORIENTED STRAND BOARD   | W/   | WITH                        |
| DD   | DIAPER DECK                 | PE   | PHOTO EYE               |      |                             |
| DIA  | DIAMETER                    | PCC  | PORTLAND CEMENT COMPANY |      |                             |
| DISC | DISCONNECT                  | PL   | PLATE                   |      |                             |
| DGA  | DENSE GRADED AGGREGATE      | PSF  | POUNDS PER SQUARE FOOT  |      |                             |
| EM   | ELECTRIC METER              | PSI  | POUNDS PER SQUARE INCH  |      |                             |
| EW   | EACH WAY                    | PT   | PRESSURE TREATED        |      |                             |
| FD   | FLOOR DRAIN                 | PV   | PHOTO VOLTAGE           |      |                             |
| FF   | FINISHED FLOOR              | REQD | REQUIRED                |      |                             |
| FG   | FINISHED GRADE              | R4S  | ROUGH FOUR SIDES        |      |                             |
| FRP  | FIBERGLASS REINFORCED PANEL | SCH  | SCHEDULE                |      |                             |
| GB   | GRAB BAR                    | SD   | SOAP DISPENSER          |      |                             |
| GLB  | GLUE LAMINATED BEAM         | SJ   | SAW JOINT               |      |                             |
| HB   | HOSE BIB/WALL HYDRANT       | SM   | SHEET METAL             |      |                             |
| HD   | HAND DRYER                  | S4S  | SURFACED FOUR SIDES     |      |                             |
| HM   | HOLLOW METAL (DOOR)         | TBD  | TO BE DETERMINED        |      |                             |
| KSI  | KIPS PER SQUARE INCH        | TD   | PAPER TOWEL DISPENSER   |      |                             |
| MBP  | MAIN BREAKER PANEL          | T&B  | TOP & BOTTOM            |      |                             |
| MD   | MAIN DISCONNECT             | TLT  | TOILET                  |      |                             |
| MIN  | MINIMUM                     | TP   | TOILET PAPER DISPENSER  |      |                             |

| SHEET | CONTENTS                      |
|-------|-------------------------------|
| G0    | COVER SHEET                   |
| G1    | GENERAL NOTES, SHEET SCHEDULE |
| A2.1  | FLOOR PLAN                    |
| A3.1  | EXTERIOR ELEVATIONS           |
| A5.1  | FOUNDATION PLAN & DETAILS     |
| A7.1  | TRUSS DETAILS                 |
| A7.2  | TRUSS DETAILS                 |
| A7.3  | TRUSS PLATE DETAILS           |
| A7.4  | COLUMN ANCHOR DETAILS         |
| A8.3  | ROOF FRAMING DETAILS          |
| A8.4  | ROOFING DETAILS               |

| Date Stamped: |          | XXXX/XX/2010 |                                   |
|---------------|----------|--------------|-----------------------------------|
| By            | Revision | Date         | Description                       |
| SK            | 1        | 02/02/10     | Revisions from Architect/Engineer |
| SK            | 2        | 04/19/10     | Revisions from Architect/Engineer |
| SK            | 3        | 04/19/10     | Revisions from Customer           |
| SK            | 4        | 06/02/10     | Revisions from Architect/Engineer |
| CR            | 5        | 02/21/11     | Revisions from Architect/Engineer |

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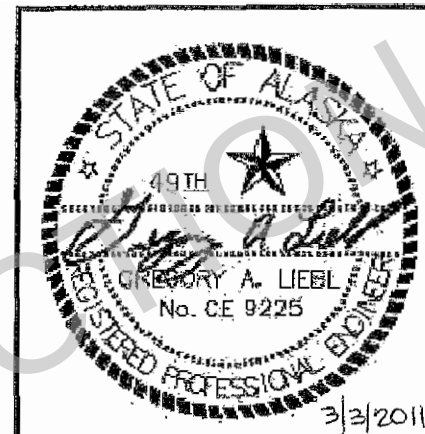
351 W. Parks Highway, Suite 200 Wasilla,  
Alaska 99654 (907) 376-7615

KIOSK #0003 &  
#0005  
THREE POST  
DOUBLE KIOSK

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
GENERAL NOTES  
SHEET SCHEDULE

Sheet No.:  
G1  
USKH W.O. 1258600



This drawing is intended to be plotted at full size 11 x 17 paper size.

Date Stamped: XX/XX/2010

| By | Revision | Date      |
|----|----------|-----------|
| SK | 1        | 3/3/2010  |
| SK | 2        | 3/4/2010  |
| SK | 3        | 3/12/2010 |
| SK | 4        | 3/20/2010 |
| SK | 5        | 3/22/2010 |
| CR |          |           |

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THREE POST  
DOUBLE KIOSK**

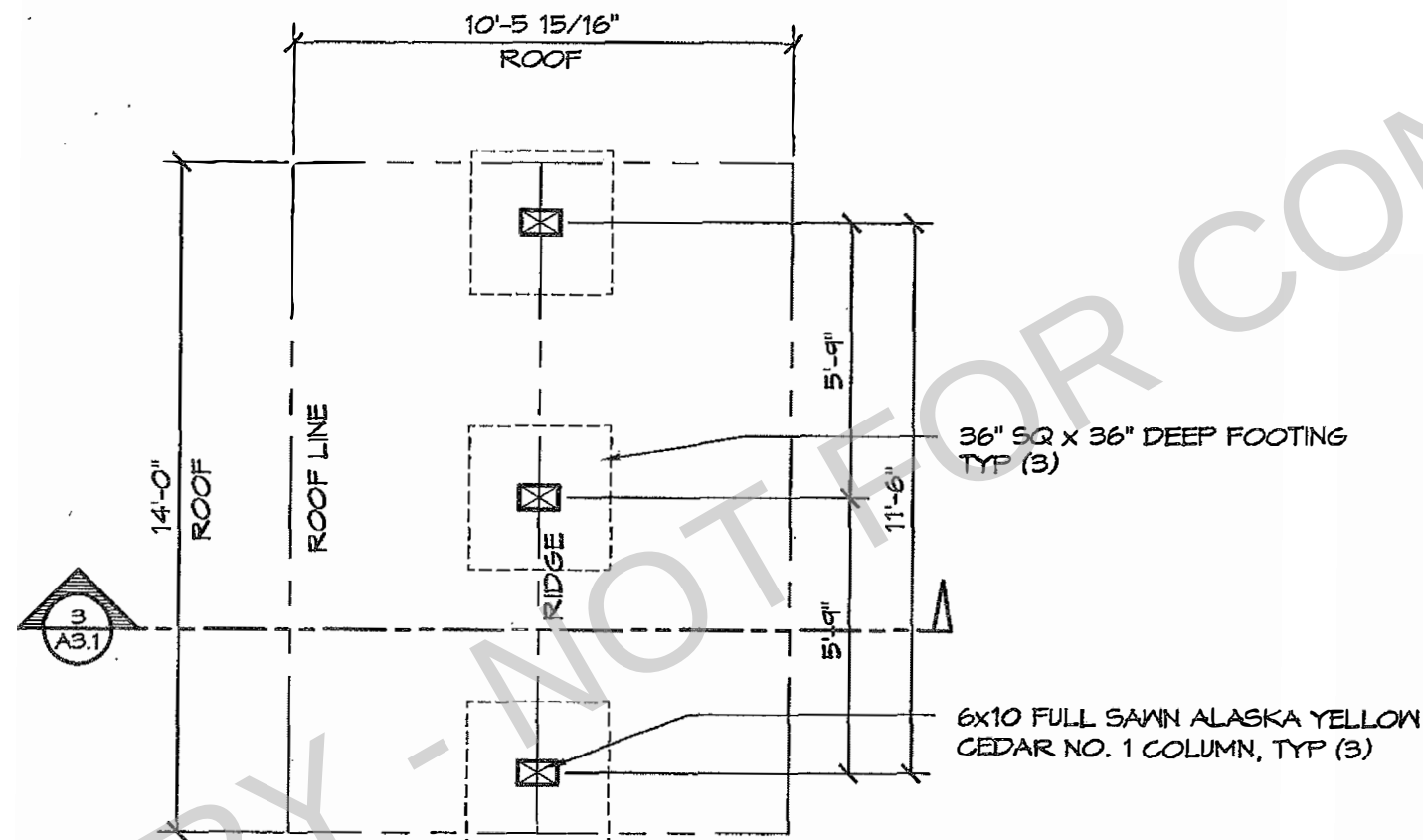
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| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
**FLOOR PLAN**

Sheet No.:

**A2.1**

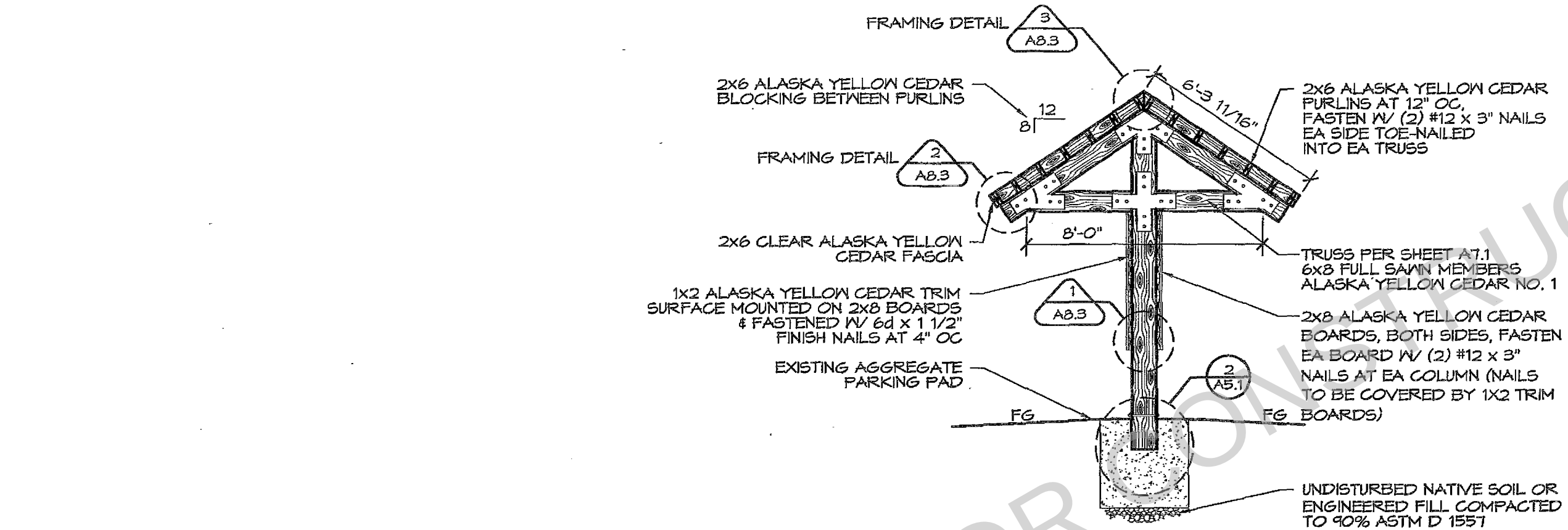
USKH W.O. 1258600



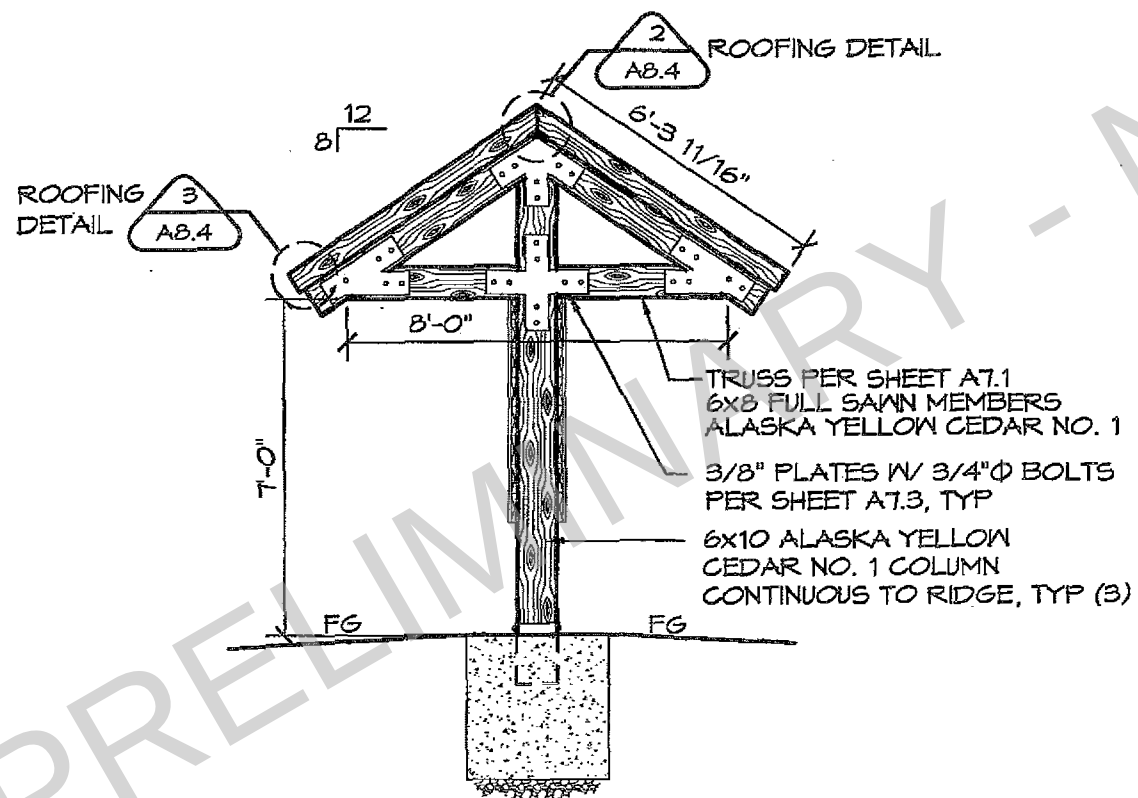
PLAN NORTH



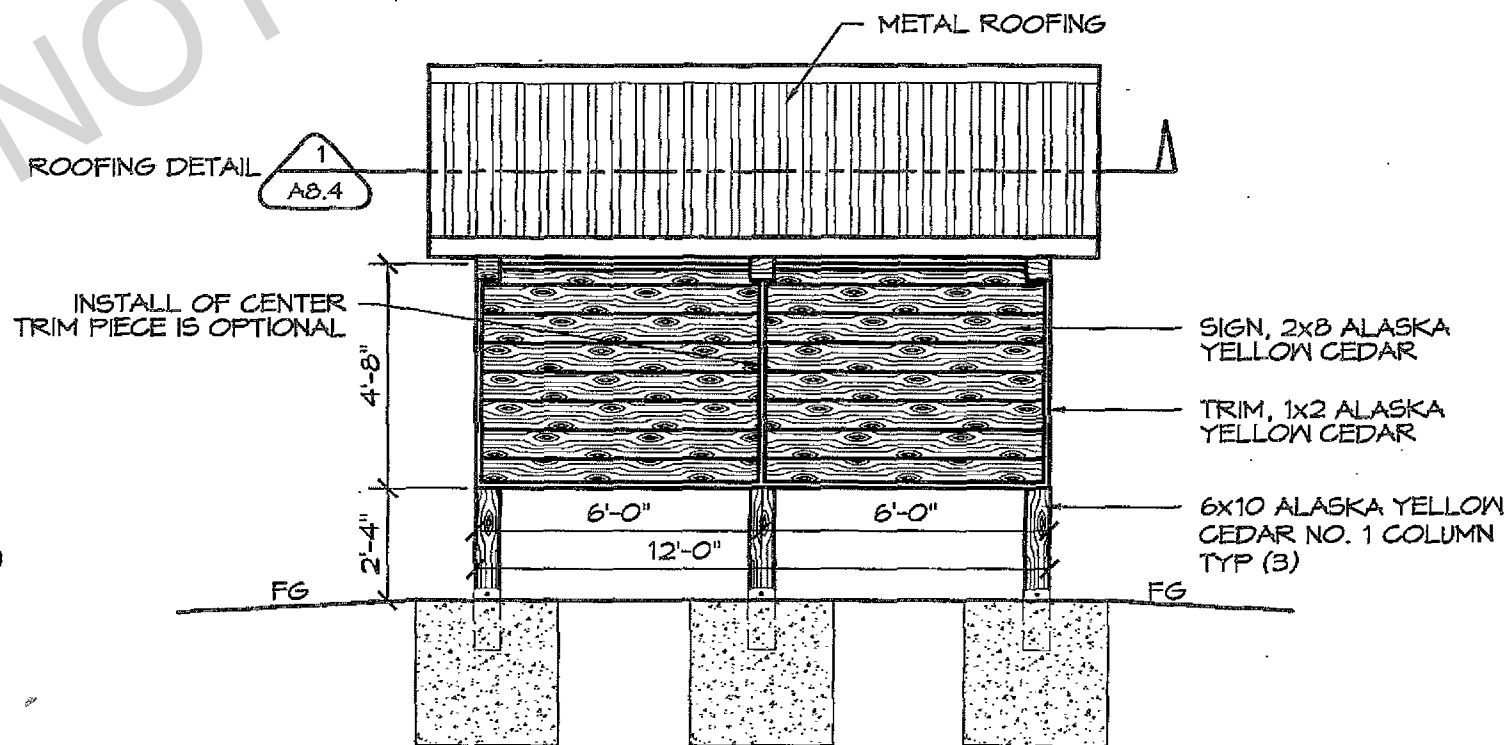
**1 FLOOR PLAN**  
SCALE: 1/4" = 1'-0"



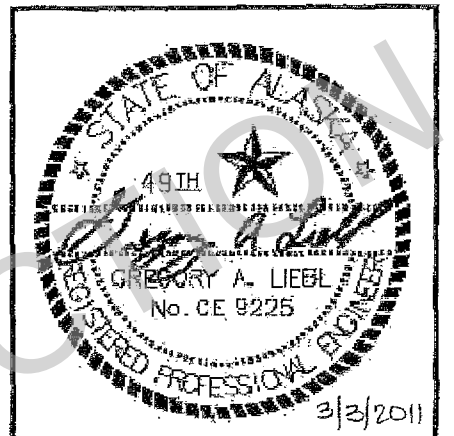
**3 SOUTH SECTION**  
SCALE: 1/4" = 1'-0"



**2 SOUTH ELEVATION**  
SCALE: 1/4" = 1'-0"



**1 EAST ELEVATION**  
SCALE: 1/4" = 1'-0"



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| Revision | By | Date     |
|----------|----|----------|
| 1        | SK | 03/03/10 |
| 2        | SK | 04/18/10 |
| 3        | SK | 07/13/10 |
| 4        | SK | 09/22/10 |
| 5        | SK | 02/21/11 |

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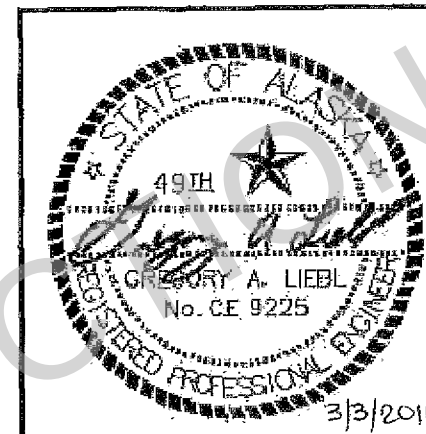
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**KIOSK #0003 & #0005**  
**THREE POST**  
**DOUBLE KIOSK**  
**Anchorage, AK**

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| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
**EXTERIOR**  
**ELEVATIONS**

Sheet No:  
**A3.1**



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Date Stamped: XXXX/XX/2010

| By       | SK                                | SK                      | SK                                | SK                                | SK                                | SK                                |
|----------|-----------------------------------|-------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Revision | Revisions from Architect/Engineer | Revisions from Customer | Revisions from Architect/Engineer | Revisions from Architect/Engineer | Revisions from Architect/Engineer | Revisions from Architect/Engineer |
| Date     | 03/03/10                          | 04/15/10                | 05/18/10                          | 06/02/10                          | 02/21/11                          |                                   |

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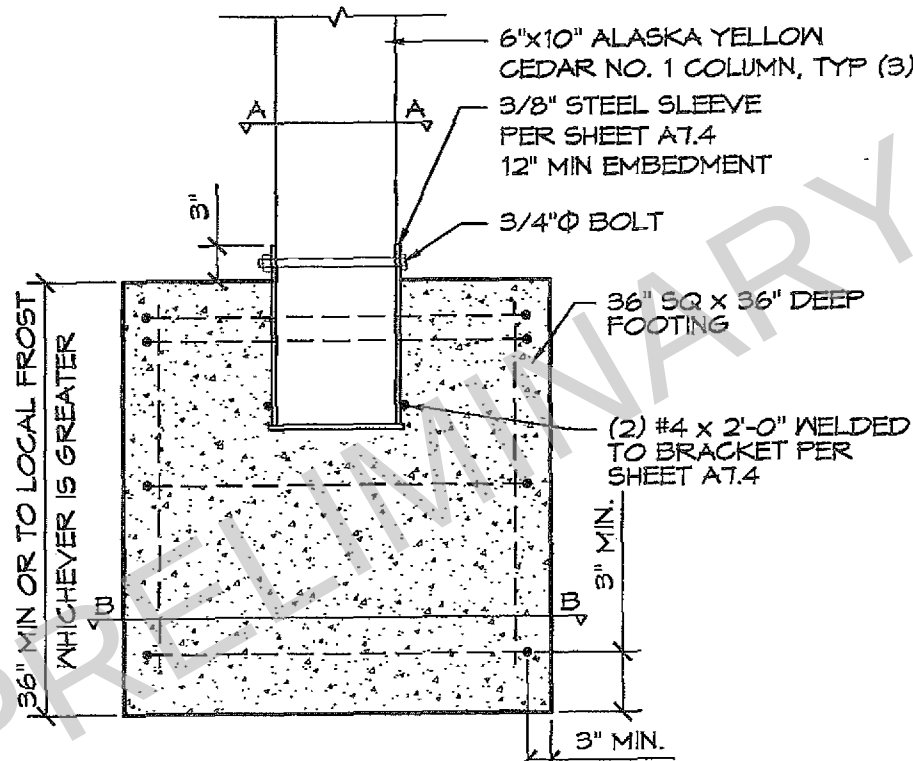
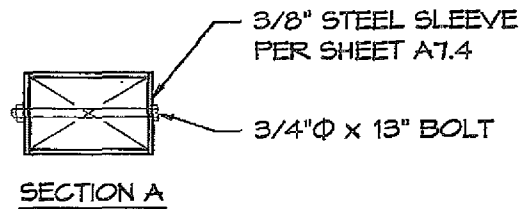
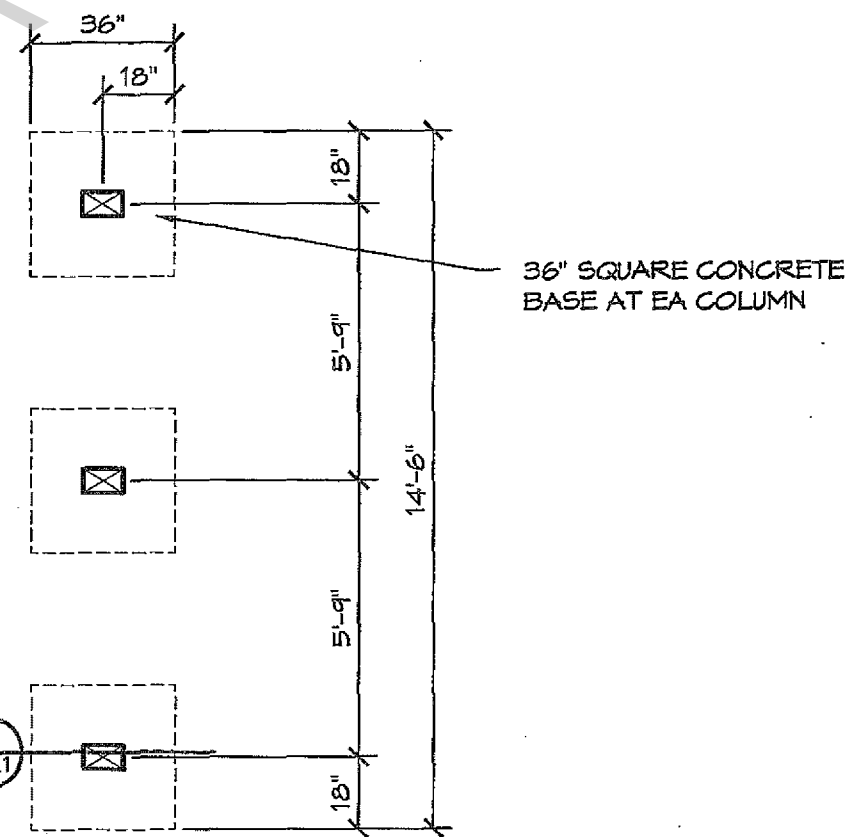
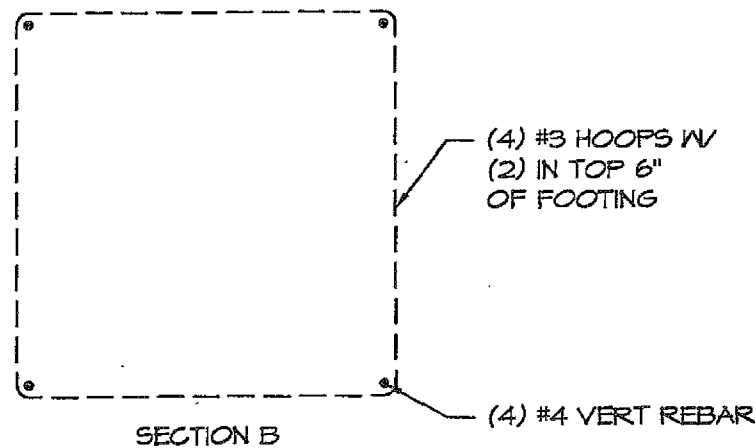
**KIOSK #0003 & #0005**  
**THREE POST DOUBLE KIOSK**  
**Anchorage, AK**

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
**FOUNDATION**  
**PLAN AND**  
**DETAILS**

Sheet No.:  
**A5.1**

USKH W.O. 1258600

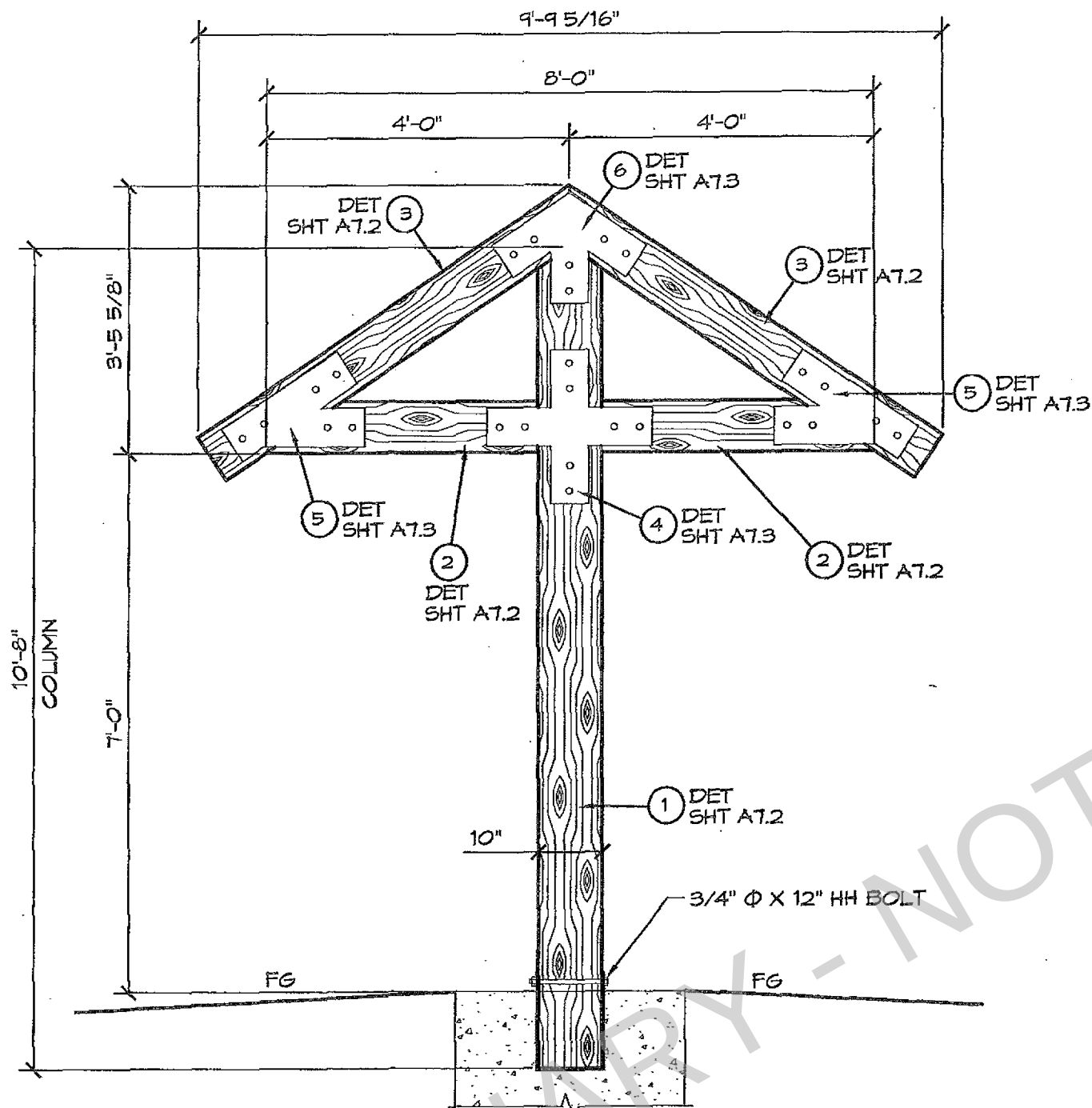


**2** ANCHOR DETAIL  
SCALE: 3/4" = 1'-0"

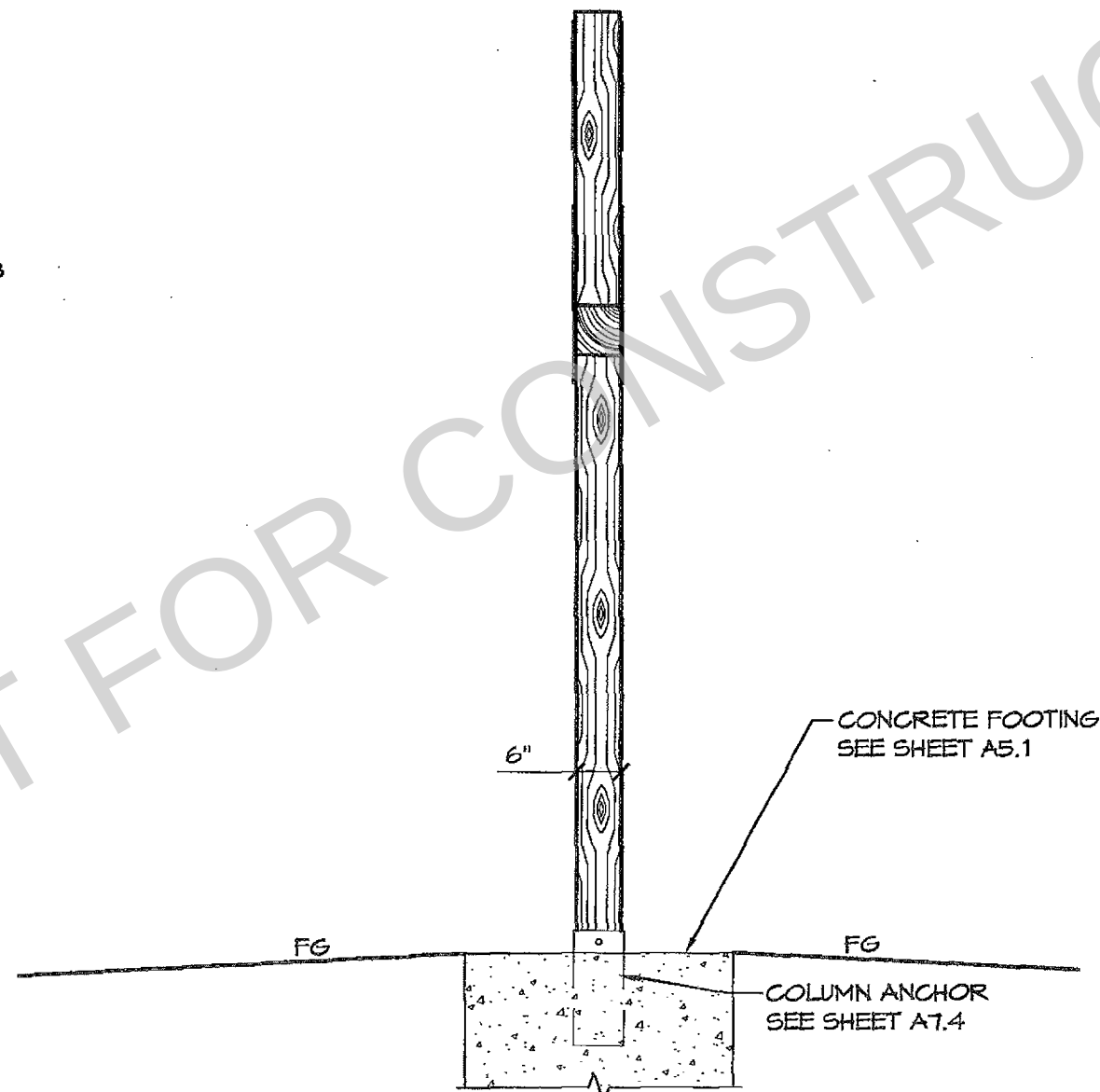
**1** FOUNDATION PLAN  
SCALE: 1/4" = 1'-0"







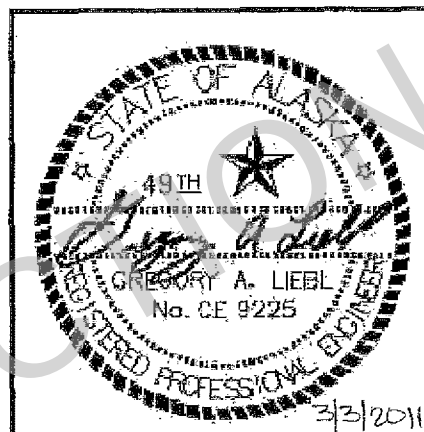
FRONT ELEVATION



SIDE ELEVATION

# 1 WOOD ROOF EXTENSION DETAIL

SCALE: 1/2" = 1'-0"  
QTY: 3



This drawing is intended to be plotted at full size 11 x 17 paper size.

Date Stamped: XX/XX/2010

| By | SK | SK | SK | SK | OR |
|----|----|----|----|----|----|
| 1  | SK | SK | SK | SK | OR |
| 2  | SK | SK | SK | SK | OR |
| 3  | SK | SK | SK | SK | OR |
| 4  | SK | SK | SK | SK | OR |
| 5  | SK | SK | SK | SK | OR |

**ROMTEC**

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**USKH**

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3031 Chilton Drive, Suite 200 Juneau,  
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351 W. Parks Highway, Suite 200 Wasilla,  
Alaska 99654 (907) 378-7815

**KIOSK #0003 & #0005  
THREE POST  
DOUBLE KIOSK  
Anchorage, AK**

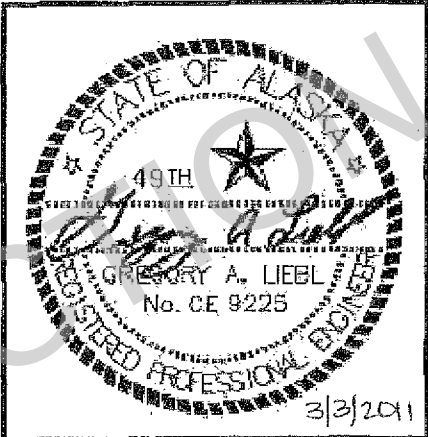
|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2008 |

Sheet Content:  
**TRUSS  
DETAILS**

Sheet No.:  
**A7.1**

USKH W.O. 1258600

| BILL OF MATERIAL |       |          |                                                              |
|------------------|-------|----------|--------------------------------------------------------------|
| ITEM             | REQ'D | PART NO. | DESCRIPTION                                                  |
| 1                | 1     |          | 6" x 10" x 10'-8" ALASKA YELLOW CEDAR #1 FULL SAWN LUMBER    |
| 2                | 2     |          | 6" x 8" x 3'-7" ALASKA YELLOW CEDAR #1 FULL SAWN LUMBER      |
| 3                | 2     |          | 6" x 8" x 5'-10 1/2" ALASKA YELLOW CEDAR #1 FULL SAWN LUMBER |



This drawing is intended to be plotted at full size 11 x 17 paper size.

| Data Stamp: |                                   | XXXX/2010 |  |
|-------------|-----------------------------------|-----------|--|
| By          | Revision                          | Date      |  |
| SK          | Revisions from Architect/Engineer | 1/2/2010  |  |
| SK          | Revisions from Client             | 2/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 3/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 4/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 5/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 6/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 7/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 8/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 9/1/2010  |  |
| SK          | Revisions from Architect/Engineer | 10/1/2010 |  |
| SK          | Revisions from Architect/Engineer | 11/1/2010 |  |
| SK          | Revisions from Architect/Engineer | 12/1/2010 |  |

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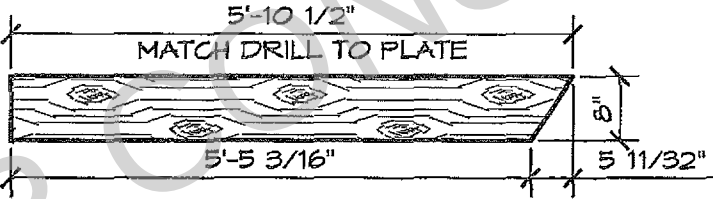
**USKH**  
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Fairbanks, Alaska 99701  
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KIOSK #0003 &  
#0005  
THREE POST  
DOUBLE KIOSK  
Anchorage, AK

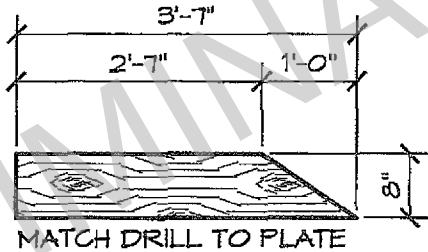
|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
TRUSS  
DETAILS

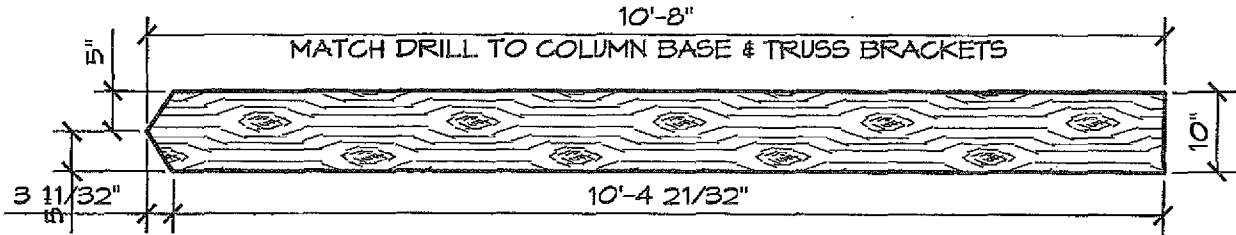
Sheet No.:  
**A7.2**



**3 6X8 CHORD DETAIL**  
SCALE: 1/2" = 1'-0"



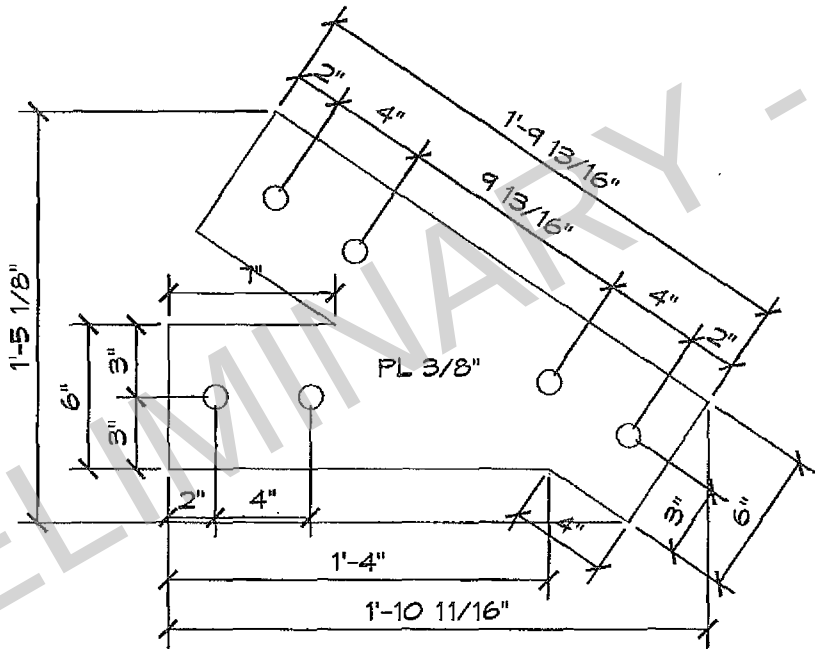
**2 6X8 CHORD DETAIL**  
SCALE: 1/2" = 1'-0"



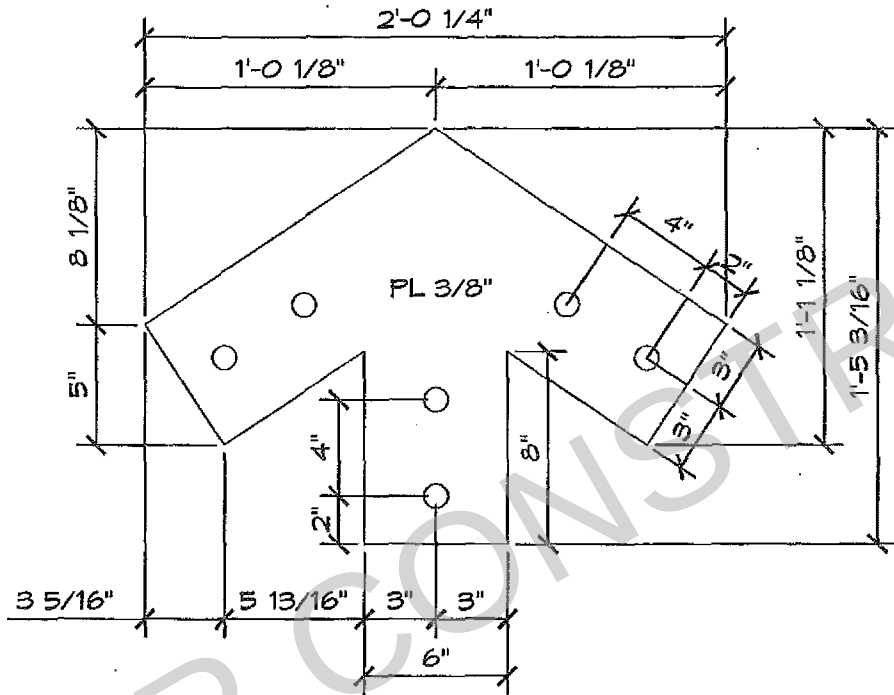
**1 6X10 COLUMN DETAIL**  
SCALE: 1/2" = 1'-0"

| BILL OF MATERIAL |       |          |                                    |
|------------------|-------|----------|------------------------------------|
| ITEM             | REQ'D | PART NO. | DESCRIPTION                        |
| 4                | 2     | BURN     | PL 3/8" x 2'-0" x 2'-2"            |
| 5                | 4     | BURN     | PL 3/8" x 1'-5 1/8" x 1'-10 11/16" |
| 6                | 2     | BURN     | PL 3/8" x 1'-5 3/16" x 2'-0 1/4"   |

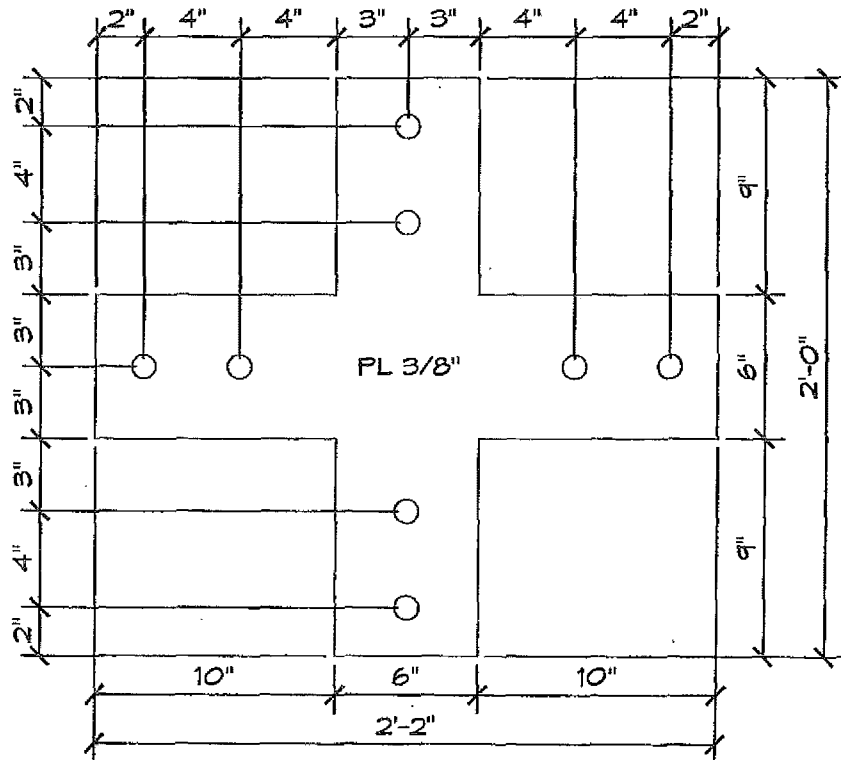
NOTE:  
POWDER COAT ALL EXPOSED STEEL BLACK



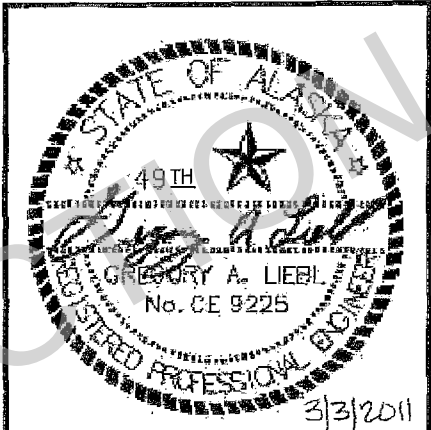
5 BRACKET DETAIL  
SCALE: 1 1/2" = 1'-0"



6 BRACKET DETAIL  
SCALE: 1 1/2" = 1'-0"



4 BRACKET DETAIL  
SCALE: 1 1/2" = 1'-0"



This drawing is intended to be plotted at full size 11 x 17 paper size.

| Date       | By | Revision                          |
|------------|----|-----------------------------------|
| 1 03/03/11 | SK | Revisions from Architect/Engineer |
| 2 04/19/10 | SK | Revisions from Customer           |
| 3 05/19/10 | SK | Revisions from Architect/Engineer |
| 4 06/02/10 | SK | Revisions from Architect/Engineer |
| 5 02/21/11 | CR | Revisions from Architect/Engineer |

**ROMTEC**  
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Alaska 99654 (907) 376-7815

KIOSK #0003 &  
#0005  
THREE POST  
DOUBLE KIOSK  
Anchorage, AK

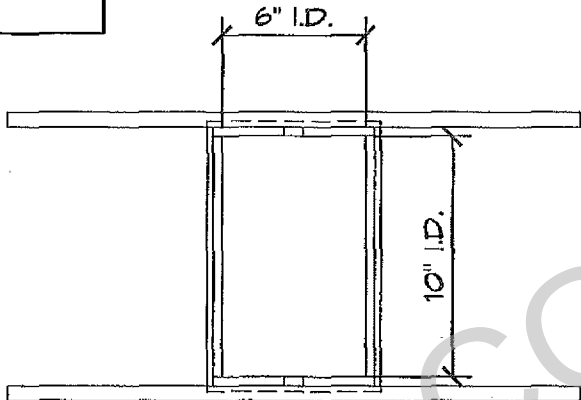
|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 08/21/2009 |

Sheet Contents:  
TRUSS PLATE  
DETAILS

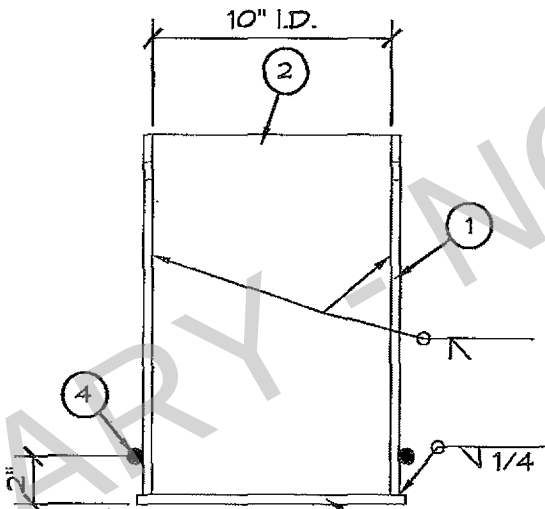
Sheet No.:  
A7.3

| BILL OF MATERIAL |       |          |                            |
|------------------|-------|----------|----------------------------|
| ITEM             | REQ'D | PART NO. | DESCRIPTION                |
| 1                | 2     | BURN     | FB 3/8" x 6 3/4" x 15"     |
| 2                | 2     | BURN     | FB 3/8" x 10" x 15"        |
| 3                | 1     | BURN     | FB 3/8" x 7 1/4" x 11 1/4" |
| 4                | 2     |          | #4 x 24" REBAR             |

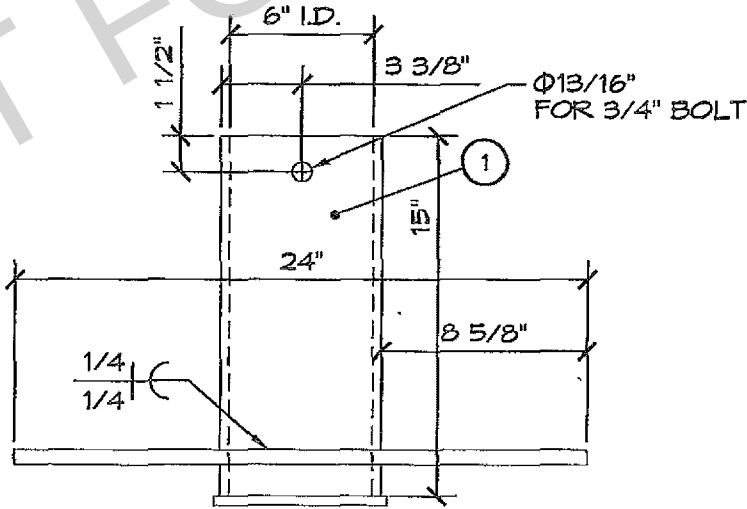
NOTE:  
POWDER COAT ALL EXPOSED STEEL BLACK



TOP



FRONT



SIDE

6"x 10" COLUMN ANCHOR

SCALE: 1 1/2" = 1'-0"

NOTE:  
MIG WELD W/ 0.035 WIRE ELCTRODE

This drawing is intended to be plotted at full size 11 x 17 paper size.

| Date     | By | Revision |
|----------|----|----------|
| 03/03/10 | SK | 1        |
| 04/18/10 | SK | 2        |
| 05/18/10 | SK | 3        |
| 06/02/10 | SK | 4        |
| 02/21/11 | SK | 5        |

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3031 Clinton Drive, Suite 200 Juneau, Alaska 99801 (907) 750-2301  
351 W. Parks Highway, Suite 200 Wasilla, Alaska 99554 (907) 376-7815

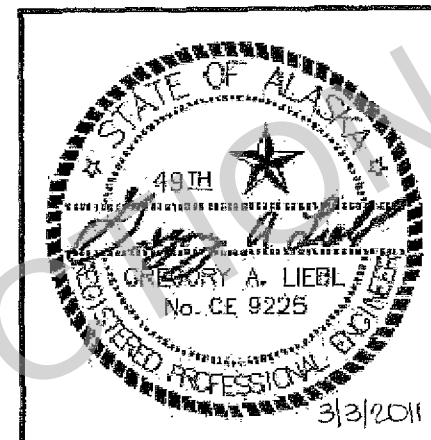
**KIOSK #0003 & #0005  
THREE POST  
DOUBLE KIOSK  
Anchorage, AK**

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
COLUMN ANCHOR  
DETAILS

Sheet No.:  
**A7.4**

USKH W.O. 1258600



|                                                                         |            |
|-------------------------------------------------------------------------|------------|
| This drawing is intended to be plotted at full size 11 x 17 paper size. |            |
| Date Stamped:                                                           | XX/XX/2010 |
| By                                                                      | GR         |
| Revision                                                                |            |
| Date                                                                    |            |

**ROMTEC**  
16240 NORTH BANK ROAD  
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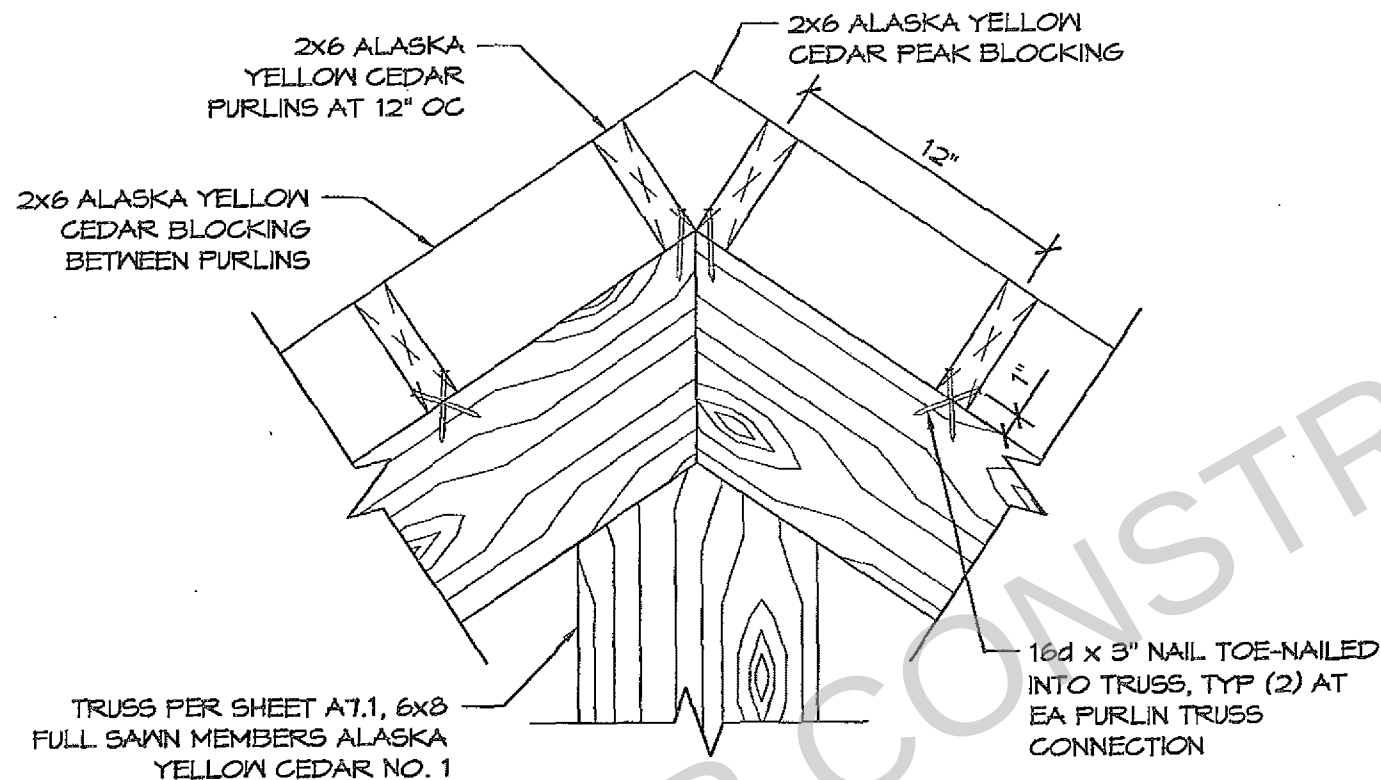
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Alaska 99801 (907) 790-2801  
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Alaska 99554 (907) 378-7615

**KIOSK #0003 & #0005**  
**THREE POST DOUBLE KIOSK**  
**Anchorage, AK**

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 09/21/2009 |

Sheet Contents:  
**ROOF FRAMING DETAILS**

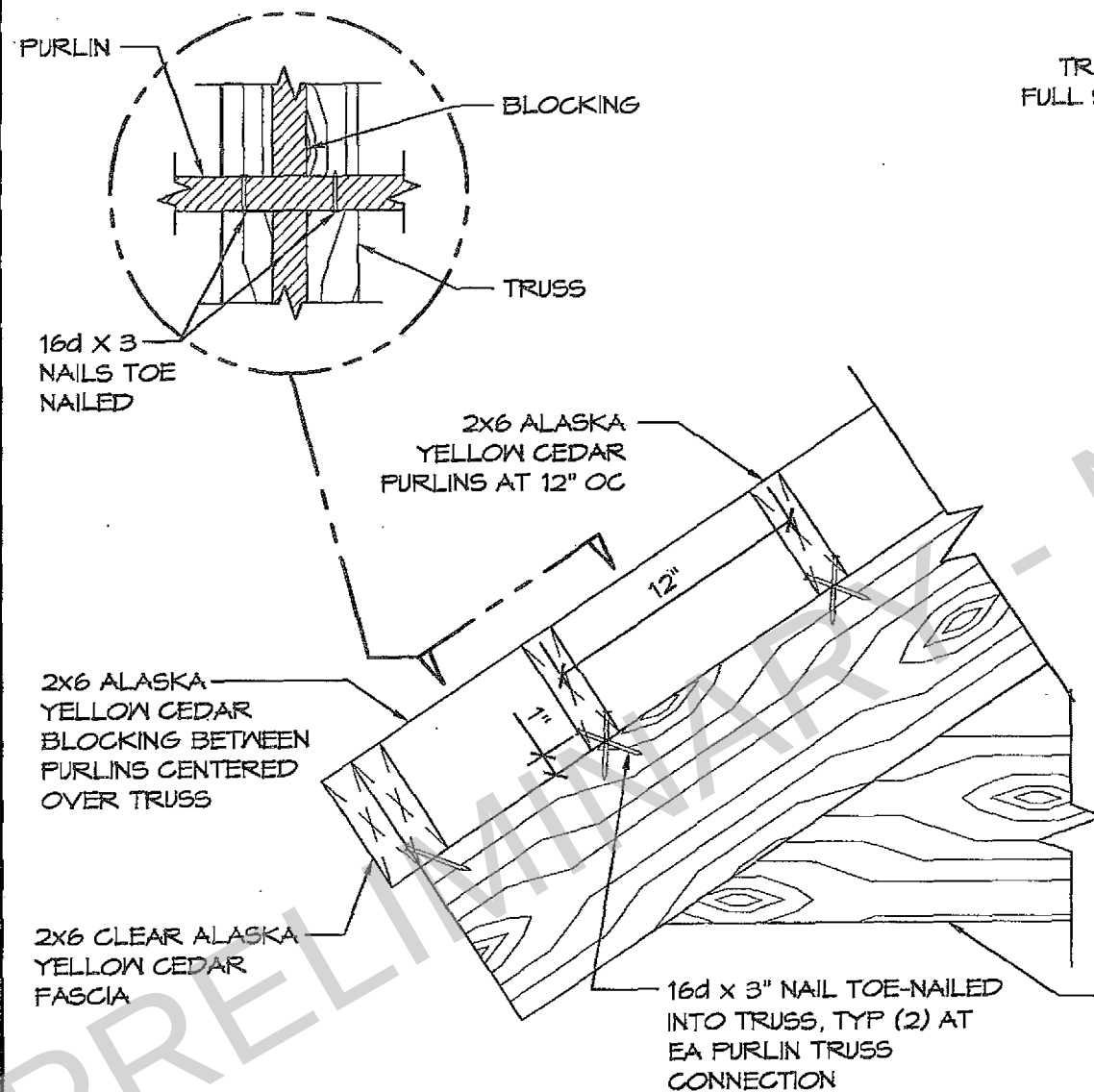
Sheet No.:  
**A8.3**  
USKH W.O. 1258600



NOTE: BRACKETS NOT SHOWN FOR CLARITY

### 3 RIDGE FRAMING DETAIL

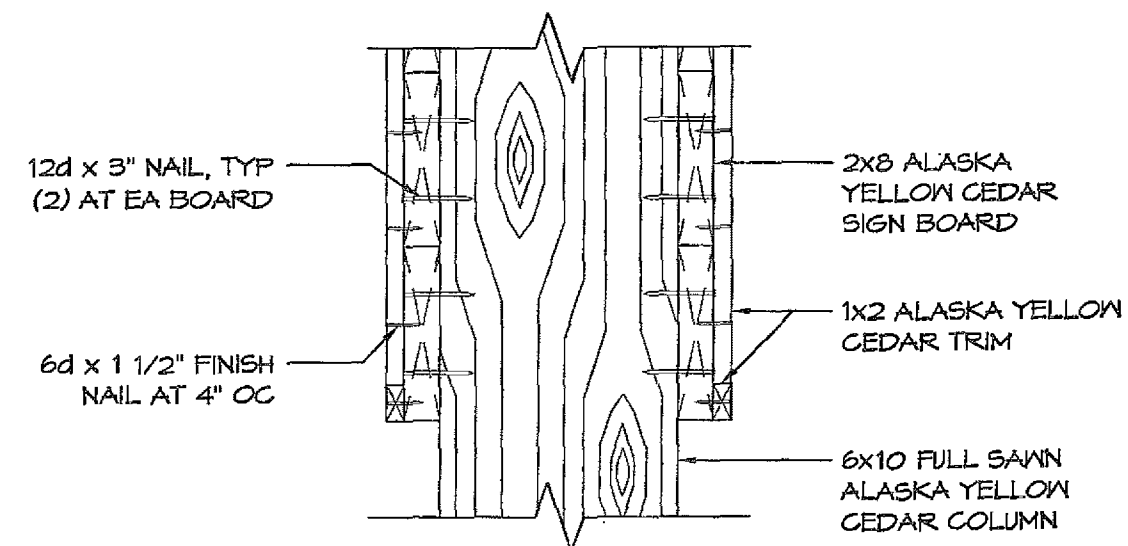
SCALE: 1 1/2" = 1'-0"



NOTE: BRACKETS NOT SHOWN FOR CLARITY

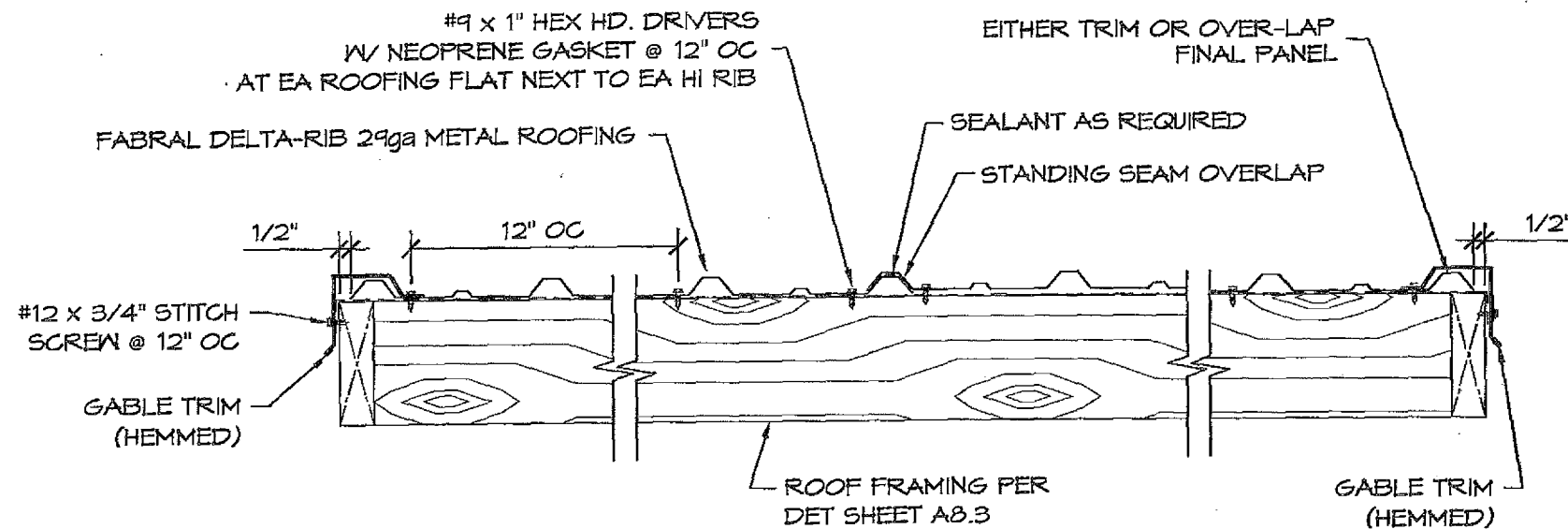
### 2 EAVE FRAMING DETAIL

SCALE: 1 1/2" = 1'-0"

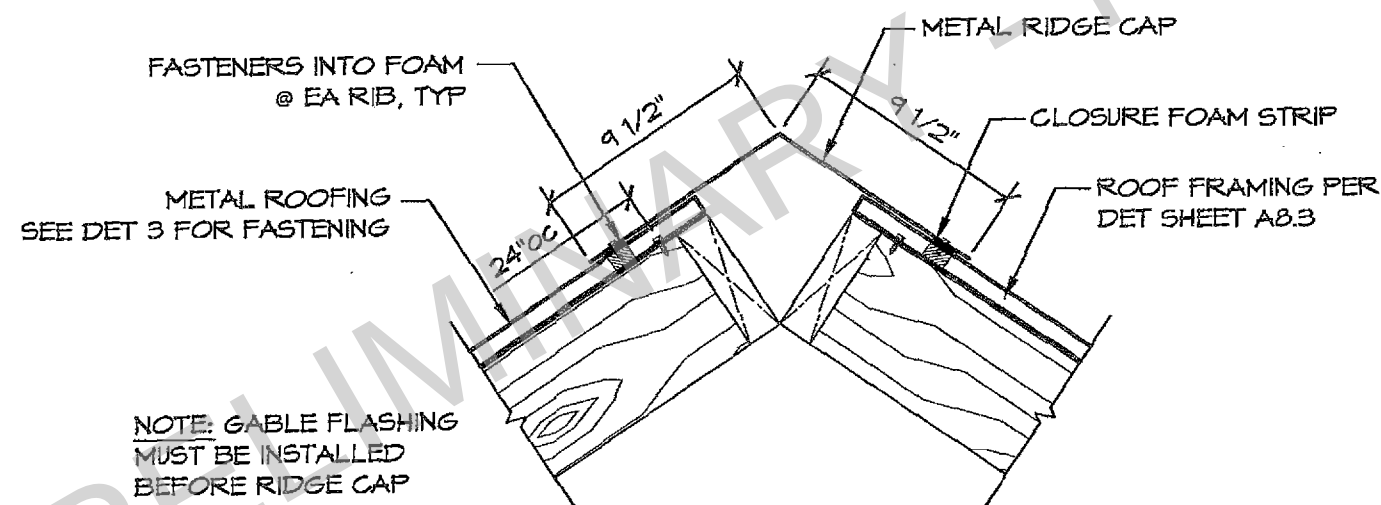


### 1 SIGN DETAIL

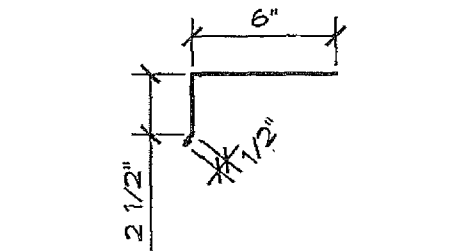
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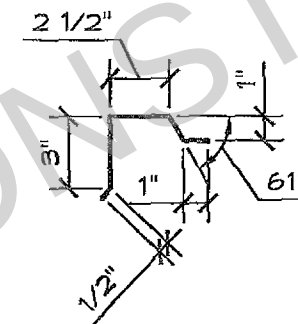
**3 GABLE ROOFING DETAIL**  
SCALE: 1 1/2" = 1'-0"



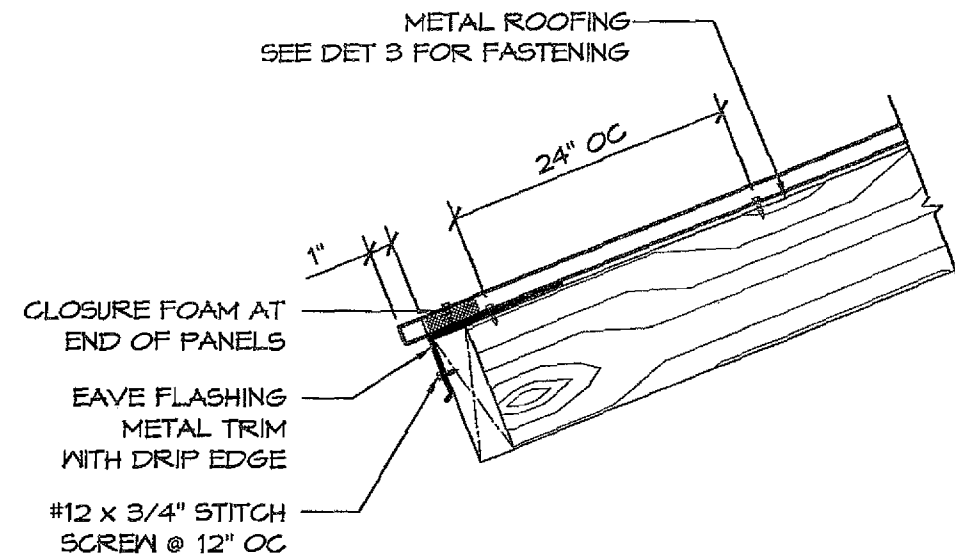
**2 EXTERIOR RIDGE ROOFING DETAIL**  
SCALE: 1 1/2" = 1'-0"



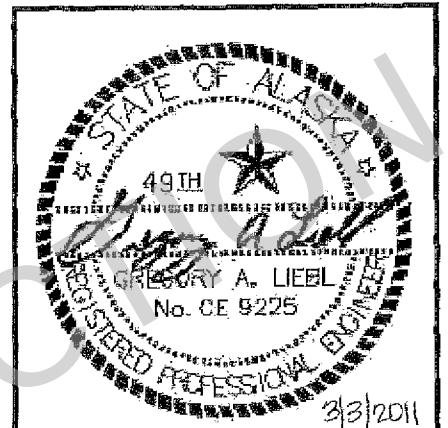
**EAVE FLASHING**



**GABLE FLASHING**



**1 EAVE ROOFING DETAIL**  
SCALE: 1 1/2" = 1'-0"



This drawing is intended to be plotted at full size 11 x 17 paper size.

| Rev | Date     | By | Revision                          |
|-----|----------|----|-----------------------------------|
| 1   | 03/03/10 | SK | Revisions from Architect/Engineer |
| 2   | 03/18/10 | SK | Revisions from Client             |
| 3   | 03/18/10 | SK | Revisions from Architect/Engineer |
| 4   | 03/02/10 | SK | Revisions from Architect/Engineer |
| 5   | 02/21/11 | GR | Revisions from Architect/Engineer |

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**KIOSK #0002**  
**TWO POST SINGLE**  
**KIOSK**  
**Anchorage, AK**

|              |            |
|--------------|------------|
| Project Mgr. |            |
| Drawn        | SK         |
| Checked      |            |
| Date         | 03/04/2010 |

Sheet Contents:  
**ROOFING**  
**DETAILS**

Sheet No.:  
**A8.4**  
USKH W.O. 1258300



c:\pw-work\0573619\cnf000901\_pdr-g.dgn [Erosion Control Plan] 24 July 2024 8:52 AM

| EROSION CONTROL QUANTITIES |                                                    |       |            |
|----------------------------|----------------------------------------------------|-------|------------|
| ITEM                       | DESCRIPTION                                        | UNITS | QUANTITIES |
| 15705-0100                 | SOIL EROSION CONTROL, SILT FENCE                   | LNFT  | 180        |
| 15705-1400                 | SOIL EROSION CONTROL, FIBER ROLL                   | LNFT  | 2,140      |
| 15706-1600                 | SOIL EROSION CONTROL, STABILIZED CONSTRUCTION EXIT | EACH  | 1.0        |

| PROJECT               | SHEET NUMBER |
|-----------------------|--------------|
| AK USFS CHUG1004 9(1) | G.1          |

LOWER TRAIL LAKE

Place both silt fence and fiber roll along lake shore ensuring ends of each extend up-grade a minimum of 1-ft vertical and 10-ft horizontal from the shoreline placement.

Approximate Ordinary High Water

Clearing/Construction limits

Soil Erosion Control, Fiber Roll (typ.)

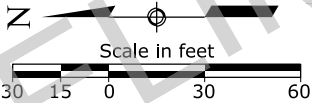
Soil Erosion Control, Fiber Roll (typ.)

Soil Erosion Control, Silt Fence

Soil Erosion Control, Stabilized Construction Exit

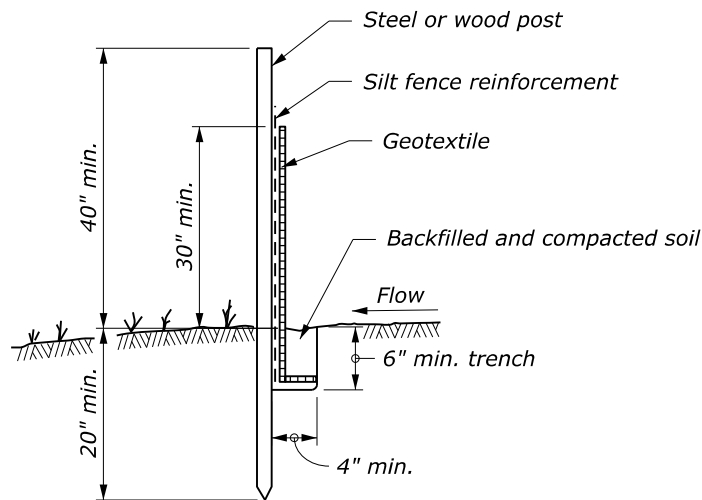
Soil Erosion Control, Fiber Roll (typ.)

ALASKA RAILROAD COMPANY RIGHT-OF-WAY

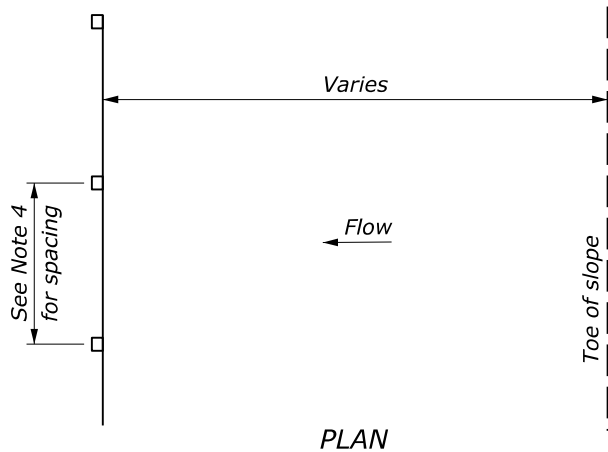


- Legend**
- Silt Fence
  - Wattles / Fiber Roll

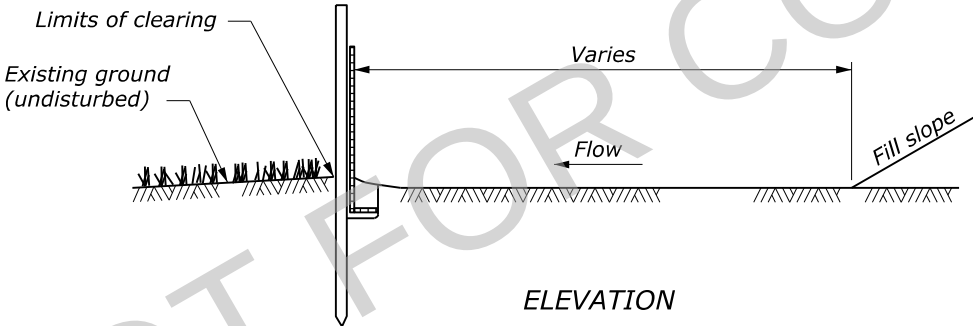
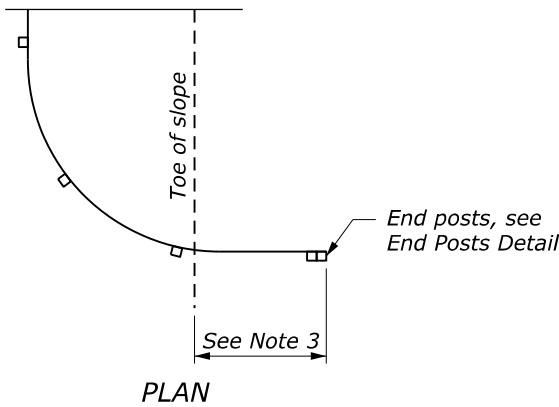
EROSION CONTROL PLAN



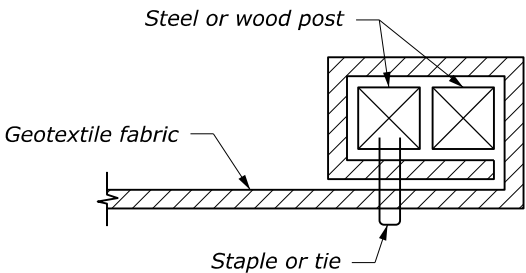
POST AND GEOTEXTILE INSTALLATION DETAIL



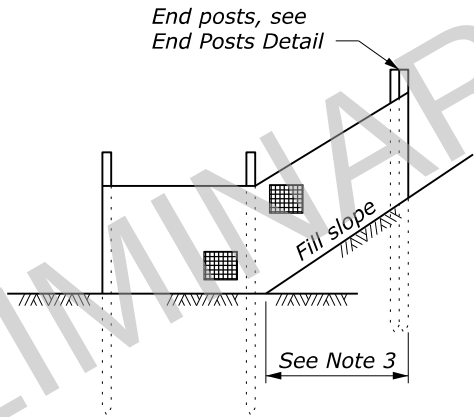
- NOTE:**
1. Alternate preassembled silt fence options will be allowed as long as specified dimensions are satisfied. Follow manufacturer's recommendations for installation procedures. All types must ensure silt fence remains attached to, and does not slide down, supporting posts.
  2. Install silt fence to follow the ground contours as closely as possible.
  3. Curve ends of silt fence upgrade to prevent water from running around the ends.
  4. 10-foot (max.) spacing with silt fence reinforcement.  
6-foot (max.) spacing without silt fence reinforcement.



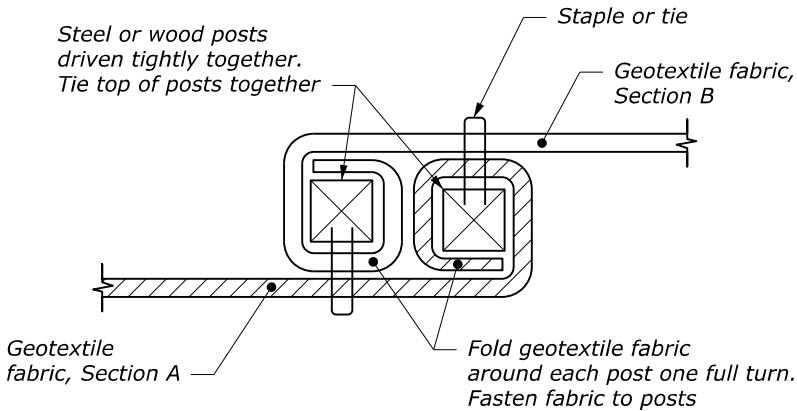
SILT FENCE INSTALLATION AT TOE OF FILL



END POSTS DETAIL



END DETAIL



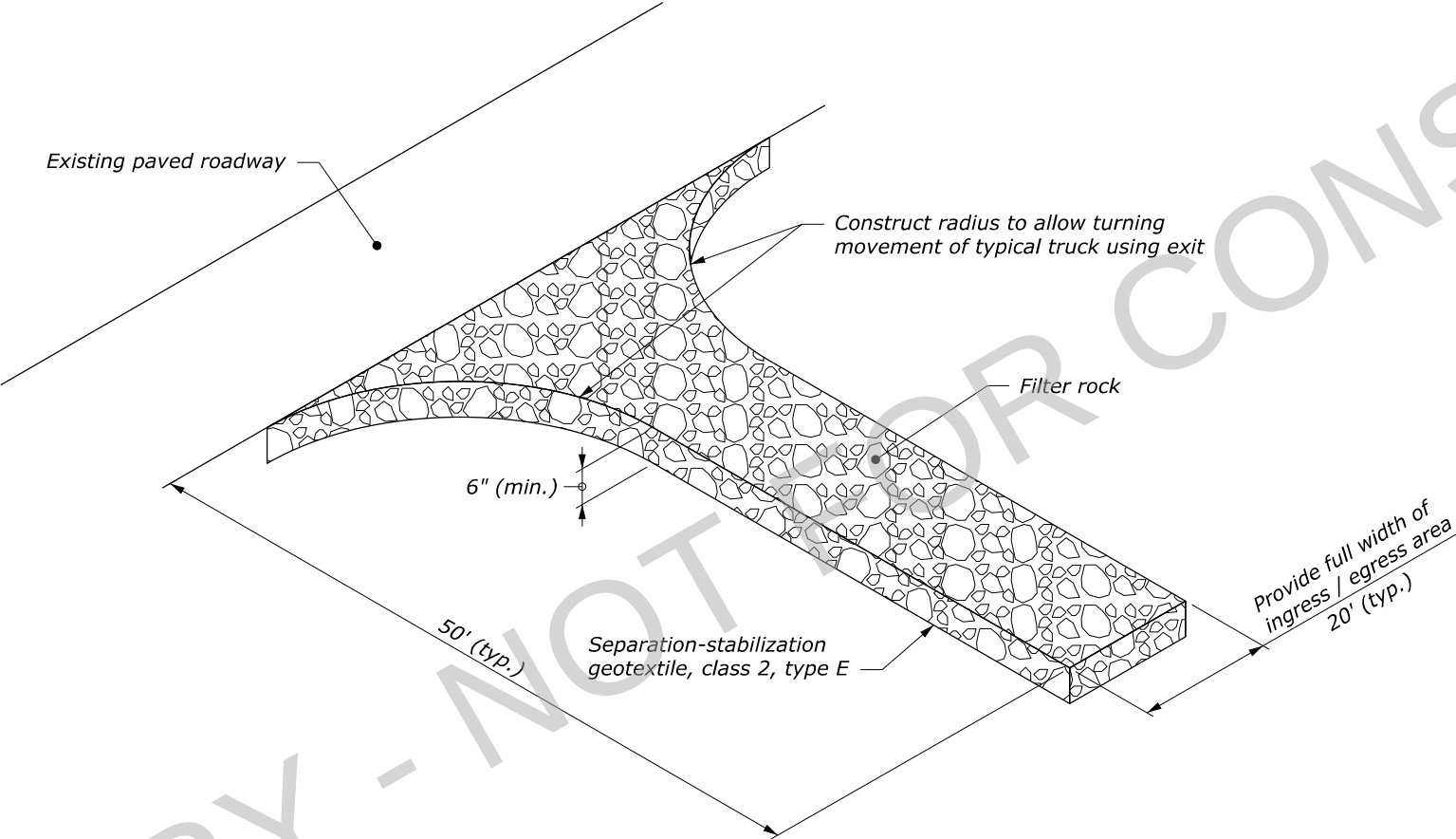
POSTS AT JOINTS

NO SCALE

|                                                                                                        |                  |
|--------------------------------------------------------------------------------------------------------|------------------|
| U.S. DEPARTMENT OF TRANSPORTATION<br>FEDERAL HIGHWAY ADMINISTRATION<br>OFFICE OF FEDERAL LANDS HIGHWAY |                  |
| WFLHD DETAIL                                                                                           |                  |
| SILT FENCE                                                                                             |                  |
| DETAIL APPROVED FOR USE 10/2016<br>REVISED:                                                            | DETAIL<br>W157-1 |

NOTE:

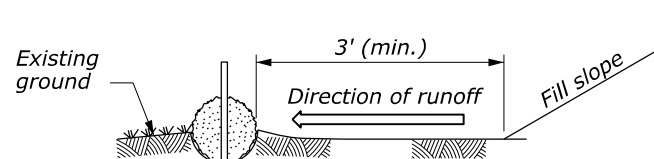
- 1. Use this entrance for construction vehicles only.
- 2. Construct drainage ditches along entrance as directed. Provide temporary drainage culvert where entrance crosses existing drainage ditches.
- 3. Minimize tracking onto paved roadway by removing built up sediment.
- 4. Adjust length to fit field conditions as approved.



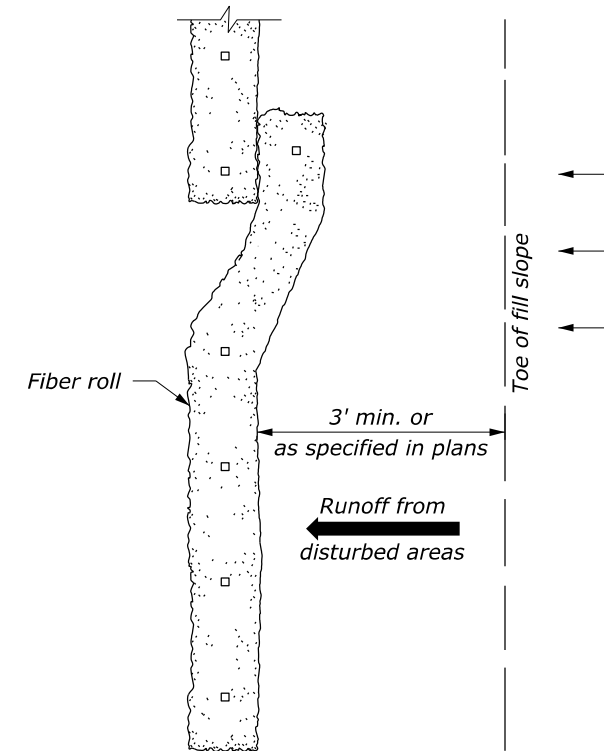
STABILIZED CONSTRUCTION EXIT

NO SCALE

|                                                                                                        |                   |
|--------------------------------------------------------------------------------------------------------|-------------------|
| U.S. DEPARTMENT OF TRANSPORTATION<br>FEDERAL HIGHWAY ADMINISTRATION<br>OFFICE OF FEDERAL LANDS HIGHWAY |                   |
| WFLHD DETAIL                                                                                           |                   |
| STABILIZED<br>CONSTRUCTION EXIT                                                                        |                   |
| DETAIL APPROVED FOR USE 7/2016<br>REVISED:                                                             | DETAIL<br>W157-19 |

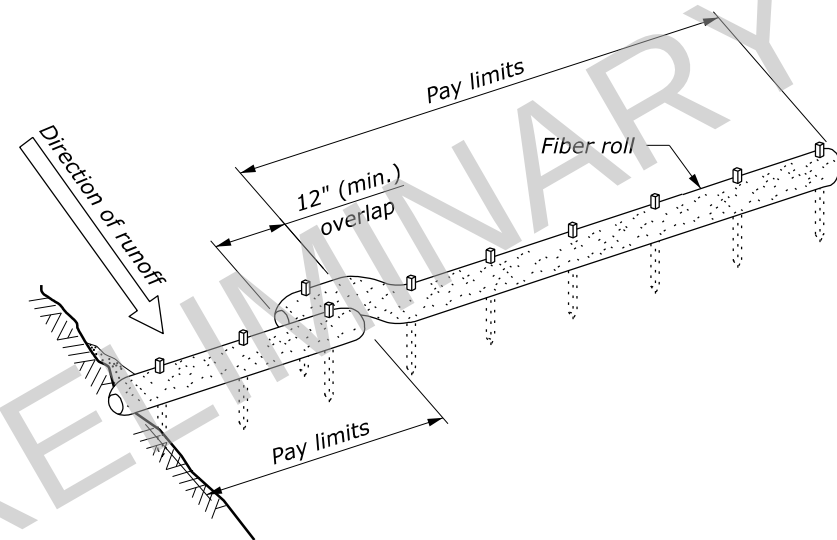


ELEVATION

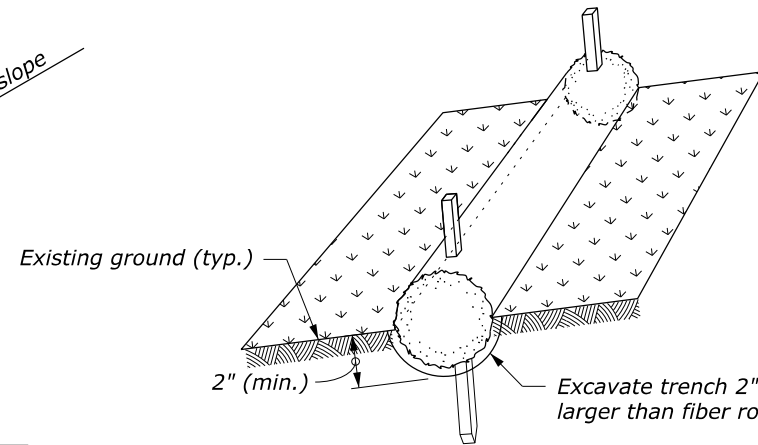


PLAN

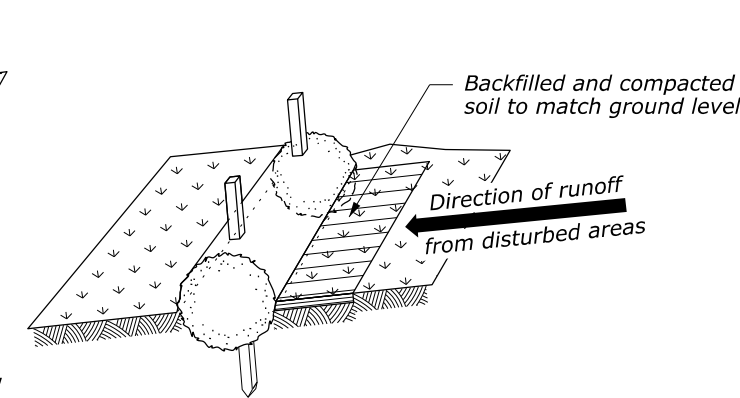
INSTALLATION BEYOND TOE OF SLOPE



ALTERNATE FIBER ROLL JOINT DETAIL  
SLOPE PROTECTION INSTALLATION



Step 1: Excavate trench  
and install fiber rolls



Step 2: Backfill soil against fiber rolls

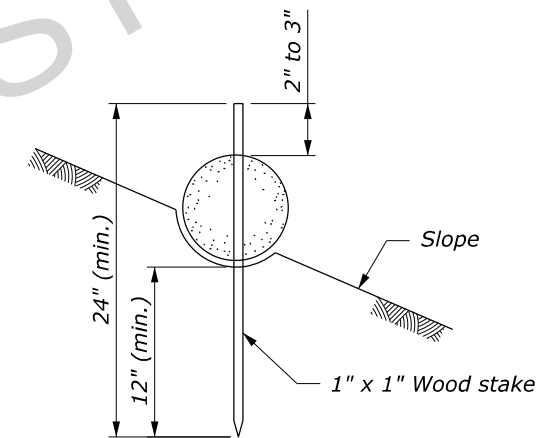
PROPERLY STAKED AND ENTRENCHED FIBER ROLL

| FIBER ROLL SPACING |              |
|--------------------|--------------|
| Slope              | Spacing (FT) |
| 1:4 or flatter     | 40           |
| 1:3                | 30           |
| 1:2                | 20           |
| 1:1                | 10           |

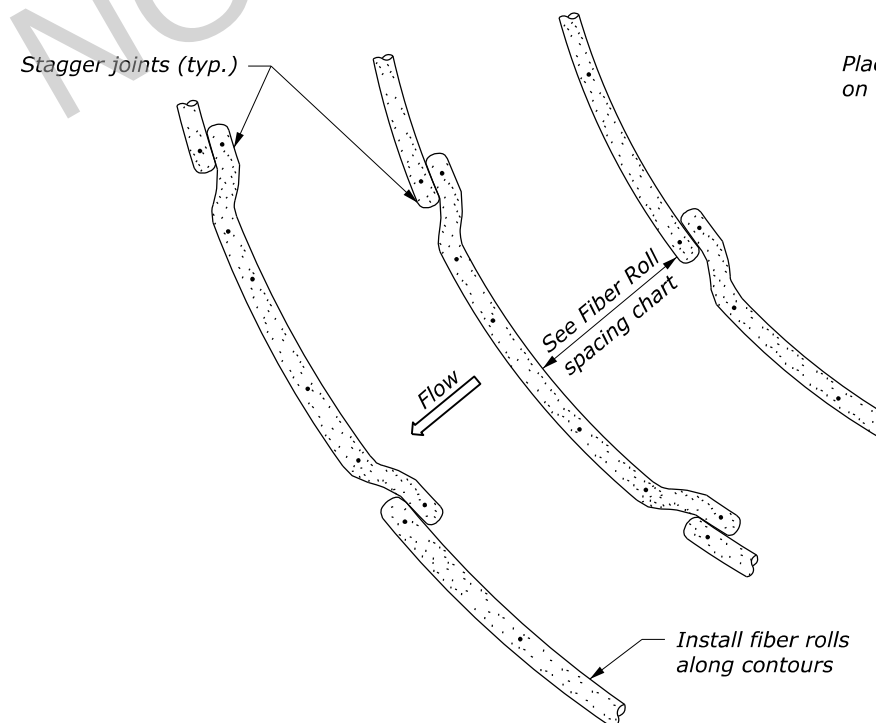
| STAKES REQUIRED        |                               |
|------------------------|-------------------------------|
| Fiber roll length (FT) | Stakes required for each roll |
| 25                     | 8                             |
| 20                     | 6                             |
| 12                     | 4                             |

NOTE:

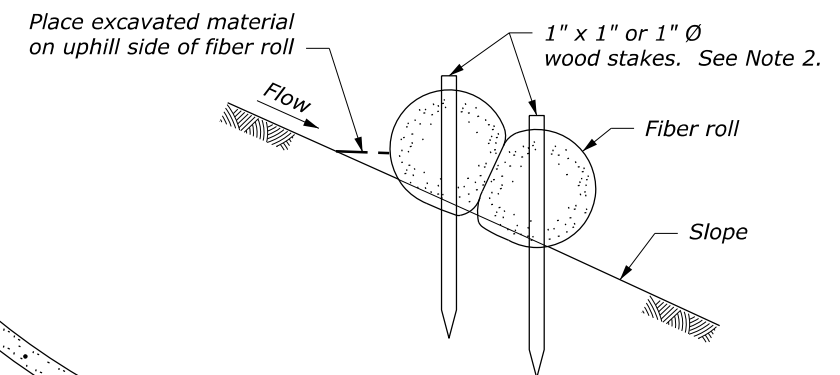
1. Drive stakes at each end and at 4-foot spacing until fiber roll is secure to slope. Live stakes may be used for permanent installations. Do not crush fiber roll while staking.
2. Overlap fiber rolls 12-inch minimum. Drive stakes at 6-inches from fiber roll end angles towards the adjacent fiber roll and space stakes at 4-foot max.



FIBER ROLL STAKING DETAIL



INSTALLATION ALONG SLOPES



FIBER ROLL LAPPING DETAIL

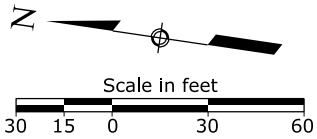
|                                                                                                        |                   |
|--------------------------------------------------------------------------------------------------------|-------------------|
| U.S. DEPARTMENT OF TRANSPORTATION<br>FEDERAL HIGHWAY ADMINISTRATION<br>OFFICE OF FEDERAL LANDS HIGHWAY |                   |
| WFLHD DETAIL                                                                                           |                   |
| FIBER ROLL                                                                                             |                   |
| DETAIL APPROVED FOR USE 10/2014<br>REVISED: 7/2016                                                     | DETAIL<br>W157-21 |

NO SCALE

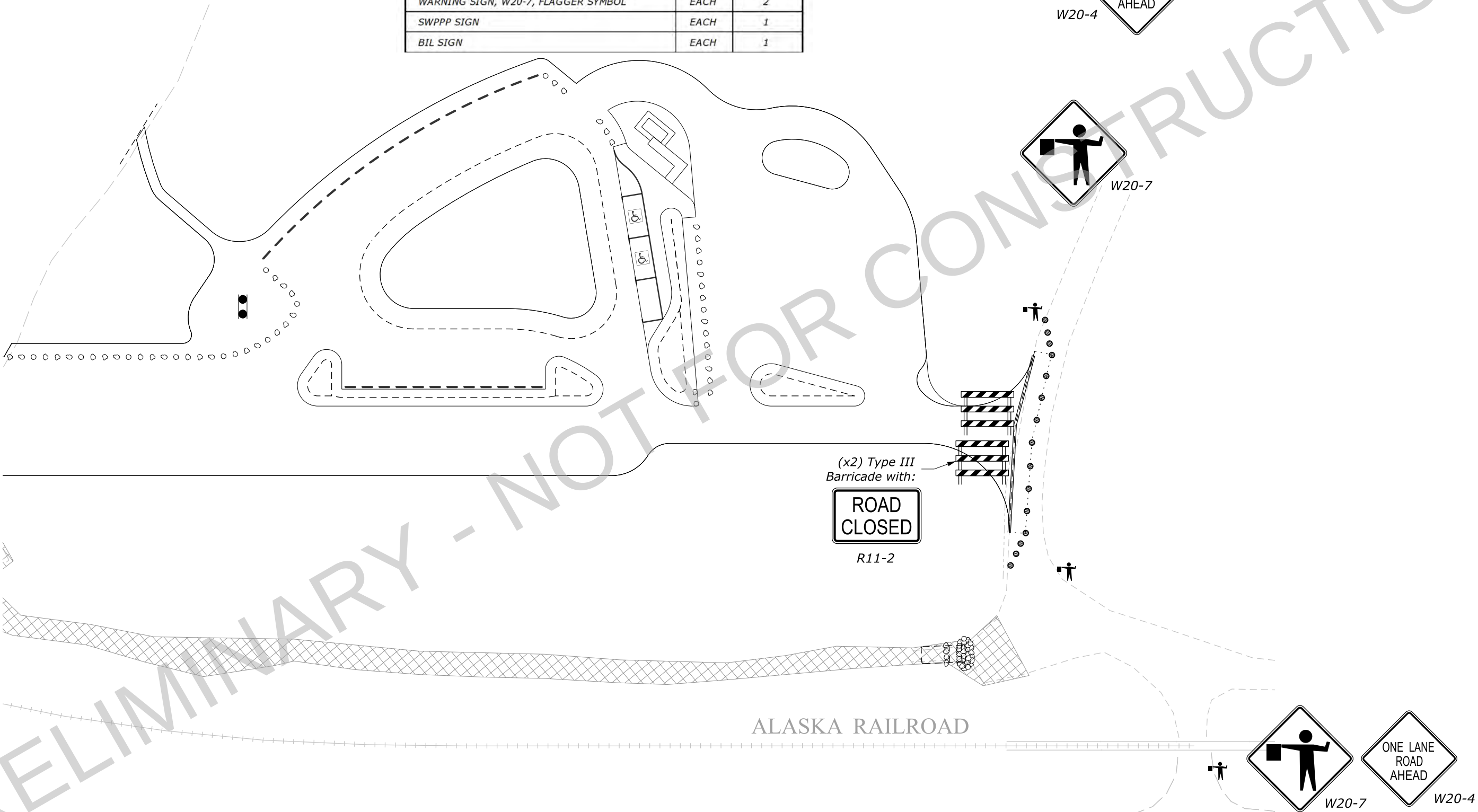
c:\pw-work\0573621\cnf00901\_pht-L.dgn [Temporary Traffic Control Plan] 9 May 2024 10:19 AM

| TEMPORARY TRAFFIC CONTROL QUANTITIES |                           |      |          |
|--------------------------------------|---------------------------|------|----------|
| ITEM                                 | DESCRIPTION               | UNIT | QUANTITY |
| 63501-0000                           | TEMPORARY TRAFFIC CONTROL | LPSM | ALL      |

| TEMPORARY TRAFFIC CONTROL<br>(INFORMATION ONLY QUANTITIES) |      |          |
|------------------------------------------------------------|------|----------|
| DESCRIPTION                                                | UNIT | QUANTITY |
| BARRICADE, TYPE III                                        | EACH | 2        |
| REGULATORY SIGN PANEL, R11-2, "ROAD CLOSED"                | EACH | 2        |
| WARNING SIGN, W20-4, "ONE LANE ROAD AHEAD"                 | EACH | 2        |
| WARNING SIGN, W20-7, FLAGGER SYMBOL                        | EACH | 2        |
| SWPPP SIGN                                                 | EACH | 1        |
| BIL SIGN                                                   | EACH | 1        |



|                       |              |
|-----------------------|--------------|
| PROJECT               | SHEET NUMBER |
| AK USFS CHUG1004 9(1) | L.1          |



TEMPORARY TRAFFIC CONTROL  
PLAN

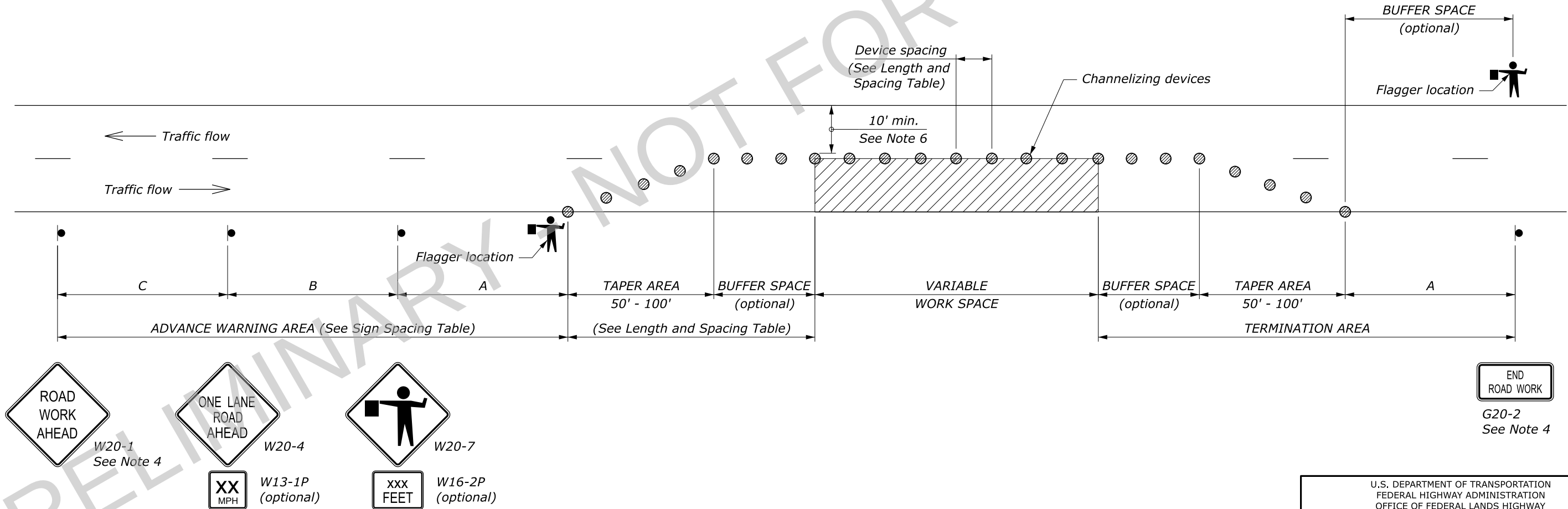
| LENGTH AND SPACING TABLE |                           |                     |                 |               |
|--------------------------|---------------------------|---------------------|-----------------|---------------|
| APPROACH<br>SPEED*       | BUFFER<br>SPACE<br>LENGTH | CHANNELIZING DEVICE |                 |               |
|                          |                           | TAPER<br>AREA       | BUFFER<br>SPACE | WORK<br>SPACE |
| MPH                      | FEET                      | SPACING IN FEET     |                 |               |
| 20                       | 115                       | 20                  | 40              | 40            |
| 25                       | 155                       | 20                  | 50              | 50            |
| 30                       | 200                       | 20                  | 60              | 60            |
| 35                       | 250                       | 20                  | 70              | 70            |
| 40                       | 305                       | 20                  | 80              | 80            |
| 45                       | 360                       | 20                  | 90              | 90            |
| 50                       | 425                       | 20                  | 100             | 100           |
| 55                       | 495                       | 20                  | 110             | 110           |
| 60                       | 570                       | 20                  | 120             | 120           |
| 65                       | 645                       | 20                  | 130             | 130           |
| 70                       | 730                       | 20                  | 140             | 140           |

\* Approach speed based on the regulatory posted speed, not the advisory speed.

| SIGN SPACING TABLE               |                                   |      |      |
|----------------------------------|-----------------------------------|------|------|
| ROAD TYPE                        | DISTANCE BETWEEN<br>SIGNS IN FEET |      |      |
|                                  | A                                 | B    | C    |
| Urban and Rural 30 MPH and less  | 100                               | 100  | 100  |
| Urban and Rural 35 MPH to 50 MPH | 350                               | 350  | 350  |
| Rural greater than 50 MPH        | 500                               | 500  | 500  |
| Expressway / Freeway             | 1000                              | 1500 | 2640 |

NOTE:

- Signs are shown for one direction of travel only. Place signs similar to those depicted for the opposite direction of travel.
- Final location and spacing of devices may be changed to fit field conditions as approved by the CO.
- For pilot car operation, mount the PILOT CAR FOLLOW ME (G20-4) sign at a conspicuous location on the rear of vehicle. Prominently display the name of the Contractor on the pilot car.
- If closure is completely within the project limits, eliminate the ROAD WORK AHEAD (W20-1) and END ROAD WORK (G20-2) signs.
- For night time flagging operation, provide floodlighting at flagger stations.
- For project specific minimum width, refer to the Special Contract Requirements, Section 156.
- Do not allow equipment, materials, or vehicles to be parked or stored in the buffer space.



NO SCALE

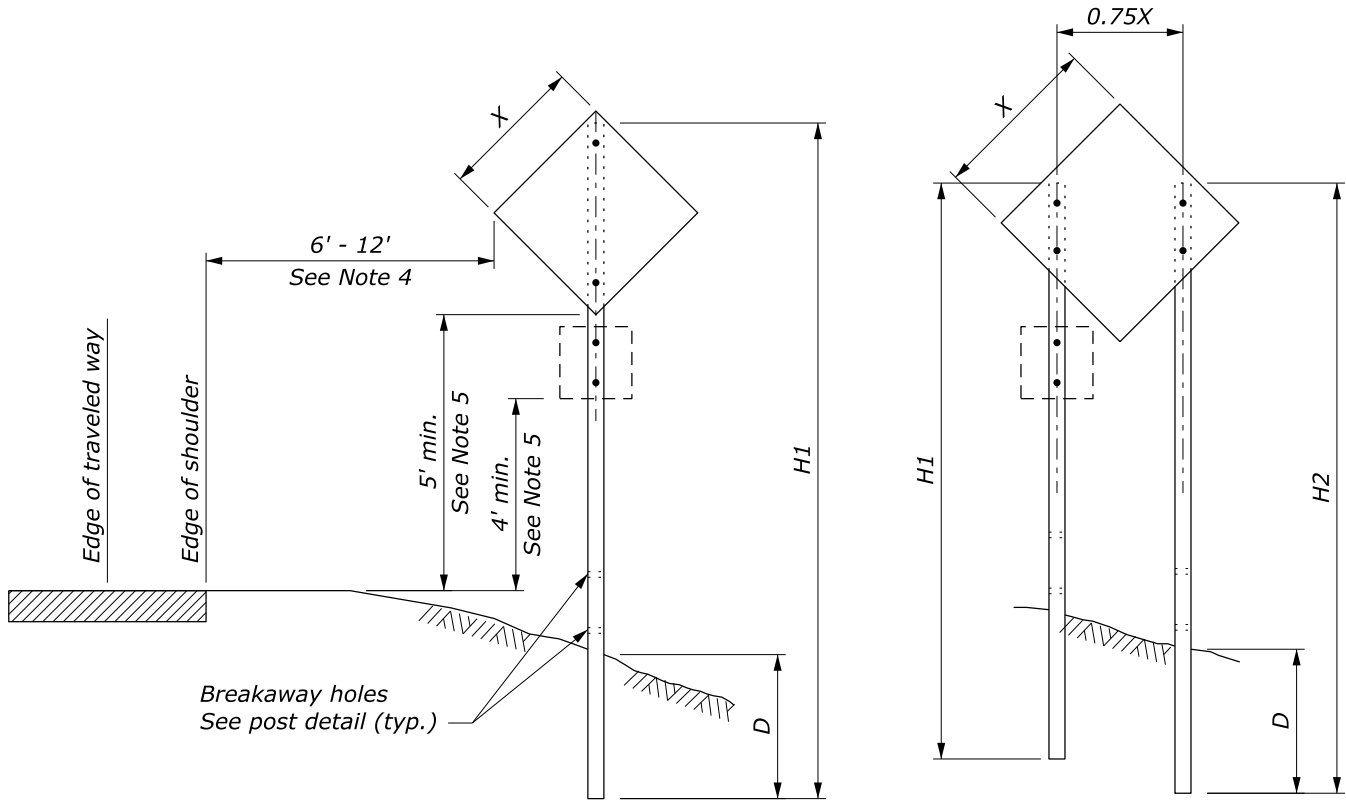


c:\pw-work\0573621\Std635-14.dgn [Std 635-14]  
3 June 2024 3:46 PM

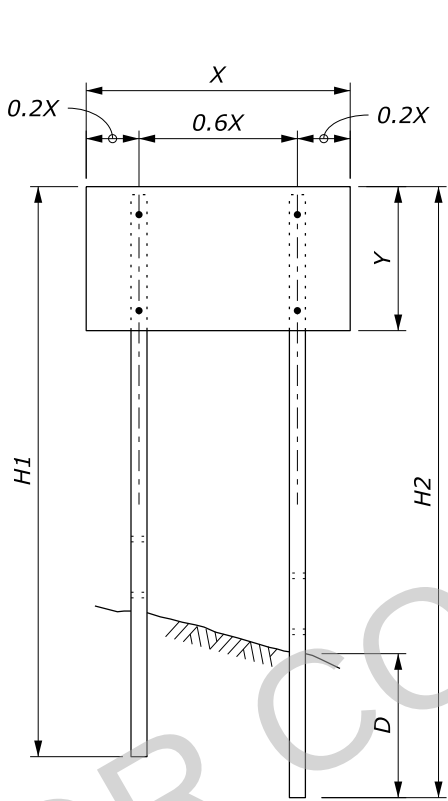
|                       |              |
|-----------------------|--------------|
| PROJECT               | SHEET NUMBER |
| AK USFS CHUG1004 9(1) | L.3          |

NOTE:

1. Attach sign panels with a minimum of two 1/4" diameter bolts per post.
2. H1 and H2 = Overall post length.  
Select post lengths to fit field conditions.
3. D = Post embedment depth for average soil conditions.
4. In areas where lateral distance is limited, a minimum lateral offset of 2 ft may be used. In areas with curbs, a minimum lateral distance of 1 ft behind the face of the curb may be used.
5. In pedestrian locations, or in areas with obstructed views, use 7 ft minimum mounting height for main sign and 6 ft minimum mounting height for secondary sign.
6. Use 7 ft minimum spacing between posts for sign posts 6" x 6" or larger.
7. State standards may be used as an alternative if approved.

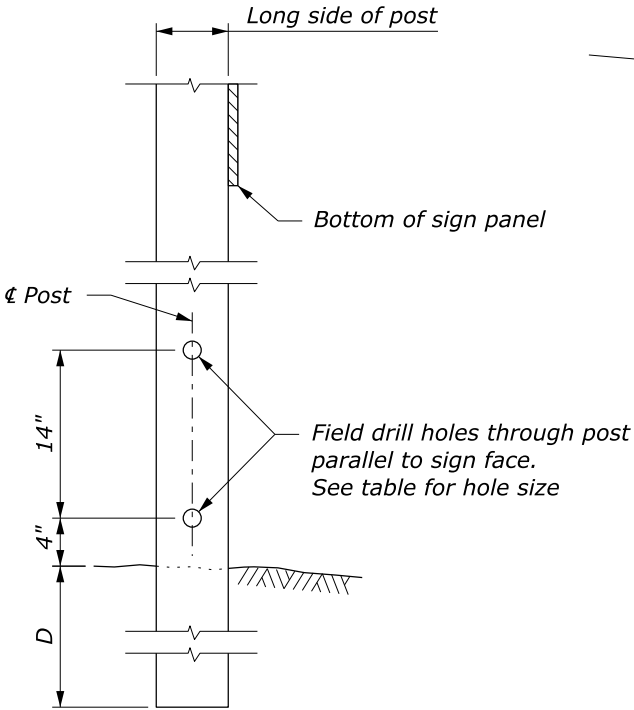


SINGLE POST SIGN

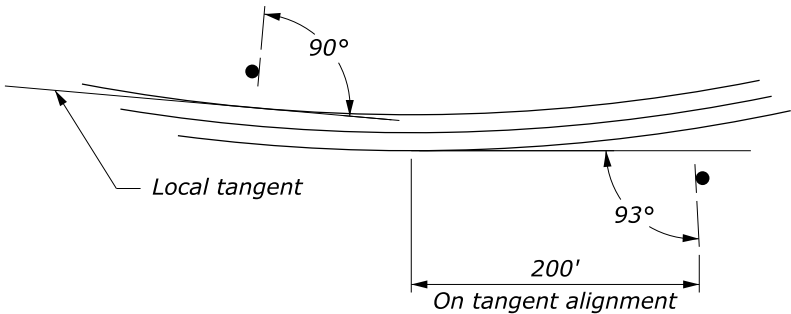


TWO POST SIGN

| WOOD POST SELECTION TABLE |             |                 |                  |          |                  |
|---------------------------|-------------|-----------------|------------------|----------|------------------|
| WIDTH "X"                 | AREA (SQFT) | NUMBER OF POSTS | POST SIZE (INCH) | D (INCH) | HOLE SIZE (INCH) |
| Diamond ≤ 36"             | < 10        | 1               | 4 x 4            | 36       | 0                |
| Other Shapes ≤ 48"        |             | 1               | 4 x 6            | 48       | 1.5              |
| Diamond ≤ 48"             | 10 - 20     | 1               | 6 x 6            | 48       | 2                |
| Diamond ≤ 48"             | 10 - 20     | 2               | 4 x 4            | 36       | 0                |
| Other Shapes ≤ 12'        | 20 - 50     | 2               | 4 x 6            | 48       | 1.5              |
| > 13'                     | 50 - 65     | 2               | 6 x 6            | 48       | 2                |
| 12' - 16'                 | 50 - 65     | 3               | 4 x 6            | 48       | 1.5              |
| > 17'                     | 65 - 95     | 4               | 4 x 6            | 48       | 1.5              |
| > 30'                     | 65 - 95     | 3               | 6 x 6            | 48       | 2                |



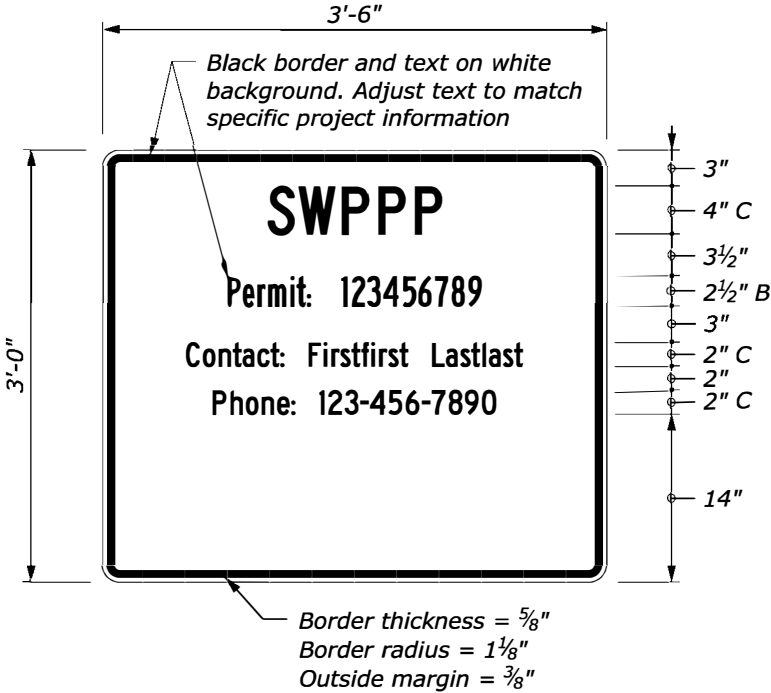
POST DETAIL



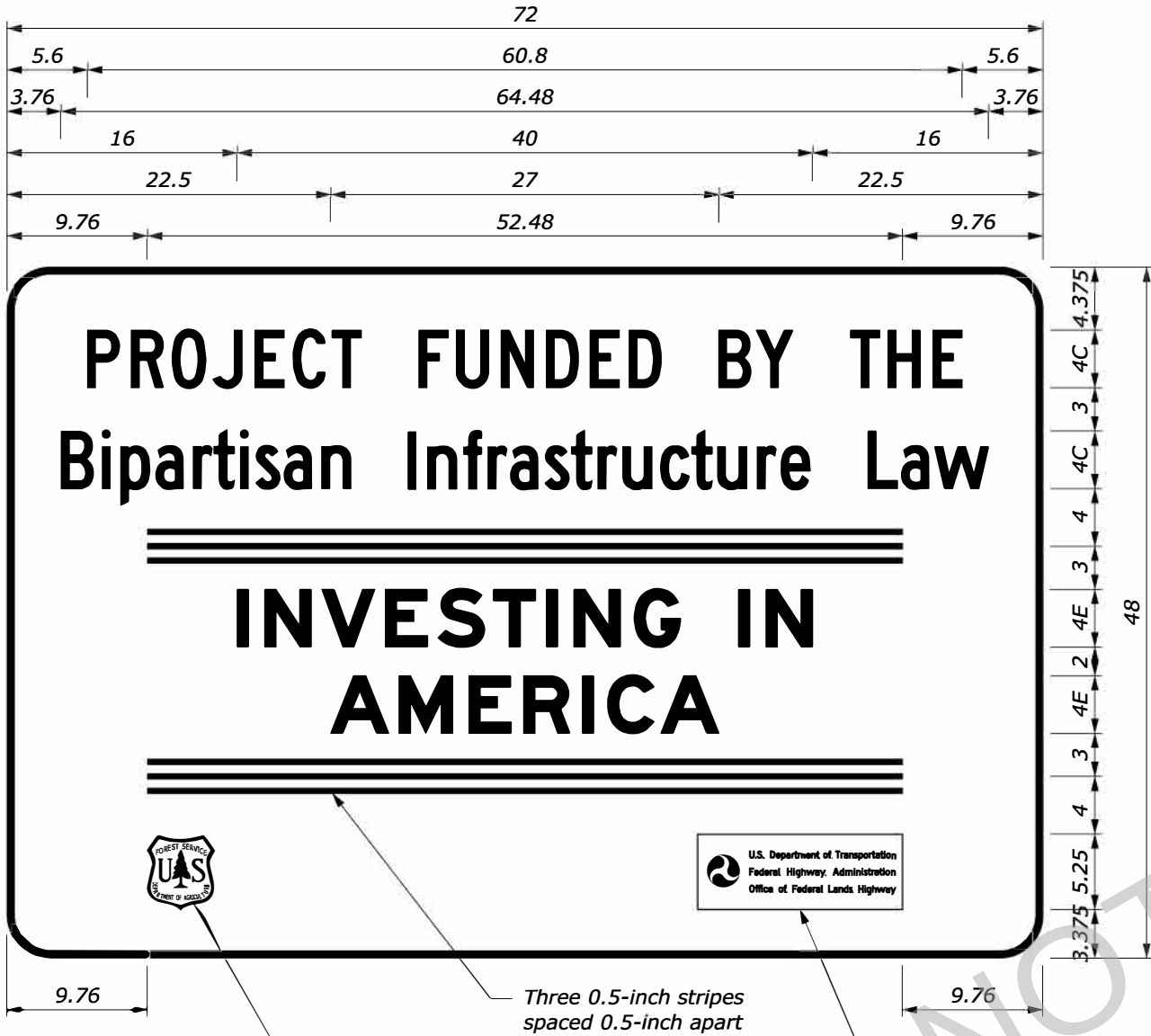
SIGN INSTALLATION ANGLE

|                                                                            |                                                             |
|----------------------------------------------------------------------------|-------------------------------------------------------------|
| U.S. DEPARTMENT OF TRANSPORTATION, FHWA<br>OFFICE OF FEDERAL LANDS HIGHWAY | FLH STANDARD<br>635-14                                      |
| TEMPORARY TRAFFIC CONTROL<br>SIGN INSTALLATION<br>WOOD POSTS               | SPECIFICATION<br>FP-24, FP-14<br>APPROVED FOR USE<br>2/2024 |

NO SCALE



**SWPPP Sign**  
Refer to Subsection 107.01A

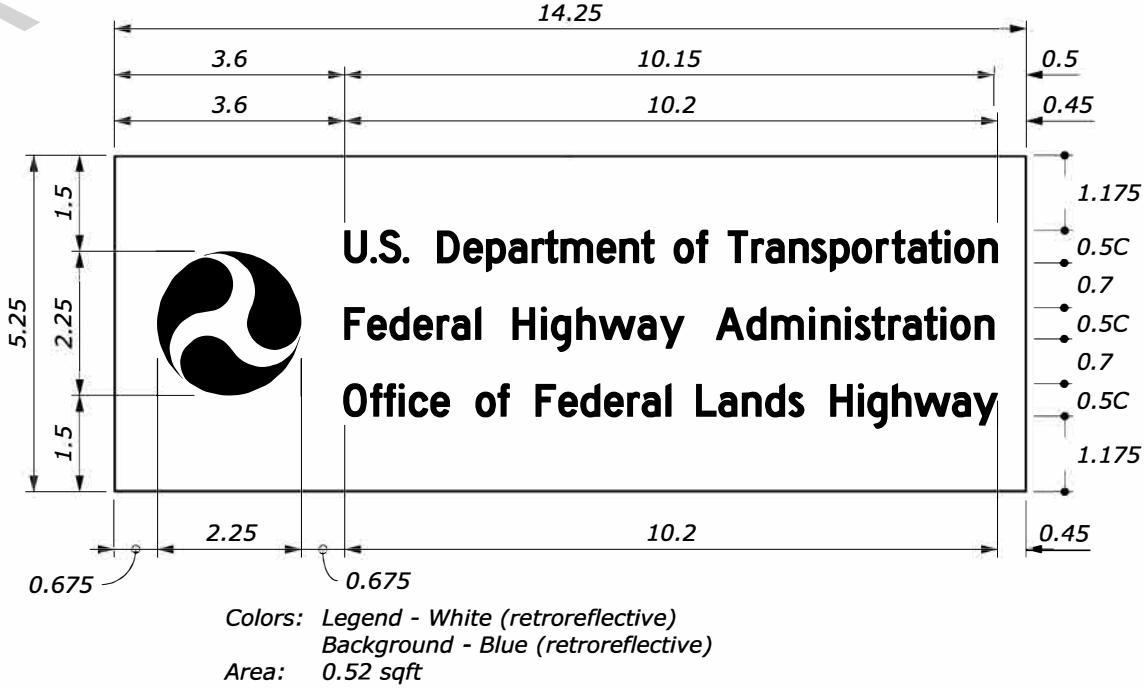


Dimensions: 2.3" Radius, 1" Border  
Colors: Legend, Border - White (retroreflective)  
Background - Green (retroreflective)  
Area: 24 sqft

**BIL PROJECT FUNDING SOURCE SIGN**



- NOTES:**
1. Unless otherwise noted, dimensions are inches.
  2. Refer to Standard 635-14 for post and sign mounting requirements.
  3. All incorporated pictographs are subject to approval. Ensure all pictographs are vector-based, vinyl-ready.

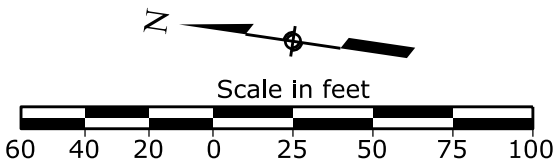
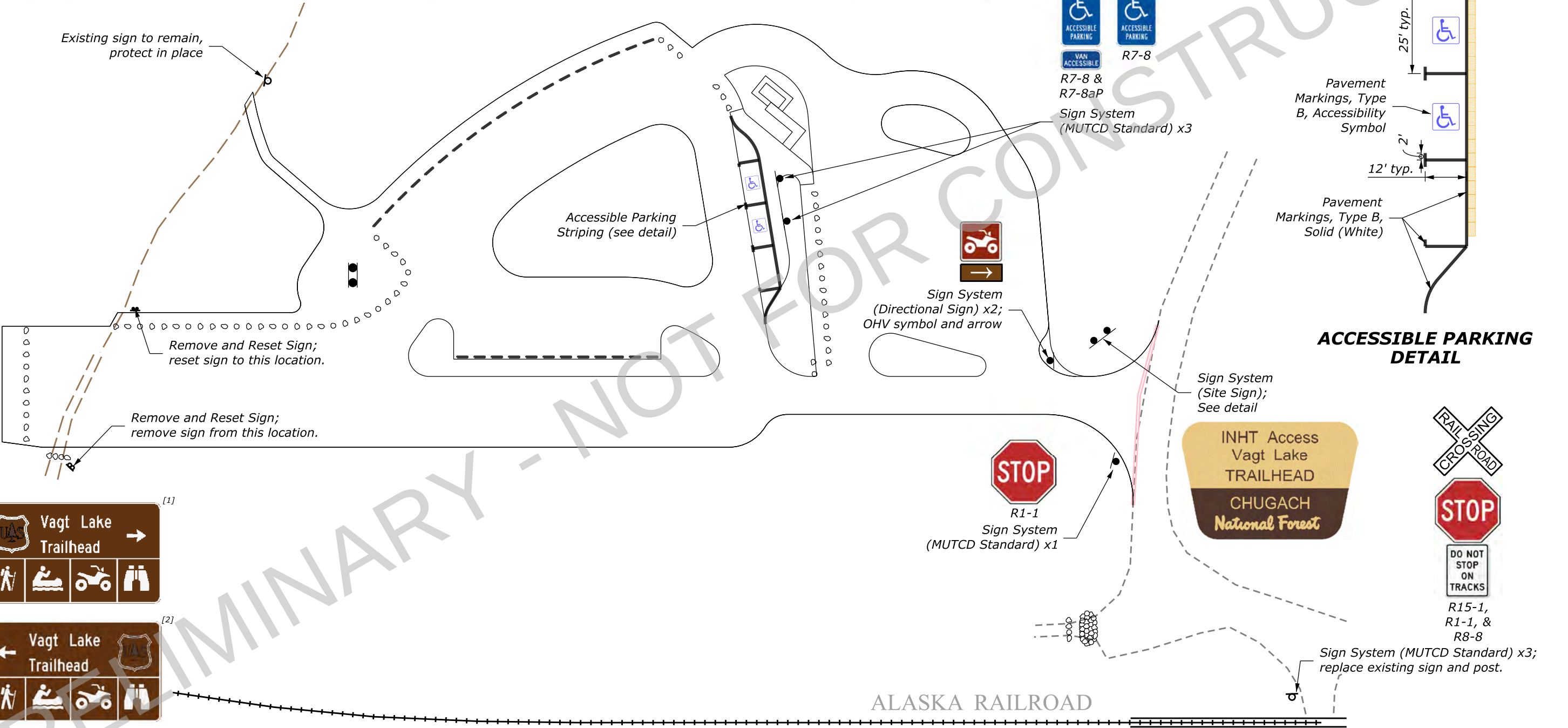


**Note:**  
Place sign conspicuously near the beginning of project, or in other location as directed by the CO, in full public view facing out away from the project toward public traffic.

**BIPARTISAN INFRASTRUCTURE  
LAW SIGN**

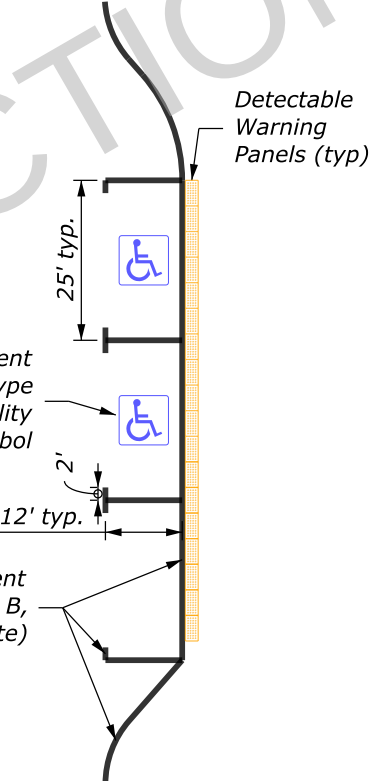
c:\pw-work\0573622\cnt000901\_01n-M.dgn [Permanent Traffic Control Plan] 24 July 2024 10:27 AM

| PERMANENT TRAFFIC CONTROL QUANTITIES |                                |       |            |            |                                                         |       |            |
|--------------------------------------|--------------------------------|-------|------------|------------|---------------------------------------------------------|-------|------------|
| ITEM                                 | DESCRIPTION                    | UNITS | QUANTITIES | ITEM       | DESCRIPTION                                             | UNITS | QUANTITIES |
| 61509-0000                           | DETECTABLE WARNING PANELS      | SQYD  | 16         | 63316-1000 | REMOVE AND RESET SIGN                                   | EACH  | 1          |
| 63301-0000                           | SIGN SYSTEM (MUTCD STANDARD)   | EACH  | 7          | 63401-0300 | PAVEMENT MARKINGS, TYPE B, SOLID (WHITE PARKING STRIPE) | LNFT  | 380        |
| 63301-0000                           | SIGN SYSTEM (DIRECTIONAL SIGN) | EACH  | 4          | 63405-0850 | PAVEMENT MARKINGS, TYPE B, ACCESSIBILITY SYMBOL         | EACH  | 2          |
| 63301-0000                           | SIGN SYSTEM (SITE SIGN)        | EACH  | 1          |            |                                                         |       |            |



| STATE | PROJECT            | SHEET NUMBER |
|-------|--------------------|--------------|
| AK    | USFS CHUG1004 9(1) | M.1          |

ACCESSIBLE PARKING DETAIL



**Notes:**

- All sign posts to be wood posts pressure treated for in-ground use. See 'Permanent Sign Installation Wood Posts' detail W633-7.
- Signs placed along highway to be set with a double-post. Meet the requirements of Alaska DOT for placement of signs along highways.

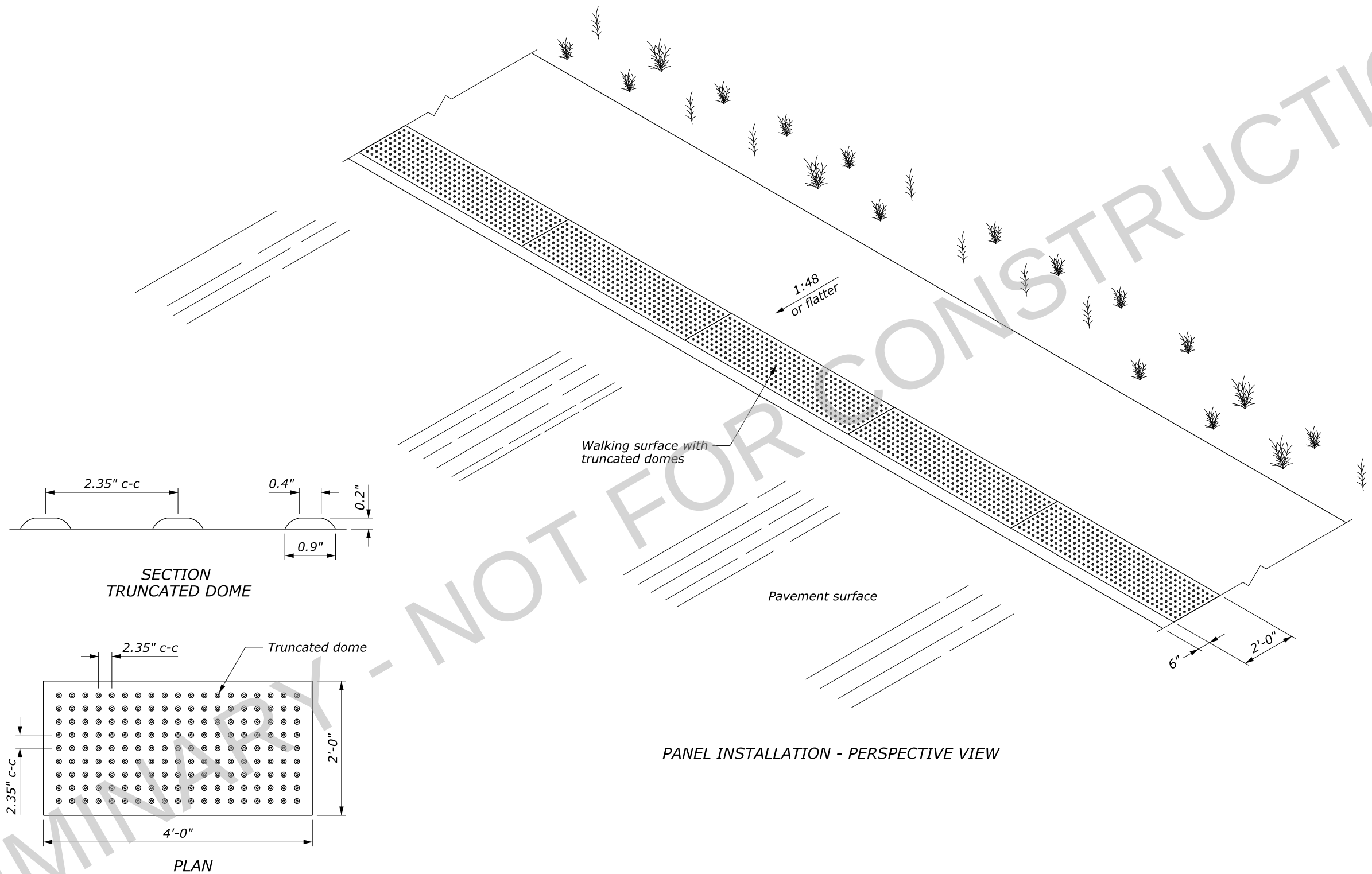
**Footnotes:**

[1] Sign System (Directional Sign); - Place sign along highway. Place sign with right arrow 1,000 ft. south of Crown Point Mine Road intersection on the roadside of the highway northbound lane.

[2] Sign System (Directional Sign); - Place sign along highway. Place sign with left arrow 1,000 ft. north of Crown Point Mine Road intersection on the roadside of the highway southbound lane.

**PERMANENT TRAFFIC CONTROL PLAN**

| STATE | PROJECT            | SHEET NUMBER |
|-------|--------------------|--------------|
| AK    | USFS CHUG1004 9(1) | M.2          |



**DETECTABLE WARNING PANEL WITH TRUNCATED DOMES**

**DETECTABLE WARNING PANEL  
DETAILS**



Recreation Site Identification Sign (RS-2)

COLOR LEGEND

(For all signs shown except DNR-1 and KING-1)

-  Brown (Federal Standard No. 20059)
-  Yellow cream (Federal Standard No. 23695)

NOTE:

- Ensure all Forest Service signing (FE, FL, RSM, P, and S signs) conforms with the current edition of EM-7100-15, Sign and Poster Guidelines for the US Forest Service.
- Legend letter series are as noted on the sign details. Standard federal logotypes are to be used for the National Forest legends on Forest Service signs.
- Refer to Chapters 7, 8, and 14 of EM-7100-15, Sign and Poster Guidelines for the US Forest Service for additional dimensions and manufacturing specifications for Forest Service signs.
- Preferred material is routed High-Density Polyethylene (HDPE). Refer to Chapter 14 of EM-7100-15 for specifications.

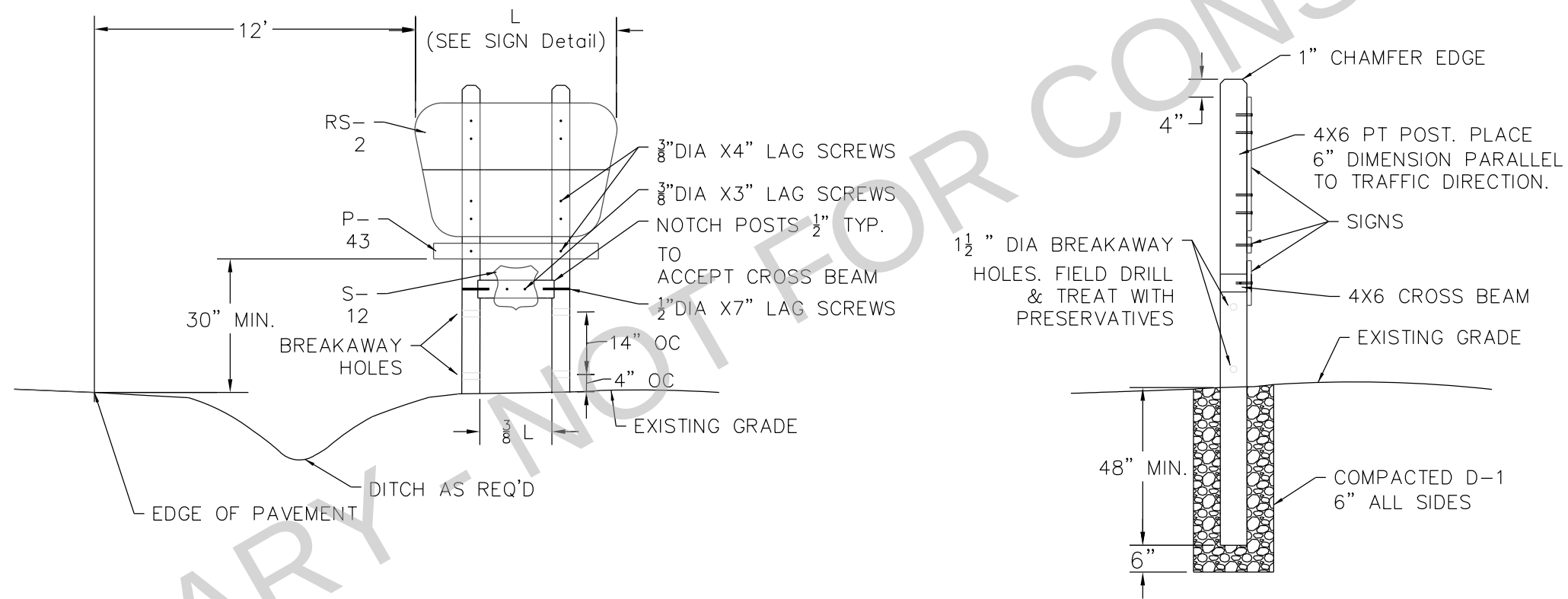


USDA Credit Line (P-43)



Forest Service Shield (S-12)

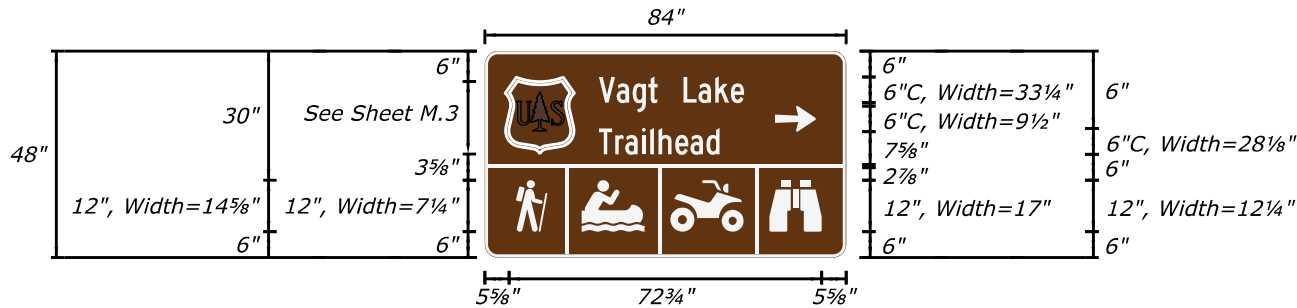




**SITE SIGN BASE DETAIL**

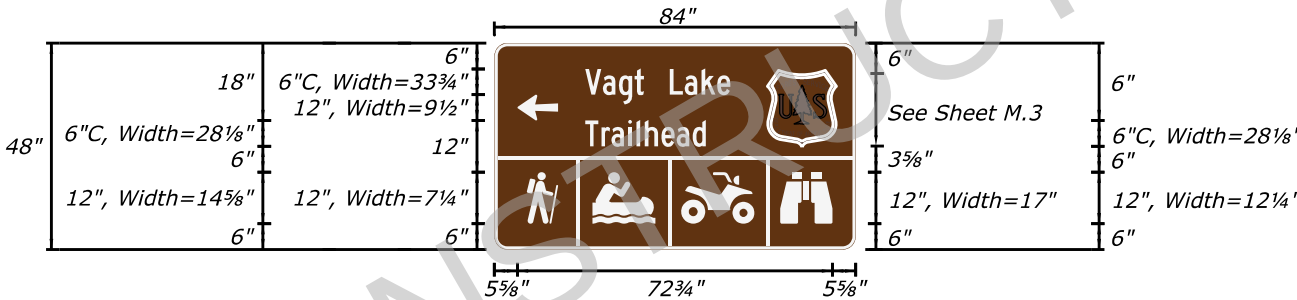
c:\pwwork\057362\Vagt Lake Custom Signs.dan [Custom Sign Details]  
9 May 2024 4:51 PM

| PROJECT               | SHEET NUMBER |
|-----------------------|--------------|
| AK USFS CHUG1004 9(1) | M.5          |



BORDER  
R=3"  
TH=0¾"  
DIVIDER  
TH=0¾"

REF-1



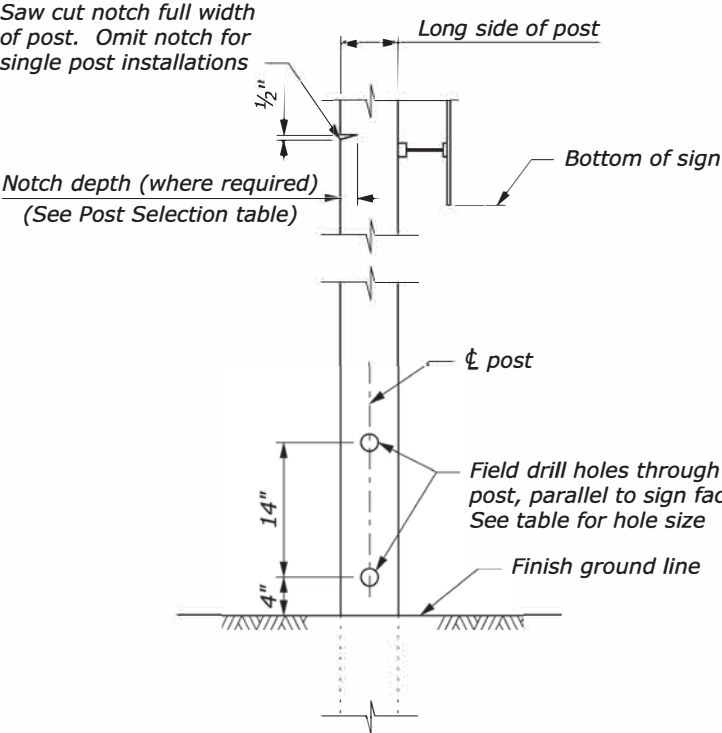
BORDER  
R=3"  
TH=0¾"  
DIVIDER  
TH=0¾"

REF-2

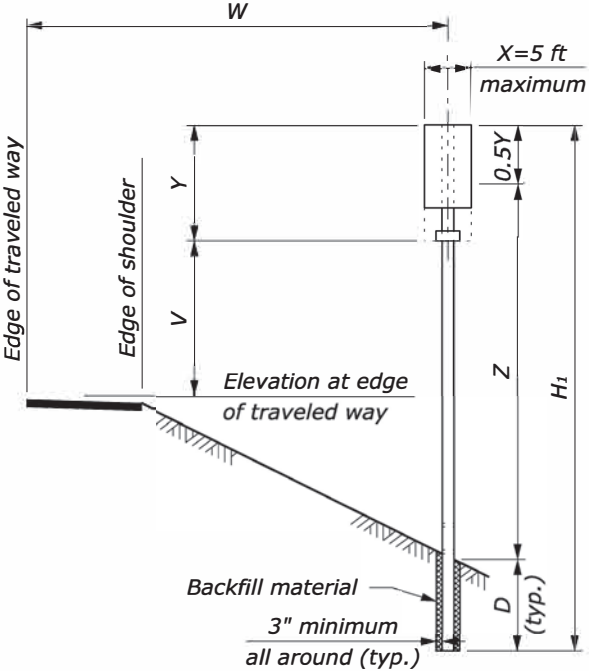
NOTE:  
1. Panel, letter, and symbol dimensions are approximate.

NO SCALE

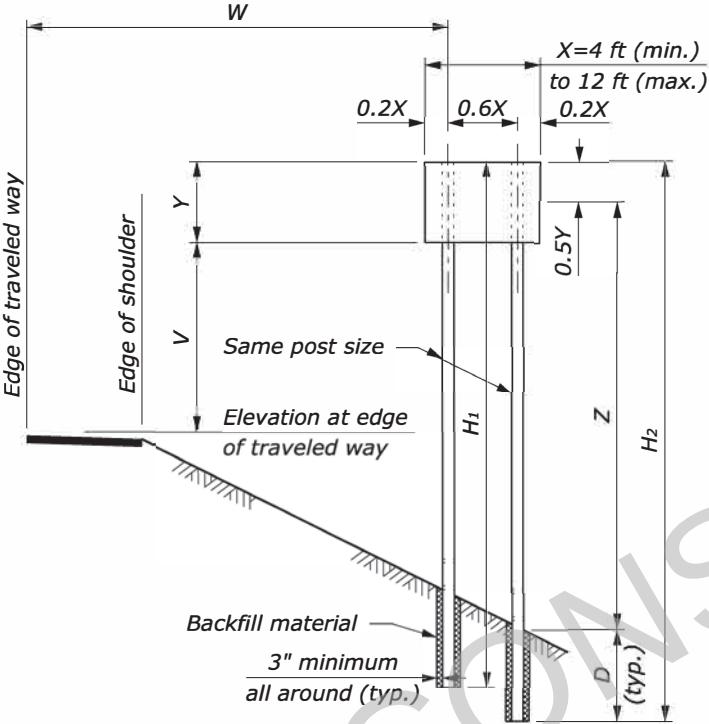
DIRECTIONAL SIGN  
DETAILS



POST DETAIL



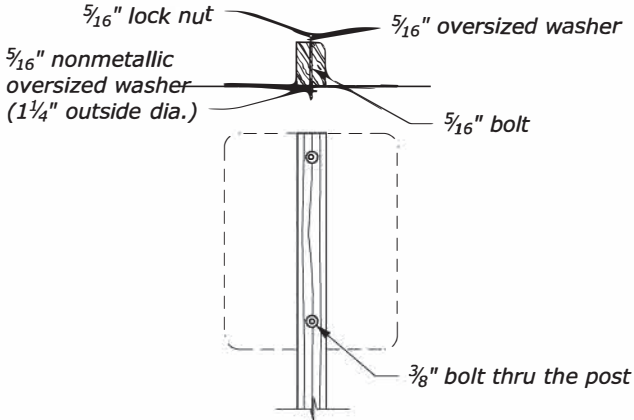
SINGLE POST SIGNS



TWO POST SIGNS

NOTE:

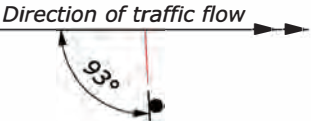
- Traffic barrier protection is required for all posts larger than 6" x 8" when located within the clear zone or if the post is vulnerable to being struck when placed outside the clear zone.
- H<sub>1</sub> thru H<sub>4</sub> indicate overall post length. Select post lengths to fit field conditions.
- D is the minimum post embedment depth for average soil conditions. See Wood Post Selection Table below.
- Z is the height from ground line to mid-height of sign at the longest post.
- For the purpose of post selection X and Y are as follows:
  - Single sign, or back to back signs: X and Y are the overall dimensions of the signs.
  - Multiple sign installations: X and Y are the dimensions of a rectangle enclosing all the signs.



TYPICAL MOUNTING FOR SIGNS WITHOUT ANGLES

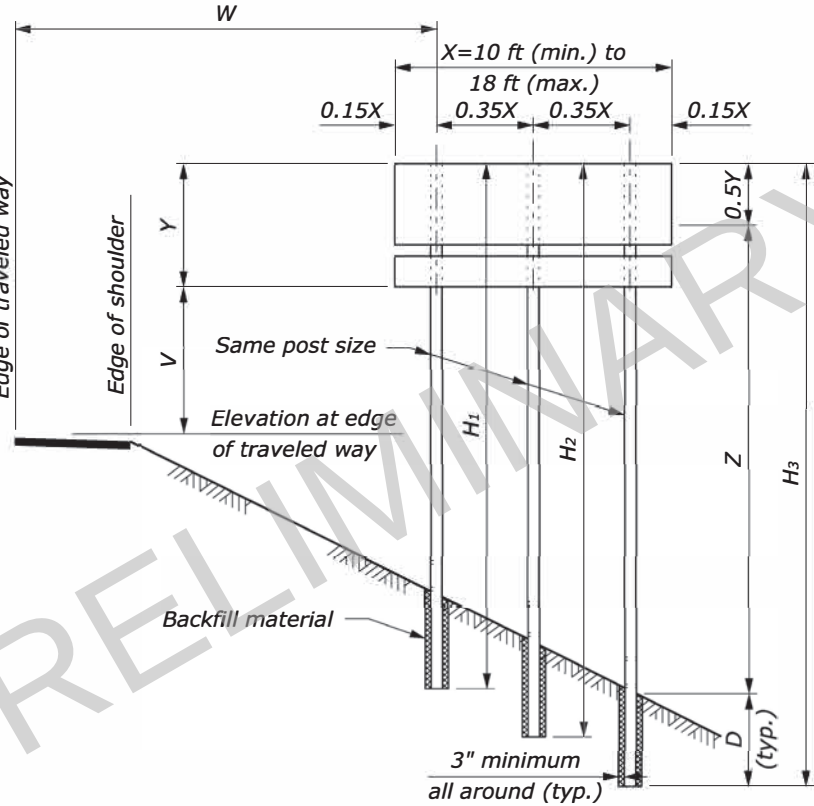
SIGN INSTALLATION ANGLE

For all retroflectorized signs where W > 25'

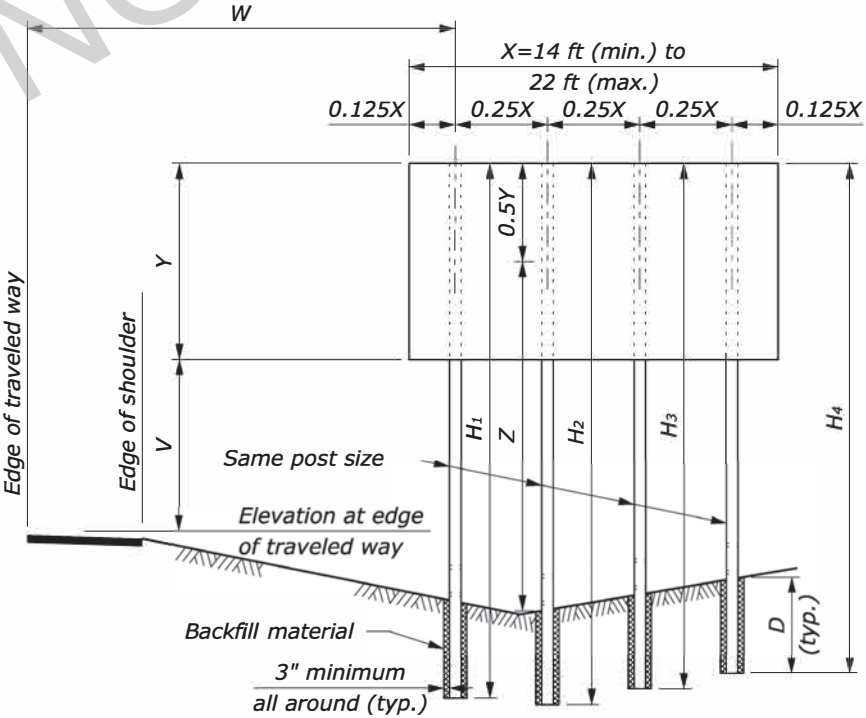


| MINIMUM DISTANCE TO SIGN        |                    |                     |
|---------------------------------|--------------------|---------------------|
| Location                        | Lateral Offset (W) | Mounting Height (V) |
| Rural Districts                 | 6 ft               | 5 ft                |
| Business or Residence Districts | 2 ft from curb     | 7 ft                |

V may be reduced by 1 foot in rural districts for a secondary sign mounted below another sign.



THREE POST SIGNS



FOUR POST SIGNS

| WOOD POST SELECTION TABLE |                          |      |      |      |       |                                        |
|---------------------------|--------------------------|------|------|------|-------|----------------------------------------|
| POST<br>SIZE<br>(inch)    | NUMBER OF POSTS          |      |      |      | D     | Notch<br>depth<br>and hole<br>diameter |
|                           | 1                        | 2    | 3    | 4    |       |                                        |
|                           | Product of X-Y-Z in CUFT |      |      |      |       |                                        |
| 4 x 4                     | 80                       | 155  | 235  | 310  | 3'-0" | -                                      |
| 4 x 6                     | 180                      | 385  | 545  | 725  | 4'-0" | 1¾"                                    |
| 6 x 6                     | 235                      | 475  | 710  | 950  | 4'-0" | 1¾"                                    |
| 6 x 8                     | 300                      | 850  | 1280 | 1700 | 4'-0" | 2½"                                    |
| 6 x 10                    | 385                      | 1180 | 1170 | 2360 | 5'-0" | -                                      |
| 8 x 10                    | 575                      | 1610 | 2410 | 3215 | 5'-0" | -                                      |
| 8 x 12                    | 775                      | 2310 | 3465 | 4620 | 6'-0" | -                                      |

Values shown are the maximum permitted. If the product of XYZ exceeds the limit for the largest post, use steel post installation.

U.S. DEPARTMENT OF TRANSPORTATION  
FEDERAL HIGHWAY ADMINISTRATION  
OFFICE OF FEDERAL LANDS HIGHWAY

WFLHD DETAIL

PERMANENT SIGN  
INSTALLATION  
WOOD POSTS

